Geometric Moments and Rational Lambert Values: Proofs and Further Results

Erdős Problem #1049

Will Cook*

September 2026

Abstract

We give complete proofs of the fixed-base Hankel asymptotic for Zudilin's normalised moments and the rational-base irrationality criterion $\log b/\log a < 0.4056830213840605\ldots$ for the Lambert series $F(a/b)$. The determinant proof separates geometric node configurations from their positive weights. The irrationality proof instead specialises the polynomial forms of Zudilin's 2004 construction after cancellation. We also examine coefficient moments, evaluated lattices and congruence conditions, distinguishing integer divisibility from nonzero remainder decay. A broader ratio-limit argument and a rational-value model are retained as unformalised remarks. None of these arguments settles $F(3/2)$.

1 Introduction

For $t > 1$, the Lambert series

$$F(t) = \sum_{n\geq 1} \frac{1}{t^n - 1} = \sum_{n\geq 1} \frac{\tau(n)}{t^n}$$

has two useful descriptions: as a sum of simple fractions and as a divisor generating series. Here $\tau(n)$ counts the positive divisors of n. The equality follows by expanding each fraction, since all terms are nonnegative; the bound $\tau(n) \leq n$ ensures convergence. Chowla conjectured that $F(t)$ is irrational for every rational $t > 1$, as recorded by Erdős [2, p. 102] and in Bloom's catalogue [26]. Erdős proved the integer case in 1948 [1].

Two constructions of Zudilin enter, for different reasons. The 2016 approximation [9] supplies positive moments, whose Hankel determinants we evaluate asymptotically. The 2004 forms [8] supply polynomial coefficients whose cancellation and rational evaluation give an irrationality region. A denominator estimate for the second construction is not an estimate for the first; each coefficient family needs its own arithmetic analysis. The shorter paper, *Hankel Determinants of Geometric Moments and Rational Lambert Values*, begins with the determinant result. Here the polynomial forms come first, making their exact arithmetic available for the later comparisons. Readers following only the

Authorship and AI use. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

determinant argument may start with Section 3: its moment calculation does not use the rational-base transfer.

For positive weights with fixed-shift ratios tending to 1 and a uniform polynomial bound on those ratios, we prove

$$\det\left(\sum_{k\geq 0} a_k q^{(i+j+1)k}\right)_{i,j<N} \sim \mathcal{M}(q)^3 q^{B_N} P^{2N} \prod_{k<N} a_k, \qquad 0<q<1,$$

where $P=(q;q)_\infty$, $B_N=\sum_{j<N} j^2$ and $\mathcal{M}(q)=\prod_{d\geq 1}(1-q^d)^{-d}$. Heine's identity expands the determinant into squared Vandermonde products. We divide by the contribution from the first N nodes and index the remaining tuples by partitions. A bound summable uniformly in N reduces their limiting sum to the unit-weight Cauchy determinant. For Zudilin's moments, two finite convolutions then give the correction $N^{-8F(1/q)}$ and its constant. We also determine the first formal term $C_N q^{B_N}$, where $C_N=(N!)^2(N+1)!/2^N$.

The formal first term fixes the rank and expands at $q=0$; the analytic asymptotic fixes q and lets the rank grow. Neither controls the coefficient denominators of an irrationality form. The ratio-limit remark extends the moment calculation to limiting adjacent-weight ratios, while Section 3.3 constructs a rational-value model with the same Hankel features. The latter shows why those features alone cannot imply irrationality. Both arguments are ordinary proofs without Lean proofs in the supplied record, and neither is used for the checked determinant asymptotic.

For the 2004 forms, we calculate the degree after cancelling the common polynomial factors. At a reduced rational base $a/b>1$, the resulting polynomials U_n, V_n have degrees W_n, W_n-1 and satisfy

$$\log\bigl(b^{W_n}(U_n(a/b)F(a/b)-V_n(a/b))\bigr)=(C_1\log b-C_0\log a)n^2+o(n^2).$$

The form inside the logarithm is positive. Thus it tends to zero when $\log b/\log a<\theta^*=C_0/C_1=0.4056830213840605\ldots$. Rationality of $F(a/b)$ would make a fixed integer multiple of every such value a positive integer, giving a contradiction. The constants and parameter direction are Zudilin's; the cancellation and rational evaluation give the region used here. It contains $31/4$ and all its positive integral powers. Bundschuh and Väänänen already treated a smaller rational-base region containing $7/2$ [6, Theorem 2, p. 177]; Section 8 checks that specialisation. The value $3/2$ lies outside both regions and remains unresolved by these arguments. It has the smallest height among reduced noninteger rationals greater than 1, where the height of r/s is $\max(r,s)$.

Section 2 gives the rational-base proof and its degree restrictions. Section 3 contains the determinant arguments and the coefficient-moment calculations. The coefficient pencil is treated for sizes $N\leq 8$ and every $p>1$; the computational ranges for shifted minors and coefficientwise positivity are specified there. Positivity for every N and convergence of the largest root remain open. Sections 4–5 apply rescaling and additive congruences to divided forms, including a quantitative bounded-fibre criterion. Sections 6–9 give the clearing, tail and exponent calculations, and Section 10 distinguishes unrestricted reformulations of irrationality from questions about specified approximation families.

All logarithms are natural, and empty products and determinants equal 1. Constants in $O_x(\cdot)$ may depend on the fixed base. We introduce the notation for each construction as it is needed. Appendix A identifies the formal sources and the separate finite computations.

Keywords. irrationality; Lambert series; rational base; Padé approximation; Lean 4.
MSC 2020. 11J72 (primary); 11J82, 68V20 (secondary).

2 The region $b^\mu < a$ and the base $31/4$

The rational-base calculation compares a gain in the remainder with the cost of making its coefficients integral. If the original coefficient has degree K_n and the cancelled coefficient has degree W_n, the leading balance is

$$b^{W_n}(a/b)^{-(K_n-W_n)} = b^{K_n}a^{-(K_n-W_n)}. \tag{1}$$

The polynomial identities and remainder estimates below justify this balance, including its lower-order factors. With $K_n/n^2 \to C_1$ and $(K_n - W_n)/n^2 \to C_0$, the cleared remainder tends to zero when $C_1 \log b < C_0 \log a$. The degree drop has two effects: it improves the real remainder and changes the denominator needed to make its coefficients integral. Both effects concern the same cancelled form. This is why one must calculate the degree before rational evaluation, rather than borrow a denominator bound from the uncancelled family.

We evaluate Zudilin's 2004 forms at a/b after cancelling their common polynomial factors. His direction $(14, 12, 14; 27)$ [8, Sec. 5, pp. 161–162] gives the constants below; the comparison between decay and the remaining degree determines the irrationality region:

$$C_1 = (14+12+14)27 - \tfrac{1}{2}(12^2+14^2+27^2) = \frac{1091}{2}, \qquad C_0 = 266 - \frac{3}{\pi^2}(225 - J),$$

$$J = \sum_{i=1}^{13}(\psi_1(u_i) - \psi_1(v_i)), \qquad \psi_1(x) = \sum_{k\ge 0}\frac{1}{(k+x)^2},$$

where ψ_1 is the trigamma function, used only for positive arguments, and the thirteen half-open intervals $[u_i, v_i)$ are $[1/14, 1/12)$, $[1/7, 1/6)$, $[3/14, 1/4)$, $[2/7, 1/3)$, $[5/14, 2/5)$, $[3/7, 7/15)$, $[1/2, 8/15)$, $[4/7, 3/5)$, $[9/14, 2/3)$, $[5/7, 11/15)$, $[11/14, 4/5)$, $[6/7, 13/15)$, $[13/14, 14/15)$. The intervals are disjoint and lie in $[1/14, 1)$, so $0 \le J \le \psi_1(1/14) - \psi_1(1) < 196$: in the defining series, $\psi_1(1/14) = 196 + \sum_{k\ge1}(k + 1/14)^{-2} < 196 + \psi_1(1)$. Since $\pi > 3$, it follows that $191 < C_0 < 266 < C_1/2$. In particular, the reciprocal constants below are positive without using decimal approximations. The accompanying rational certificate gives

$$\begin{aligned} 77.94318445520543 < J < 77.94318449473569,\\ 221.30008815898543 < C_0 < 221.30008817100119,\\ 0.40568302137302 < \theta^* := C_0/C_1 < 0.40568302139506. \end{aligned}$$

Put $\mu = 1/\theta^*$. These parameter ratios, intervals and constants, including $C_1/C_0 = 2.46497868\ldots$, occur in [8, p. 162]; that ratio is the integer-base exponent bound in its Theorem 1, not a new optimised constant. The displayed interval endpoints are rounded outwards from rational bounds; no digits beyond their certified accuracy are required.

The proof below needs J in its combinatorial form as well. Put

$$\omega(\xi) = \max\{0,\ \lfloor 14\xi\rfloor + \lfloor 13\xi\rfloor - \lfloor 12\xi\rfloor - \lfloor 15\xi\rfloor,\ 2\lfloor 14\xi\rfloor - \lfloor 13\xi\rfloor - \lfloor 15\xi\rfloor\},$$

the weight that [8, p. 162] attaches to the six-tuple $c = (13, 14, 12, 14, 15, 13)$ for the chosen parameter ratios; it gives the exponents $\nu_l = \omega(n/l)$ of the source's (22) and enters its limit (26). Zudilin lists the support of ω in the same place; the next lemma verifies that list by exact evaluation. Since $14 + 13 = 12 + 15$ and $2 \cdot 14 = 13 + 15$, each floor difference is unchanged by $\xi \mapsto \xi + 1$. Thus ω has period 1, and the lemma also determines every value $\omega(n/l)$ needed for the cyclotomic exponents.

Lemma 2.1 (the weight is an indicator)**.** *On* $[0, 1)$ *the function* ω *takes only the values* 0 *and* 1*, and* $\omega = 1$ *exactly on the union of the thirteen half-open intervals listed above.*

Proof. Each of the three expressions inside the maximum is constant on every interval between consecutive elements of $\{k/c : c \in \{12, 13, 14, 15\},\ 0 \le k \le c\}$, since those are the only points at which one of the four floor functions changes. There are 48 such intervals in $[0, 1)$; evaluating ω at the midpoint of each gives the value 1 on the thirteen listed half-open intervals and 0 elsewhere, and the thirteen listed intervals are unions of consecutive cells. Each floor function is right-continuous, so a cell has the same value at its left endpoint as at its midpoint. This also verifies the half-open endpoint convention. All evaluations use exact rational arithmetic. □

Lemma 2.1 and termwise integration of the positive series $-\psi_1'(\xi) = \sum_{k\ge 0} 2(k+\xi)^{-3}$ give the two forms of J used below,

$$J = \int_0^1 \omega(\xi) \sum_{k\ge 0} \frac{2}{(k+\xi)^3}\, d\xi = \sum_{i=1}^{13} (\psi_1(u_i) - \psi_1(v_i)),$$

the first being the integral $\int_0^1 \omega\, d(-\psi_1)$ of [8, Lemma 2, p. 155, and (26), p. 162]. Since ω vanishes near 0, both are finite.

For $b = 1$ the logarithmic ratio is zero, so the condition includes every integer base. For $b > 1$ it requires $a > b^{2.46497868\ldots}$, substantially more than $a > b$. There are still infinitely many admissible coprime numerators for each fixed denominator. The base $31/4$ is included, whereas $3/2$ is excluded. Taking positive integral powers does not change the ratio. At equality with the cutoff, the estimates give no conclusion.

Theorem 2.2 (rational-base region)**.** *Let* $a > b \ge 1$ *be coprime integers with*

$$b^{\mu} < a, \qquad \textit{equivalently} \qquad \frac{\log b}{\log a} < \theta^* = 0.40568302138406054\ldots$$

Then $F(a/b) = \sum_{m\ge 1} ((a/b)^m - 1)^{-1}$ *is irrational.*

The proof has three steps. We cancel the common polynomial factors using Zudilin's integrality argument [8, pp. 156–161]. We then compute the degree of the remaining polynomials, using his cyclotomic limits and the reciprocal-interval method of [7, Lemma 1, p. 466]. Finally, we evaluate at a/b and show that the positive remainder still decays after multiplication by the required power of b. These steps occupy Sections 2.1–2.4. The construction, direction and constants are Zudilin's; the rational specialisation and its degree and remainder estimates are proved below.

2.1 The polynomial forms and their common factors

We construct U_n, V_n as integral polynomials before evaluating them. Cancelling their common factors at this stage lowers the degree and hence the power of b required at a rational base a/b.

For each $n \geq 1$, the coefficient formulas below are rational functions of an indeterminate X. We evaluate them at real $x > 1$ and put $q = x^{-1}$ when using the convergent series. Set

$$a_0 = 14n + 1, \quad a_1 = 12n + 1, \quad a_2 = 14n + 1,$$
$$\beta_n = 27n + 2, \quad N = 15n, \quad M_n = 266n^2 + 34n + 1.$$

The symbol β_n is the parameter b of [8, Sec. 3, p. 155], the exponent in the lower parameter q^b of the Heine series (13) on p. 157; the letter b continues to denote the denominator of the base. In this construction alone, $N = 15n$ is the cyclotomic cutoff. The exponent M_n is the integer of that paper's (16) on p. 157. Write $(z;q)_m = \prod_{j=0}^{m-1}(1 - zq^j)$ for the q-Pochhammer symbol (the shifted q-factorial). Zudilin's positive remainder is

$$H_n(x) = \sum_{t\geq 0} \frac{(q^{t+1};q)_{a_1-1}}{(q;q)_{a_1-1}} \frac{(q;q)_{\beta_n-a_2-1}}{(q^{a_2+t};q)_{\beta_n-a_2}} q^{a_0 t}. \tag{2}$$

The denominator length $\beta_n - a_2$ agrees with the gamma expression (7) and residues (8) of [8, p. 156]. The unnumbered display of $R(T)$ on that page instead has length $\beta_n - a_2 - 1$; we use the normalisation fixed by the numbered identities. Then [8, (8)–(11), pp. 156–157] gives $H_n = A_n F - B_n$ with $A_n, B_n \in \mathbb{Q}(X)$. We need their explicit formulas to compute degrees and bound coefficients. Let $\begin{bmatrix} m \\ r \end{bmatrix}_X = \prod_{j=1}^{r}(1 - X^{m-r+j})/(1 - X^j)$ for $0 \leq r \leq m$. This Gaussian binomial, or q-binomial, polynomial has degree $r(m-r)$ and nonnegative coefficients summing to $\binom{m}{r}$. For $a_2 \leq k \leq \beta_n - 1$ put

$$c_{n,k}(X) = (-1)^{a_1+a_2+k+1} X^{e_{n,k}} \begin{bmatrix} k-1 \\ a_1-1 \end{bmatrix}_X \begin{bmatrix} \beta_n - a_2 - 1 \\ \beta_n - k - 1 \end{bmatrix}_X, \tag{3}$$

$$e_{n,k} = \frac{a_1(a_1-1)}{2} - \frac{(\beta_n - a_2)(\beta_n - a_2 - 1)}{2} + \frac{(\beta_n - k)(\beta_n - k - 1)}{2}.$$

Then

$$A_n(X) = \sum_{k=a_2}^{\beta_n-1} c_{n,k}(X) X^{a_0 k}, \tag{4}$$

$$B_n(X) = \sum_{k=a_2}^{\beta_n-1} c_{n,k}(X) X^{a_0 k} \Bigg(\sum_{l=1}^{k-a_1} \frac{1}{X^l - 1} + \sum_{j=1}^{a_0-1} \frac{X^{-j(k-a_1)}}{X^j - 1} \Bigg). \tag{5}$$

Put

$$D_N(X) = \prod_{l=1}^{N} \Phi_l(X), \qquad \nu_{n,l} = \omega(n/l), \qquad \Omega_n(X) = \prod_{l=2}^{N} \Phi_l(X)^{\nu_{n,l}}, \tag{6}$$

with Φ_l the lth cyclotomic polynomial; by Lemma 2.1 every $\nu_{n,l}$ is 0 or 1, and $\nu_{n,l} = 0$ for $l > 15n$ because the intervals all begin at or above $1/14$.

Zudilin's Lemma 7 [8, p. 161, (23)] applies at the parameters above: the tuple is $c = n \cdot (13, 14, 12, 14, 15, 13)$ with maximum $m(c) = 15n$ and difference $s(c) = n > 0$, and the source's (14) on p. 157 holds because $12n+1 \le 14n+1$ and $26n+2 \le 27n+2 \le 28n+2$. Its conclusion is the inclusion

$$U_n := X^{-M_n} \frac{D_N}{\Omega_n} A_n \in \mathbb{Z}[X], \qquad V_n := X^{-M_n} \frac{D_N}{\Omega_n} B_n \in \mathbb{Z}[X]. \tag{7}$$

Two points about that citation matter. First, the conclusion used is the one in $\mathbb{Z}[X]$, not integrality at integer arguments: an integer-valued polynomial such as $X(X-1)/2$ shows that the two statements differ. The source's proof supplies the polynomial form. Its Lemma 4 (pp. 157–159) clears the two rational coefficients separately by Gaussian-binomial identities in $\mathbb{Z}[X]$. Lemma 5 (p. 160) makes the normalised series invariant under the order-twelve group generated by Heine's parameter transformation and the exchange of a_1 and a_2. For the divisibility argument, however, only its order-six subgroup preserving $s > 0$ is used: one application of Heine's transformation reverses the sign of s. Under that subgroup the normalising factorial product takes the three forms displayed in the source. Comparing their cyclotomic valuations gives the maximum of three terms in ω. Lemma 7 (p. 161) removes the resulting common cyclotomic factors by polynomial divisibility; the divisor Ω_n is monic. These steps take place before the integer specialisation in (24) on p. 162.

Second, the module inclusion gives a polynomial coefficient pair, and uniqueness identifies it with the displayed cancelled source pair. If two pairs differed, subtracting their identities would express F as a rational function of X. This is impossible: as $h \downarrow 0$, splitting the original Lambert sum $F(e^h) = \sum_{m \ge 1} (e^{hm} - 1)^{-1}$ at $L = \lfloor 1/h \rfloor$ gives an initial sum $h^{-1} \sum_{m \le L} m^{-1} + O(L)$, since $y^{-1} - 1 \le (e^y - 1)^{-1} \le y^{-1}$ for $0 < y \le 1$. The remaining sum is at most $e^{-h(L+1)}/((1 - e^{-1})(1 - e^{-h})) = O(h^{-1})$. Hence $F(e^h) = h^{-1} \log(1/h) + O(h^{-1})$, so $(X-1)F(X) \to \infty$ and $(X-1)^2 F(X) \to 0$ as $X \downarrow 1$, which no rational function does. Thus the source's module inclusion yields the specific pair in (7), not merely some unspecified polynomial representation of the same function. This functional nonrationality says nothing about an individual value. For $x > 1$, write the cancelled remainder as

$$\Lambda_n(x) = U_n(x) F(x) - V_n(x) = x^{-M_n} \frac{D_N(x)}{\Omega_n(x)} H_n(x).$$

2.2 Exact degrees

The final summand of A_n has strictly larger degree than all the others. We use it to determine the leading coefficient as well as the degree. After cancellation, this will give the least common power of the base denominator that clears the pair.

Let $d_{n,k} = \deg(c_{n,k}(X)X^{a_0 k})$. The Gaussian factors are monic of degrees $(a_1-1)(k-a_1)$ and $(\beta_n-k-1)(k-a_2)$. Adding these to the monomial exponent gives

$$d_{n,k} = e_{n,k}+a_0k+(a_1-1)(k-a_1)+(\beta_n-k-1)(k-a_2) = \frac{-k^2+80kn+3k-340n^2-26n}{2}.$$

Taking the difference at consecutive indices, we obtain

$$d_{n,k+1} - d_{n,k} = 40n+1-k > 0 \qquad (a_2 \le k \le \beta_n - 2),$$

the inequality because $k \le \beta_n - 2 = 27n$ and $27n < 40n+1$. The summand of top degree is therefore the one at $k = \beta_n - 1$ alone, no cancellation occurs there, and

$$\begin{aligned} K_n &:= \deg A_n = d_{n,\beta_n-1} = \frac{1091n^2+81n+2}{2}, \\ W_n &:= \deg U_n = K_n - M_n + \sum_{l\le N}\varphi(l) - \sum_{2\le l\le N}\nu_{n,l}\varphi(l), \end{aligned} \tag{8}$$

the second equality because $\deg D_N = \sum_{l\le N}\varphi(l)$ and $\deg\Omega_n = \sum_{2\le l\le N}\nu_{n,l}\varphi(l)$, and because U_n is a polynomial by (7), so subtracting M_n from the degree of $D_N A_n/\Omega_n$ is legitimate. In particular $U_n \neq 0$. There is also no integer leading-coefficient factor to cancel: the top summand has leading coefficient $(-1)^{a_1+a_2+\beta_n} = (-1)^n$, and both Gaussian factors and D_N/Ω_n are monic. Thus the leading coefficient of U_n is $(-1)^n$.

The net decrease in degree is

$$K_n - W_n = M_n - \deg(D_N/\Omega_n).$$

Dividing by X^{M_n} lowers the degree by M_n, but multiplying by the remaining cyclotomic factors increases it by $\deg(D_N/\Omega_n)$. This is why the remainder estimate below involves $K_n - W_n$, rather than M_n alone.

For a reduced fraction a/b, the unit leading coefficient also gives the exact denominator of $U_n(a/b)$, not just an upper bound. For every prime $p \mid b$,

$$b^{W_n}U_n(a/b) \equiv (-1)^n a^{W_n} \not\equiv 0 \pmod{p}.$$

No prime dividing b cancels from the cleared numerator, so the reduced denominator is exactly b^{W_n}. In particular every common integer divisor of the cleared row is coprime to b. This excludes cancelling a factor of b from this row, not cancellation at other primes or a saving from a different pair.

The formal arithmetic checks cover the value of $2M_n$ from the source's formula (16), the degree difference $d_{n,k+1} - d_{n,k} = 40n+1-k$, its positivity for $a_2 \le k \le \beta_n - 2$, and the top degree $2K_n = 1091n^2+81n+2$. These statements check the polynomial degree calculation, not the analytic estimates used later.

For the second coefficient, fix n and let $x \to \infty$. For $x \ge 2$, every finite product in (2) lies between $\prod_{j\ge1}(1-2^{-j}) > 0$ and 1, independently of t. Also $\sum_{t\ge0} x^{-a_0 t} \le 2$. These bounds give $H_n(x) = O(1)$ for the whole sum, not just each summand. Also $F(x) = x^{-1} + O(x^{-2})$, because $\tau(1) = 1$ and $\sum_{m\ge2} m x^{-m} = O(x^{-2})$. The monic normalising

factor at infinity gives $\Lambda_n(x) = O(x^{W_n-K_n})$. Since $K_n \geq 2$ and U_n has leading coefficient $(-1)^n$,

$$V_n(x) = U_n(x)F(x) - \Lambda_n(x) = (-1)^n x^{W_n-1} + O(x^{W_n-2}).$$

The degree formula gives $W_n \geq K_n - M_n = (559n^2 + 13n)/2 > 0$. Polynomial integrality was established earlier, so the displayed asymptotic proves the exact statement

$$\deg V_n = W_n - 1, \qquad \operatorname{lc} V_n = (-1)^n. \tag{9}$$

In particular V_n is not the zero polynomial. Applying the same prime-by-prime reduction at its own degree shows that $V_n(a/b)$ has reduced denominator exactly b^{W_n-1}. Thus b^{W_n} is the least common clearing denominator, although the second coordinate alone needs one fewer power of b. In the common normalisation,

$$\gcd(b^{W_n}V_n(a/b), b^{W_n}) = b.$$

Indeed $b^{W_n}V_n(a/b)$ is b times an integer coprime to b. These facts concern the fixed source pair, not all approximants. The degree calculation fixes n and varies x; the decay calculation fixes x and varies n.

2.3 The cyclotomic limit

We prove the cyclotomic limits of Zudilin [8, Lemmas 1–2, p. 155] by summing over reciprocal intervals, following [7, Lemma 1, p. 466]. The summatory totient estimate handles each finite set of intervals. A bound on the remaining small indices permits passage to their infinite union.

Write $\Sigma_n = \sum_{2\leq l\leq N} \nu_{n,l}\varphi(l)$. The elementary summatory estimate is $\sum_{l\leq y}\varphi(l) = 3\pi^{-2}y^2 + O(y\log(2y))$. Fix one half-open interval $[u, v)$ of Lemma 2.1. The condition $\{n/l\} \in [u, v)$ holds exactly on the blocks

$$\frac{n}{k+v} < l \leq \frac{n}{k+u}, \qquad k = 0, 1, 2, \dots,$$

so the summatory estimate gives, for each fixed k, $n^{-2}\sum_{l \text{ in block } k}\varphi(l) \to 3\pi^{-2}((k+u)^{-2} - (k+v)^{-2})$. For $L \geq 1$, all blocks with $k \geq L$ lie below $n/(L+u)$. For every $y \geq 0$,

$$\sum_{l\leq y}\varphi(l) \leq y^2.$$

For $y < 1$ the sum is empty; otherwise it is at most $\lfloor y\rfloor(\lfloor y\rfloor + 1)/2 \leq y^2$. Hence the normalised contribution of the omitted blocks is at most $(L+u)^{-2}$, uniformly in n. This is the bound needed to pass from finitely many blocks to all of them. Taking $n \to \infty$ first and then $L \to \infty$ justifies the infinite block sum, and summing the thirteen half-open intervals gives $n^{-2}\Sigma_n \to 3J/\pi^2$. With $\sum_{l\leq 15n}\varphi(l) = 3\pi^{-2}225n^2 + O(n\log n)$, equation (8) yields

$$\frac{K_n}{n^2} \to C_1, \qquad \frac{K_n - W_n}{n^2} = \frac{M_n - \sum_{l\leq N}\varphi(l) + \Sigma_n}{n^2} \to 266 - \frac{3}{\pi^2}(225 - J) = C_0, \tag{10}$$

and hence $W_n/n^2 \to C_1 - C_0$. These are the rational-base degree limits built on Zudilin's cyclotomic limits [8, Lemmas 1–2, p. 155; (26), p. 162].

For the real size estimate, fix $x > 1$. The Möbius product formula and reindexing the divisors give

$$\sum_{l\le N} |\log \Phi_l(x) - \varphi(l)\log x| \le \sum_{d\le N} \lfloor N/d\rfloor [-\log(1-x^{-d})] \le N \sum_{d\ge 1} \frac{-\log(1-x^{-d})}{d}.$$

The bound $-\log(1-x^{-d}) \le x^{-d}/(1-x^{-1})$ shows that the series on the right is finite for each fixed $x > 1$. Since $\nu_{n,l} \in \{0,1\}$, this proves the bound

$$\log \frac{D_N(x)}{\Omega_n(x)} = \Big(\sum_{l\le N} \varphi(l) - \Sigma_n\Big)\log x + O_x(N). \tag{11}$$

Here $N = 15n$, so the error is $O_x(n) = o(n^2)$. The bracket equals $W_n - K_n + M_n$ by (8). No uniform constant as $x \downarrow 1$ is asserted.

2.4 Positive linear forms with integer coefficients

We now specialise at $x = a/b > 1$. Put $q = 1/x$ and $P_q = (q;q)_\infty$. Every finite q-Pochhammer product in (2) has factors $1 - q^j$ with $j \ge 1$, so it lies between P_q and 1. Each of the two ratios therefore lies between P_q and P_q^{-1}. Summing the resulting geometric bounds gives

$$P_q^2 \le H_n(x) \le \frac{P_q^{-2}}{1-q^{a_0}} \le \frac{P_q^{-2}}{1-q},$$

whence $H_n(x) > 0$ and $\log H_n(x) = O_x(1)$ uniformly in n. The cyclotomic factors are positive on $(1,\infty)$, so the cancelled remainder $\Lambda_n(x)$ is positive as well, by (7). By (8) and (9) both polynomial degrees are at most W_n, so

$$\widehat{U}_n = b^{W_n} U_n(a/b), \qquad \widehat{V}_n = b^{W_n} V_n(a/b)$$

are integers, and $\widehat{\Lambda}_n := b^{W_n}\Lambda_n(a/b) = \widehat{U}_n F(a/b) - \widehat{V}_n$ is positive and lies in $\mathbb{Z}F(a/b) + \mathbb{Z}$. This follows by multiplying each polynomial value by b^{W_n}. The integer coefficients and the earlier cancellation of a common cyclotomic factor are separate inputs. Combining the size estimates,

$$\begin{aligned}\log \widehat{\Lambda}_n &= W_n \log b - M_n \log x + \Big(\sum_{l\le N} \varphi(l) - \Sigma_n\Big)\log x + O_x(n)\\ &= K_n \log b - (K_n - W_n)\log a + o(n^2),\end{aligned}$$

the second equality by $\log x = \log a - \log b$ and (8). With (10) this proves

$$\lim_{n\to\infty} \frac{\log \widehat{\Lambda}_n}{n^2} = C_1 \log b - C_0 \log a. \tag{12}$$

Proof of Theorem 2.2. The hypothesis $\log b/\log a < \theta^* = C_0/C_1$ makes the right side of (12) negative, so $\widehat{\Lambda}_n > 0$ and $\widehat{\Lambda}_n \to 0$. Suppose $F(a/b) = P/Q$ with integers P and $Q \ge 1$. Then $Q\widehat{\Lambda}_n = P\widehat{U}_n - Q\widehat{V}_n$ is a positive integer for every n and tends to 0, which is impossible. □

Above the threshold (12) says that these same homogenised forms grow like e^{cn^2} with $c > 0$, and at equality the limit decides nothing. Both statements are about the displayed family.
The hypothesis $b \geq 1$ admits $b = 1$, where the statement reduces to the known integer-base theorem. Coprimality fixes the reduced representation of the base. It is used in the exact-denominator assertion above, but is not needed for the sufficient irrationality implication once the displayed logarithmic inequality holds. Negative bases are excluded, because the proof uses positivity for real bases greater than 1.

Theorem 2.3 (the base 31/4 outside the Bundschuh–Väänänen region)**.** *$F(31/4)$ is irrational, and so is $F((31/4)^r)$ for every integer $r \geq 1$. Here*

$$\frac{\log 4}{\log 31} = 0.4036981731641997\ldots < \frac{81}{200} < \theta^*,$$

while

$$4^{\mu} = 30.483515\ldots < 31 < 4^{\mu_{\mathrm{BV}}} = 32.369642\ldots$$

with $\mu_{\mathrm{BV}} = 2\pi^2/(\pi^2 - 2) = 2.508284761994\ldots$, so 31/4 lies outside the region $\log b/\log a < 1/2 - 1/\pi^2 = 0.3986788163576622\ldots$ of [6, Thm. 2, p. 177] and inside the region of Theorem 2.2.

Proof. The comparison $\log 4/\log 31 < 81/200$ is the integer certificate $4^{200} < 31^{81}$, which Lean checks, and Lean also checks that it places 31/4 in the region $\log b/\log a < 81/200$. The ratio $\log b/\log a$ is invariant under $(a, b) \mapsto (a^r, b^r)$, which places every power of 31/4 in the same region, and $(31^r, 4^r)$ are coprime. Lean also checks that 31/4 lies outside the earlier region. That exclusion also has a two-line rational certificate: $31^2 < 4^5$ gives $\log 4/\log 31 > 2/5$, and $\pi^2 < 10$ gives $1/2 - 1/\pi^2 < 2/5$, so

$$\frac{1}{2} - \frac{1}{\pi^2} < \frac{2}{5} < \frac{\log 4}{\log 31} < \frac{81}{200} < \theta^*.$$

The remaining comparison $81/200 < \theta^*$ is proved in Section 2.5. Theorem 2.2 then applies. □

Corollary 2.4 (an irrationality measure uniform over powers)**.** *For coprime $a > b \geq 1$ with $\theta = \log b/\log a < \theta^*$ and every integer $r \geq 1$,*

$$\mu_{\mathrm{irr}}(F((a/b)^r)) \leq \frac{1-\theta}{\theta^* - \theta}.$$

Here $\mu_{\mathrm{irr}}(\xi)$ is the supremum of the exponents ν for which $|\xi - p/q| < q^{-\nu}$ has infinitely many reduced rational solutions. In particular, $\mu_{\mathrm{irr}}(F((31/4)^r)) < 301$ for every $r \geq 1$.

Proof. For the irrationality exponent, bound A_n in (4): its $O(n)$ Laurent monomials multiply two Gaussian binomial polynomials, each with nonnegative coefficients summing to at most 2^{27n+2}. Thus its absolute coefficient sum is $\exp(O(n))$, and its largest exponent K_n gives $|A_n(x)| \leq x^{K_n}\exp(O(n))$ for fixed $x > 1$. The cyclotomic estimate in Section 2.4 bounds the multiplier $x^{-M_n}D_N(x)/\Omega_n(x)$ by $x^{W_n - K_n}\exp(O_x(n))$. Therefore

$$|U_n(x)| \leq x^{W_n}\exp(O_x(n)) \qquad (x > 1 \text{ fixed}).$$

Set $\xi = F(a/b)$, $Q_n = b^{W_n} U_n(a/b)$ and $P_n = b^{W_n} V_n(a/b)$. With

$$\alpha = (C_1 - C_0)\log a, \qquad \tau = C_0 \log a - C_1 \log b > 0,$$

we have $\log(Q_n\xi - P_n) = -\tau n^2 + o(n^2)$ and $|Q_n| \le \exp(\alpha n^2 + o(n^2))$.

For integers A, B, p, q with $q > 0$, if $L = A\xi - B$ and $2q|L| \le 1$, then

$$|L| \le |A|\,|\xi - p/q|.$$

When $Ap - Bq = 0$ this is equality. Otherwise the nonzero integer $Ap - Bq$ gives $1/q \le |L| + |A|\,|\xi - p/q|$, which proves the inequality. Fix $0 < \eta < \tau$. For all sufficiently large n, the estimates above give

$$e^{-(\tau+\eta)n^2} \le Q_n\xi - P_n \le e^{-(\tau-\eta)n^2}, \qquad |Q_n| \le e^{(\alpha+\eta)n^2}.$$

For every sufficiently large denominator q, the upper remainder bound suggests choosing $n = \lceil \sqrt{\log(2q)/(\tau - \eta)} \rceil$. Then $2q(Q_n\xi - P_n) \le 1$, so the preceding integer argument applies to $(A, B) = (Q_n, P_n)$ for every numerator p. Also $Q_n \ne 0$: otherwise $Q_n\xi - P_n$ would be an integer strictly between 0 and 1. The lower remainder bound and the coefficient bound now give

$$|\xi - p/q| \ge e^{-(\alpha+\tau+2\eta)n^2} = q^{-(\alpha+\tau+2\eta)/(\tau-\eta)-o(1)},$$

since $n^2 = \log(2q)/(\tau - \eta) + O_\eta(\sqrt{\log q} + 1)$.

Let $\eta \downarrow 0$. The resulting bound is $1 + \alpha/\tau = (1-\theta)/(\theta^* - \theta)$. Taking a common power multiplies α, τ by r, leaving this quotient unchanged; the constants in the approximation inequality may depend on r. For $31/4$, the exact rational interval calculation below gives

$$0.40568302137302 < \theta^* < 0.40568302139506,$$
$$0.40369817316419 < \frac{\log 4}{\log 31} < 0.40369817316420.$$

For each trigamma difference, sum $k = 0, \dots, 255$ exactly and bound the remaining decreasing positive summand $f_{u,v}(x) = (x+u)^{-2} - (x+v)^{-2}$ by its integral from 256 to infinity and that integral plus $f_{u,v}(256)$. Bound π with Machin's identity and alternating arctangent series, and logarithms with the positive arctanh series and a geometric tail bound. Outward rational interval operations then place $(1-\rho)/(\theta^* - \rho)$, where $\rho = \log 4/\log 31$, between 300.4269130 and 300.4269164, hence below 301. All endpoints are rational. For the bound 301 alone, the coarser inequalities $\rho < 0.4036982$ and $\theta^* > 0.40568$ suffice:

$$\frac{1-\rho}{\theta^* - \rho} < \frac{1 - 0.4036982}{0.40568 - 0.4036982} = \frac{2981509}{9909} < 301.$$

Here the quotient is increasing in ρ and decreasing in θ^* because $\rho < \theta^* < 1$. The finer enclosure also certifies the displayed decimal approximation. Lean checks the coarser calculation, including $2981509/9909 < 301$, and proves enclosures of θ^* and ρ sharper than the finer ones displayed above (Section 2.5 and Theorem 2.3); the 256-summand calculation is an exact computation outside Lean. □

2.5 The rational bracket around θ^*

Rational bounds for the trigamma differences locate θ^* without a numerical approximation to the full sums. Their first terms already place $31/4$ inside the region and $3/2$ outside it. A sharper enclosure then gives the irrationality-exponent bound 301.

For the lower bound, keep only the $k = 0$ term of each of the thirteen differences $\psi_1(u_i) - \psi_1(v_i)$. All later terms are positive, so

$$J \geq \sum_{i=1}^{13} \Big(\frac{1}{u_i^2} - \frac{1}{v_i^2} \Big) = \frac{2015640690251}{25971865920} > \frac{776}{10}.$$

Since $\pi > 157/50$ and $225 - J < 225$,

$$C_0 > 266 - \frac{3\,(225 - 776/10)}{(157/50)^2} = \frac{5451134}{24649} > \frac{88371}{400} = \frac{81}{200}\,C_1,$$

so $81/200 < \theta^*$. For the upper bound the thirteen half-open intervals are disjoint and ordered and ψ_1 is positive and decreasing, so the sum telescopes below its first term:

$$J < \psi_1(1/14) = 196 + \sum_{k \geq 1} \frac{1}{(k + 1/14)^2} < 196 + \frac{\pi^2}{6} < 198 < 225,$$

whence $C_0 < 266$ and $\theta^* < 532/1091 < 1/2$. So

$$\frac{81}{200} < \theta^* < \frac{1}{2}.$$

Lean checks the definitions of C_0, C_1 and θ^* together with the lower bound on J, rational lower bound on C_0, and the two comparisons $81/200 < \theta^*$ and $\theta^* < 1/2$. It also checks that $31/4$ belongs to the region, that every positive integral power does too, and that $3/2$ is excluded. These comparisons use the defined constant; the analytic estimates belong to the proof of Theorem 2.2, which Lean also checks. Lean also proves $0.40568302138406054100 \leq \theta^* \leq 0.40568302138406054104$ and enclosures of J and C_0 (evidence record, Theorem 2.2). The sharper interval in the preceding proof comes from the stated exact-arithmetic certificate with 256 summands per trigamma difference. It does not rely on the historical thousand-term decimal evaluation reported in Section 2.7.

The shorter interval above suffices for the bound 301 but does not certify all digits printed in the cutoff. Lean's enclosure of θ^* implies the following bounds, and a separate integer-arithmetic calculation reproduces them independently of Lean:

$$0.4056830213840605403 < \theta^* < 0.4056830213840605417,$$
$$2.4649786835749750334 < \mu < 2.4649786835749750415.$$

Here is the complete error bound used in that calculation. For each of the thirteen intervals, put $f_{u,v}(k) = (k+u)^{-2} - (k+v)^{-2}$. Round each of the first $M = 65536$ positive rational summands down to a multiple of 10^{-50}. Their true sum is between this rounded sum and that sum plus $M10^{-50}$. Since $f_{u,v}$ is positive and decreasing, the remaining tail lies between its integral from M to infinity and that integral plus $f_{u,v}(M)$. The integral is

$(M+u)^{-1}-(M+v)^{-1}$. Summing these rational intervals and using Machin's alternating-series bounds for π gives the displayed bounds on θ^* and its reciprocal. The coefficient $225-J$ is positive, so the interval endpoints for C_0 have the claimed order. This verifies the displayed decimal truncations; the program and its recorded result are public. The proofs use the exact definitions and never these decimal values.

2.6 Earlier criteria and changes of direction

Zudilin's parameters and constants give a rational-base condition that can be compared directly with the earlier height criterion. We make that comparison before varying the parameter direction. A single family valid at every real base greater than 1 is subject to the degree restriction proved below.

Attribution. Bundschuh and Väänänen proved the irrationality of $F(a/b)$ on the region $\log b/\log a < 1/2 - 1/\pi^2 = 0.3986788163576622\ldots$ [6, Thm. 2, p. 177; hypotheses pp. 175–176], and every base of Theorem 2.2 with $b \le 3$ is already theirs. Their printed hypothesis for $\alpha = -1$ is $\lambda < (1/2 + 1/\pi^2)^{-1}$ with $\lambda = \log h(q)/\log|q|$. Here the source uses q for the base, not its reciprocal, and

$$E_q(z) = \prod_{j\ge1}(1 + zq^{-j}), \qquad L_q(z) = \frac{E_q'(z)}{E_q(z)} = \sum_{j\ge1}\frac{1}{q^j + z}.$$

For $q = a/b > 1$, $E_q(-1) > 0$ and $L_q(-1) = F(a/b)$; in particular $\alpha = -1$ is not one of the excluded zeros $-q^j$. Since a, b are coprime and $a > b > 0$, the rational height is $h(q) = a$. Thus $\lambda = \log a/\log(a/b) = 1/(1 - \log b/\log a)$, and the printed hypothesis is equivalent to the displayed inequality. Duverney proved another, strictly smaller, region for the same series [14, Théorème 2, p. 174]. Zudilin remarked that his generalized q-logarithm results can be given at non-integer rational bases under an assumption $\log|r| > c\log|s|$ for a computable $c > 0$, without computing a value [9, Sec. 2, p. 4]. Zudilin's 2004 paper supplies the forms and the exponent μ under the standing hypothesis $p = 1/q \in \mathbb{Z} \setminus \{0, \pm1\}$, and it states no rational-base result [8, Sec. 2, p. 154; Thm. 1]. The contribution here is the rational specialisation of the 2004 forms, with the denominator accounting and the limit passage carried out in full, which identifies the printed μ as an admissible c for F itself, together with the region that constant defines and its application to $31/4$.

Where the earlier criterion stops. Both regions are cut out by the same quantity. The published cutoff $1/2-1/\pi^2$ of [6, Thm. 2, p. 177] is the reciprocal of $\mu_{\mathrm{BV}} = 2\pi^2/(\pi^2-2)$, the constant Van Assche later recovered as an integer-base irrationality-exponent bound for $F(p)$ [18, Thm. 1, p. 10], and the cutoff θ^* of Theorem 2.2 is the reciprocal of the smaller integer-base bound $\mu = C_1/C_0$ printed at [8, p. 162]. For Zudilin's family this reciprocal relation uses both the exact degree limit $d = C_1 - C_0$ and the stronger evaluation bound $|U_n(p)| \le p^{W_n}\exp(O_p(n))$ proved in Corollary 2.4, together with decay exponent $\sigma = C_0$. A general exponential coefficient-height bound would contribute an additional term to the integer-base exponent estimate; it cannot simply be discarded. The discussion following Theorem 2.5 makes that distinction explicit. The rational-base proof clears the denominators of the already cancelled polynomials. It does not infer value irrationality merely by taking the reciprocal of a published exponent bound. The

bases gained are exactly the strip $s^{\mu} < r \le s^{\mu_{\mathrm{BV}}}$, which is empty for $s = 2$ and $s = 3$ and is first occupied at $s = 4$ by the single coprime numerator 31. The integer comparisons and rational bracket in Section 2.5 establish membership in the region. The irrationality conclusion in Theorem 2.3 also uses the constructed forms and their proved analytic estimates. This application does not improve either inherited integer-base exponent bound.

What a larger region would require. At $3/2$ the logarithmic parameter is

$$\frac{\log 2}{\log 3} = 0.6309297535714574\ldots,$$

which exceeds θ^* by $0.2252467\ldots$. Theorem 2.2 gives no conclusion when $\log b/\log a \ge \theta^*$. The same argument can apply to another family once its polynomial integrality, degrees and fixed-base remainder estimates are proved. A smaller ratio of the resulting constants C_1/C_0 would enlarge the reciprocal-cutoff region. For example, $C_1/C_0 \le 2.4234$ would give a cutoff greater than 0.4126 and admit $29/4$. An integer-base irrationality-exponent bound alone does not supply the polynomial data needed for this specialisation. The existing formal proofs establish irrationality and measure bounds for the constructed forms; they do not supply a different family whose divided remainder is small and nonzero at $3/2$.

The next restriction concerns one polynomial family whose estimates hold at every fixed $x > 1$, with the same leading constants but possibly base-dependent error terms. This quantifier is essential: the proof chooses a large integer base at which remainder decay would outpace coefficient growth. Estimates only at $3/2$, or for a different family at each base, do not permit that choice. The 2004 family satisfies the hypotheses. The conclusion concerns forms in $F(x)$ and 1, not simultaneous forms in several independent target values. Here height means the largest absolute coefficient. To compare it with the sum of the absolute coefficients, note that for a polynomial of degree at most d,

$$\max_j |p_j| \le \sum_j |p_j| \le (d+1) \max_j |p_j|.$$

For a nonzero polynomial of degree $d = O(n^2)$, the logarithms of these norms differ by $O(\log n)$. They have the same leading quadratic growth rate, but the literal zero-height conditions are not identical: $1 + X$ has maximum coefficient 1 and coefficient sum 2. In particular, when $h = 0$, the displayed bound $hn^2(1 + o(1))$ is zero, not an arbitrary $o(n^2)$ term. The proof below also works with the additive bound $hn^2 + o(n^2)$; its evaluation estimate absorbs the $O(\log n)$ difference. Alternatively, either norm convention implies the hypothesis for the other norm after replacing h by any larger positive constant. The degree conclusion is independent of that replacement. In the rational-base conclusions, write a/b with integers $a > b \ge 1$. The estimates hold without coprimality; a reduced representation gives the smaller denominator-clearing factor.

Theorem 2.5 (a degree restriction for estimates valid at every base)**.** *Let $(U_n, V_n) \in \mathbb{Z}[x]^2$ be a sequence such that, for constants $\sigma, \delta > 0$ and $h \ge 0$ independent of n and of the base,*

1. $\Lambda_n(x) := U_n(x)F(x) - V_n(x) \ne 0$ *for every real $x > 1$;*

2. $\deg U_n, \deg V_n \le \delta n^2(1+o(1))$;
3. *the coefficient heights satisfy*

$$\log \max(H(U_n), H(V_n)) \le hn^2(1+o(1)),$$

where $H(P)$ is the largest absolute value of a coefficient of P;

4. *the remainders satisfy*

$$\log|\Lambda_n(x)| = -\sigma n^2 \log x\,(1+o(1))$$

for every real $x > 1$.

Put $d_n := \max(\deg U_n, \deg V_n)$. Then $\sigma \le \delta$, and for every fixed rational base $a/b > 1$,

$$\limsup_{n\to\infty} n^{-2}\log|b^{d_n}\Lambda_n(a/b)| \le \delta\log b - \sigma\log(a/b).$$

Consequently the homogenised forms tend to zero whenever $\log b/\log a < \sigma/(\sigma+\delta)$, a sufficient region whose cutoff is at most $1/2$. If the actual degrees satisfy $d_n/n^2 \to d$, then $d \ge \sigma$ and the limit exists and equals $d\log b - \sigma\log(a/b)$; in that case the forms tend to zero below $\log b/\log a = \sigma/(\sigma+d)$ and their absolute values tend to infinity above it, so the exact-degree case has no decaying homogenised forms at $3/2$.

Proof. The coefficient-height bound controls evaluation at an integer base. For every fixed $p > 1$,

$$|U_n(p)|, |V_n(p)| \le (d_n+1)\max(H(U_n), H(V_n))p^{d_n} \le \exp((h+\delta\log p)n^2 + o(n^2)).$$

Here $\log(d_n+1) = o(n^2)$ by the degree bound. This is where the coefficient-height hypothesis is needed.

Suppose $\sigma > \delta$. To make remainder decay beat coefficient growth, choose an integer $p \ge 2$ with $(\sigma-\delta)\log p > h$ and put $u_n = U_n(p)$, $v_n = V_n(p)$ and $\ell_n = u_nF(p) - v_n$. The integer determinant eliminates $F(p)$; the evaluation bound and hypothesis (4) give

$$u_nv_{n+1} - u_{n+1}v_n = u_{n+1}\ell_n - u_n\ell_{n+1} = o(1).$$

Indeed, each product on the right has absolute value at most $\exp((h+(\delta-\sigma)\log p)n^2 + o(n^2))$. The expression on the left is an integer, so it is eventually zero. Also $u_n \ne 0$ eventually, since $u_n = 0$ would make $\ell_n = -v_n$ a nonzero integer of absolute value less than 1. Thus v_n/u_n is eventually a fixed rational number r. If $F(p) = r$, then $\ell_n = 0$; otherwise $|\ell_n| = |u_n||F(p)-r| \ge |F(p)-r|$. Both contradict the nonzero remainders tending to zero. Hence $\sigma \le \delta$.

For a fixed rational base $a/b > 1$, taking logarithms gives

$$n^{-2}\log|b^{d_n}\Lambda_n(a/b)| = \frac{d_n}{n^2}\log b - \sigma\log(a/b) + o(1).$$

Hypothesis (2) yields the stated upper limit and sufficient region. Since $\sigma \le \delta$, its cutoff is at most $1/2$. If $d_n/n^2 \to d$, apply the preceding integer-base argument with $d+\varepsilon$ in place of δ for every $\varepsilon > 0$ to obtain $d \ge \sigma$. The displayed identity then gives the exact limit. Its sign is that of $(\sigma+d)\log b - \sigma\log a$, which proves both assertions away from equality. At $3/2$ this sign is positive because $\log 2/\log 3 > 1/2 \ge \sigma/(\sigma+d)$. □

Corollary 2.6 (nondecay when $b < a < b^2$)**.** *Under the hypotheses of Theorem 2.5, for positive integers a, b with $b < a < b^2$, the undivided forms $b^{d_n}\Lambda_n(a/b)$ do not tend to zero. No limit of d_n/n^2 is assumed.*

Proof. Write $x = a/b$ and suppose $C_n = b^{d_n}\Lambda_n(x) \to 0$. Since $C_n \neq 0$ and $|C_n| \leq 1$ eventually, logarithms and the remainder asymptotic at x give the degree bound

$$d_n \log b \leq -\log|\Lambda_n(x)| = \sigma \log x\, n^2 + o(n^2).$$

Set $c = \sigma \log x/\log b$. Since $1 < x < b$, we have $0 < c < \sigma$, and the displayed inequality gives $d_n \leq (c+\varepsilon)n^2$ eventually for each $\varepsilon > 0$. Apply Theorem 2.5 to the same polynomial family with degree upper rate c: its height bound, nonvanishing and all-base remainder asymptotics are unchanged. This gives $\sigma \leq c$, contradicting $c < \sigma$. □

This strengthens the failure of a sufficient-cutoff test to an exclusion of decay under the stated all-base hypotheses. It does not supply an actual-degree limit or assert divergence. If $d_n/n^2 \to d$, the separate exact-degree result gives the stronger conclusion $|b^{d_n}\Lambda_n(a/b)| \to \infty$ in the same strict region. Equality $a = b^2$ remains unclassified. The corollary concerns the undivided forms; it does not exclude base-dependent content division or other irrationality methods outside its hypotheses.
A degree bound alone would not supply the evaluation estimate: even degree-zero polynomials can have arbitrarily large integer coefficients. This explains the height hypothesis (3), without establishing that it cannot be weakened or omitted. At $\log b/\log a = \sigma/(\sigma+d)$, the normalised logarithm is zero and the hypotheses decide neither behaviour. Theorem 2.5 is a restriction on the stated families, not on all Padé constructions.

There is also a distinction between the degree cutoff and an irrationality-exponent estimate at a fixed integer base $p \geq 2$. The hypotheses above give

$$|U_n(p)| \leq \exp((h+\delta\log p)n^2 + o(n^2)), \qquad |\Lambda_n(p)| = \exp(-\sigma\log p\, n^2 + o(n^2)).$$

Apply the integer-form argument in the proof of Corollary 2.4, with coefficient growth rate $h + \delta\log p$ and remainder decay rate $\sigma\log p$. It gives

$$\mu_{\mathrm{irr}}(F(p)) \leq 1 + \frac{\delta}{\sigma} + \frac{h}{\sigma\log p}.$$

Thus the reciprocal $\sigma/(\sigma+\delta)$ of the degree expression $1+\delta/\sigma$ is not, under these hypotheses alone, the reciprocal of the exponent bound furnished by that argument. To obtain the latter identification it suffices to prove the stronger evaluated estimate $|U_n(p)| \leq \exp(\delta\log p\, n^2 + o(n^2))$. This is separate from bounding the polynomial's coefficients by $\exp(O(n^2))$.

For the family of Section 2.1 the fourth hypothesis is the size estimate proved there and the second is (8). The third is proved next, so Theorem 2.5 applies to that family.

Lemma 2.7 (coefficient heights of the constructed polynomials)**.** *There is a constant h with $\log\max(H(U_n), H(V_n)) \leq hn^2$ for every $n \geq 1$, where U_n and V_n are the polynomials of (7).*

Proof. For a polynomial or Laurent polynomial P, write $\|P\|$ for the sum of the absolute values of its coefficients. Then $\|PQ\| \le \|P\|\,\|Q\|$; for an ordinary polynomial, $H(P) \le \|P\|$. This convention includes the Laurent monomials in $c_{n,k}$.

By Lemma 2.1 every $\nu_{n,l}$ is 0 or 1, so $D_N/\Omega_n = \prod_{l\in S} \Phi_l$ over a subset $S \subseteq \{1,\dots,N\}$. A monic polynomial with roots $\zeta_1,\dots,\zeta_d$ on the unit circle has coefficient norm at most 2^d: expanding its linear factors bounds the sum of the absolute coefficients by $\prod_{j=1}^{d}(1+|\zeta_j|) = 2^d$. Hence $\|\Phi_l\| \le 2^{\varphi(l)}$ and

$$\Big\|\frac{D_N}{\Omega_n}\Big\| \le 2^{\sum_{l\le N}\varphi(l)} \le 2^{225n^2}.$$

For A_n, the Gaussian binomial $\begin{bmatrix} m \\ r \end{bmatrix}_X$ has nonnegative coefficients summing to $\binom{m}{r}$, so $\|c_{n,k}\| \le \binom{k-1}{a_1-1}\binom{\beta_n-a_2-1}{\beta_n-k-1} \le 2^{2\beta_n}$ by (3), and summing the at most β_n terms of (4) gives $\|A_n\| \le \beta_n 2^{2\beta_n}$. Hence $\|U_n\| \le \|D_N/\Omega_n\|\,\|A_n\| \le e^{O(n^2)}$ and $H(U_n) \le e^{O(n^2)}$.

For V_n, bound it on the circle $|z| = 2$ and use Cauchy's estimate $H(V_n) \le \max_{|z|=2}|V_n(z)|$: each coefficient satisfies $|v_i| \le 2^{-i}\max_{|z|=2}|V_n(z)|$. The bound also holds for the zero polynomial. On that circle $|z^l - 1| \ge 2^l - 1 \ge 1$, so each of the at most $\beta_n + a_0$ inner terms of (5) has modulus at most 1; also $|c_{n,k}(z)z^{a_0k}| \le \|c_{n,k}\|\,2^{K_n}$ and $|D_N(z)/\Omega_n(z)| \le \|D_N/\Omega_n\|\,2^{225n^2} \le e^{O(n^2)}$, while $|z^{-M_n}| \le 1$. There are $O(n)$ outer summands, each with $O(n)$ inner terms. After multiplication by the normalising factor, each term is bounded by $e^{O(n^2)}$, since $K_n = O(n^2)$. Summing the $O(n^2)$ terms preserves this bound. Hence $\max_{|z|=2}|V_n(z)| \le e^{O(n^2)}$ and $H(V_n) \le e^{O(n^2)}$. □

With Lemma 2.7, Zudilin's family satisfies all four hypotheses of Theorem 2.5, with exact degree limit $d = C_1 - C_0$ by (10), and decay exponent $\sigma = C_0$ because $\log\Lambda_n(x) = -(K_n - W_n)\log x + o(n^2)$ for each fixed real $x > 1$ by the size estimate of Section 2.4. For it $\sigma/(\sigma+d) = \theta^*$ and $(\sigma+d)/\sigma = \mu$. The stronger bound $|U_n(p)| \le p^{W_n}\exp(O_p(n))$ from Corollary 2.4 removes the extra height term at each integer base. Within this family the rational-base threshold is therefore the reciprocal of the proved integer-base irrationality-exponent bound.

Examples of rational bases. Among reduced a/b with $1 \le b < a \le 60$, the region of Theorem 2.2 has 137 members, of which 78 are non-integral. The two largest admitted logarithmic ratios belong to $53/5$ and $31/4$, respectively 0.000313 and 0.001985 below θ^*; the smallest excluded ratio is that of $52/5$, namely 0.4073243836 …, which exceeds the cutoff by 0.001641. A calculation with rational bounds for logarithms and for the cutoff checks all 1101 reduced fractions in the stated range, these rounded margins and the four fixed-denominator comparisons below (program, result). The bases in this region and outside the region of [6] are those in the strip $s^{\mu} < r \le s^{\mu_{\mathrm{BV}}}$, which is empty for $s = 2$ and $s = 3$, is $\{31\}$ for $s = 4$, and is $\{53, 54, 56\}$ for $s = 5$. The strip is infinite: its width $s^{\mu_{\mathrm{BV}}} - s^{\mu}$ eventually exceeds s, so for every large enough denominator it contains an integer congruent to 1 modulo s. Hence $31/4$ is the member of the strip with least denominator and least numerator.

A certified finite parameter search. The parameter ratios $(14, 12, 14; 27)$ are Zudilin's [8, p. 162]. They maximise C_0/C_1 in the following finite box: positive integer parameters

$(\alpha_0, \alpha_1, \alpha_2; \beta)$ with every entry at most 30, greatest common divisor 1, and

$$\alpha_1 \le \alpha_2, \qquad \alpha_1 + \alpha_2 < \beta \le \alpha_0 + \alpha_2.$$

These are the source's parameter inequalities [8, Sec. 5, p. 161], with a bound imposed for enumeration. The definitions of C_0, C_1 and their floor function are reproduced in Section 10. There are exactly 37,533 tuples. Exactly two attain the maximum: $(14, 12, 14; 27)$ and $(15, 12, 13; 26)$. Both give θ^*, although they are distinct primitive directions.

The enumeration and comparisons are certified by rational intervals. The fractions j/c with $1 \le c \le 30$ partition $[0, 1)$ into 278 half-open intervals. Every floor function occurring in the search is constant on each interval, and the first interval contributes zero. On each remaining interval $[u, v)$, the integral is a sum of positive differences $(k+u)^{-2} - (k+v)^{-2}$, weighted by 0 or 1. With $M = 256$, the tail lies between

$$(M+u)^{-1} - (M+v)^{-1} \quad \text{and} \quad (M+u)^{-1} - (M+v)^{-1} + (M+u)^{-2} - (M+v)^{-2}.$$

This follows by comparing the positive decreasing summand with its integral. Outward rounding of rational numbers and Machin's bounds for π give, for each of the two maximisers,

$$0.40568302137302 < C_0/C_1 < 0.40568302139506.$$

Every other tuple has $C_0/C_1 < 0.40563943278333$. The intervals therefore separate these two candidates from all other tuples, without relying on floating-point ordering.

Their exact equality is not inferred from overlapping intervals. Write ω_A, ω_B for their step functions in the order just listed, and let J_A, J_B be their integrals against $d(-\psi_1)$ on $[0, 1]$. The floor formulas give

$$\omega_B(u) - \omega_A(u) = \lfloor 13u \rfloor + \lfloor 15u \rfloor - 2\lfloor 14u \rfloor.$$

For every positive integer c, telescoping the c intervals and splitting the defining trigamma series into residue classes gives

$$\int_0^1 \lfloor cu \rfloor \, d(-\psi_1(u)) = \sum_{j=1}^{c-1} \psi_1(j/c) - (c-1)\psi_1(1) = \frac{c(c-1)\pi^2}{6}.$$

Hence $J_B - J_A = \pi^2/3$. Both directions have $m = 15$ and $C_1 = 1091/2$, while the quadratic part of C_0 before the $3/\pi^2$ correction is 266 for A and 265 for B. Its decrease by 1 exactly cancels the increase $3(J_B - J_A)/\pi^2 = 1$. Thus C_0 also agrees.

A calculation compares every tuple of the box with interval bounds, and a second implementation repeats it with 512 summands before the tail and a different enumeration order; both are in one program, with its recorded result. The calculation proves optimality only in the displayed finite box. No bound on a direction outside that box follows, and this calculation is not needed for the irrationality theorem.

The two-coordinate forms in 1 and $F(p)$ should be distinguished from simultaneous approximation to two target values. Postelmans and Van Assche prove the $\mathbb{Q}$-linear independence of $1, \zeta_q(1), \zeta_q(2)$ for $q = 1/p$ with integer $p \ge 2$ [30, Thm. 1.3, p. 3]. Their confluent multiple little q-Jacobi construction has two orthogonality conditions; its

common integer normalisation and nonvanishing argument are in Section 6, especially (6.1)–(6.4). That theorem treats two target values and retains the integer inverse-base restriction. It supplies neither the coefficient pairs nor the all-base estimates assumed in Theorem 2.5; it is not an application at 3/2.

2.7 Finite calculations

The polynomial reconstruction below determines the forms for $n = 1, 2, 3, 4$. Together with the exact comparison of all directions in the preceding finite box, it provides small cases against which to check the general formulas. The two calculations are available, along with the other finite calculations, in the reproduction guide, and a single script reruns them and compares each with its recorded result. The historical high-precision evaluations were not rerun. None of these finite calculations proves a statement for all indices, and none is a Lean proof. The tests for $N \le 8$ and the rational interval certificate are described separately.

The public reproduction guide gives the command and dependency pin for the direction-search program. Run it with `--bound 30` and the output path named there; adding `--check` compares the result without replacing the stored search results. The stored record specifies mpmath 1.3.0 at thirty decimal digits. Its exact enumeration count is 37,533, but its ordering by C_0/C_1 is numerical and has no interval certificate in that historical record. The separate rational certificate above verifies the maximisers in the stated finite box; the archived program itself still uses numerical comparisons.

An exact integer calculation (program, every coefficient) reconstructs the forms of Section 2.1 directly in $\mathbb{Z}[X]$ from the displayed source formulas. It checks division by Ω_n and X^{M_n} with zero remainder and computes all coefficients of U_n, V_n. The exact degrees are

n	1	2	3	4
K_n	587	2264	5032	8891
W_n	333	1315	2944	5220
$\deg D_N$	72	278	628	1102
$\deg \Omega_n$	25	94	219	380.

In each case $\deg V_n = W_n - 1$ and both leading coefficients are $(-1)^n$. Independently expanding $F(1/q) = \sum_{j\ge1} \tau(j) q^j$ checks the vanishing coefficients below order $K_n - W_n$ in $U_n(1/q)F(1/q) - V_n(1/q)$ and its next twelve coefficients against the positive source expression. Exact rational evaluation at 31/4, 3 and 7/2 also checks homogenisation; at the two noninteger bases, the power b^{W_n-1} fails to clear $U_n(a/b)$, as predicted by the general leading-coefficient argument.

A separate historical numerical report evaluated the remainder identity at those three bases to relative accuracy below 10^{-39} at $n = 3$. It reported positivity of $\widehat{\Lambda}_n$ and agreement with the bounds on H_n in Section 2.4. These high-precision real evaluations were not reproduced by the exact polynomial calculation just described.

The historical constant evaluation used ω, the thirteen intervals of Lemma 2.1 and forty-digit trigamma values. It reported

$$J = 77.94318447500909\ldots, \qquad C_0 = 221.30008816500502\ldots,$$

and $C_1/C_0 = 2.46497868357497\ldots$, agreeing with the source's printed precision [8, p. 162]. The rational interval estimate stated earlier, rather than these rounded digits, supports the numerical inequality.

The main term $K_n \log b - (K_n - W_n)\log a$ at $31/4$ is negative for every $1 \le n \le 400$, as verified by exact rational logarithm enclosures and exact totient sums (program, result). Dividing by n^2 gives, to three decimal places, $-58.478, -10.280, -4.297, -3.863$ at $n = 1, 10, 100, 400$. This finite sign check does not assert negativity at every index. The proved limit is $C_1 \log 4 - C_0 \log 31 = -3.718\ldots$; the elementary convergence-rate bound is $O(\log^2(n+2)/n)$. Indeed, summing the $O(y \log(2y))$ endpoint errors over the floor blocks with $k \le n$ gives $O(n \log^2(n+2))$; the remaining main-term tail is $O(1)$ since its summands are $O(n^2/k^3)$. The linear terms of M_n add $O(n)$. The stronger rate $O(1/n)$ does not follow from these estimates. Only the limit enters Theorem 2.2.

For orientation, the all-rank order and leading-coefficient formulas of Theorem 3.4 give the following first seven values:

N	1	2	3	4	5	6	7
ord	0	1	5	14	30	55	91
lc	1	6	108	4320	324000	40824000	8001504000.

3 Power comparisons and Hankel determinants

Three integer comparisons will be used to estimate residue counts and scalar exponents. The first gives $\log 3/\log 2 < 65/41$ and shows that 65 is the least integer q with $3^{41} < 2^q$; the inequality $2^{129} < 3^{82}$ checks the failure of the subset count at depth 41 one row earlier. They determine numerical thresholds, leaving the coefficient and remainder hypotheses to be checked for any family of approximants. Readers interested first in the determinant asymptotic may begin at Section 3.1.

Theorem 3.1 (sharp power bracket)**.** *One has*

$$2^{64} < 3^{41} < 2^{65}.$$

Consequently

$$\frac{41}{65} < \frac{\log 2}{\log 3}, \qquad \frac{\log 3}{\log 2} < \frac{65}{41}.$$

Proof. Direct evaluation gives

$$\begin{aligned} 2^{64} &= 18446744073709551616, \\ 3^{41} &= 36472996377170786403, \\ 2^{65} &= 36893488147419103232. \end{aligned}$$

Taking logarithms of the upper bound gives $41 \log 3 < 65 \log 2$. Dividing this inequality by $65 \log 3$ and by $41 \log 2$, respectively, gives the two stated logarithmic bounds; both divisors are positive. □

The upper power bound gives $\log 2/\log 3 > 41/65$, a sharper lower bound than $81/200$. In particular, $41/65 - 2/5 = 3/13$. Combining this with $1/2 - 1/\pi^2 < 2/5$ gives the first gap below; the same logarithmic lower bound is then compared with the rectangular expression.

For $\rho \geq 0$ and $\sigma \geq 1 + \rho$, define

$$\Theta_{\mathrm{HP}}(\rho, \sigma) = \frac{(1+\rho^2)/2 + \sigma - 3\sigma^2/\pi^2}{(1+\rho)^2/2 + \sigma(1+\rho) + (1+\rho^2)/2 + \sigma}.$$

The denominator is positive on this domain.

Corollary 3.2 (gaps between the stated logarithmic thresholds)**.** *For every $\rho, \sigma \in \mathbb{R}$ with $0 \leq \rho$ and $1 + \rho \leq \sigma$,*

$$\frac{3}{13} < \frac{\log 2}{\log 3} - \left(\frac{1}{2} - \frac{1}{\pi^2}\right), \qquad \frac{3}{13} < \frac{\log 2}{\log 3} - \Theta_{\mathrm{HP}}(\rho, \sigma),$$

where Θ_{HP} is the rectangular exponent threshold. Moreover

$$\frac{\log 3/\log 2 - 1}{3} < \frac{8}{41}.$$

Proof. The first inequality follows from $41/65 - 2/5 = 3/13$, Theorem 3.1, and $1/2 - 1/\pi^2 < 2/5$. The polynomial calculation in Section 10 proves $\Theta_{\mathrm{HP}}(\rho, \sigma) \leq 1/2 - 1/\pi^2$ on the stated domain, which gives the second inequality. The final inequality is a direct rearrangement of $\log 3/\log 2 < 65/41$. □

The next inequalities compare two proposed savings in the denominator exponent with the exponent $4N^3 - 3N^2$ before division. Here E denotes the exponent saved. The statement assumes the bounds $E \leq N^3 - N$ or $E \leq 2N^3 - N$; it does not derive them for a polynomial family. Under either bound, the saving is less than $39/41$ of the original exponent. Thus these bounds alone cannot justify the reduction required by this model.

Theorem 3.3 (bounds for two proposed degree savings)**.** *For every integer $N > 0$,*

$$41(N^3 - N) < 39(4N^3 - 3N^2).$$

For every integer $N \geq 2$,

$$41(2N^3 - N) < 39(4N^3 - 3N^2).$$

Hence the same strict inequalities hold with the left side replaced by $41E$ whenever, respectively, $E \leq N^3 - N$ or $E \leq 2N^3 - N$.

Proof. After subtraction, the first inequality is

$$N(115N^2 - 117N + 41) > 0,$$

whose quadratic factor is $115(N-1)^2 + 113(N-1) + 39$, positive for $N \geq 1$. The second becomes

$$N(74N^2 - 117N + 41) > 0.$$

Its quadratic factor is $74(N-2)^2 + 179(N-2) + 103$, positive for $N \geq 2$. The assertions for E follow by monotonicity. □

An application of these degree bounds must separately prove that its proposed divisor satisfies one of the stated bounds. These inequalities do not rule out additional factors caused by cancellation in a determinant, a different choice of integral forms, or another proof of irrationality at 3/2.

3.1 The sharp q-order of the normalised Hankel determinant

We begin with the formal expansion of Zudilin's normalised Hankel determinant at $x = z = 1$. Determining its first nonzero term is a calculation in power series in q; an interpretation as a divisor after integer evaluation would additionally require an integral normalisation and a divisibility argument. Zudilin constructs the determinant by combining the Padé-type approximations of Bundschuh and Zudilin [12] with Bézivin's method as developed in [13]; see [9, Sec. 2, p. 3]. His passage from the q-order to the size of the determinant borrows the proofs of [13, Lemma 2, p. 12, and Prop. 3, p. 13], as he notes in [9, Sec. 4, p. 7].

The source writes $\ell_p(x,z) = x\sum_{r\ge1} z^r/(p^r - x)$, so $\ell_p(1,1) = F(p)$. Here $p = q^{-1}$ and x, z are auxiliary parameters, not the base. Write N for the Hankel rank. With q a formal variable, the moments and determinant in question are

$$v_m^* = \sum_{t\ge0} q^{(m+1)t}\frac{(q;q)_m^3(q^{t+1};q)_m}{(q^{m+t+1};q)_{m+1}}, \qquad V_N^* = \det(v_{i+j}^*)_{0\le i,j<N}.$$

All product denominators have constant term 1, so their inverses exist in $\mathbb{Z}[[q]]$. The order of a nonzero series is the least exponent of q with a nonzero coefficient.

Theorem 3.4 (the first nonzero term at every rank)**.** *For every rank N, the normalised Hankel determinant V_N^* of [9, Sec. 4, (6), p. 6], evaluated at $x = z = 1$, has*

$$\operatorname{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6}, \qquad \textit{leading coefficient} \quad \frac{(N!)^2(N+1)!}{2^N}.$$

The source proves the order inequality and also gives $|V_N^*(q)| \le |q|^{N^3/3}\exp(O(N^2))$, referring to additional estimates for the passage from formal order to analytic size [9, Sec. 4, p. 7]. Equality in the order bound rules out a further initial power of q at $x = z = 1$; it does not itself prove a fixed-q estimate. For $0 < q < 1$, the positive-measure argument below gives both positivity and a two-sided comparison with the leading term, with logarithmic error $O_q(N)$. It refines the subleading control, not the cubic exponent of q.

The row identity. We choose row operations whose first surviving coefficients can be computed explicitly. The auxiliary lemma applies to more general products than the moments above; its use here is to identify the coefficient that remains after each row operation.

Work over $A = \mathbb{Z}[[q]]$. Let $H(X) = 1 + \sum_{s\ge1} a_s(q)X^s$. For nonnegative integers m, t, put

$$W_m(t) = q^{(m+1)t}\prod_{r=1}^{m} H(q^r), \qquad D_j = \prod_{r=0}^{j-1}(I - q^r\mathcal{N}), \qquad E(m,j) = mj - \frac{j(j-1)}{2},$$

where $\mathcal{N}$ is the backward shift $(\mathcal{N}f)_m = f_{m-1}$ in the index m. Thus D_j is Zudilin's backward-difference operator in [9, Sec. 4, (7), p. 6]. Write $\bar{H} = H \bmod q$ and $h_r = [X^r]\bar{H}(X)^{-1}$, with $h_r = 0$ for $r < 0$ and $h_0 = 1$.

The lemma allows arbitrary coefficients $a_s(q) \in \mathbb{Z}[[q]]$; the only normalisation imposed on H is its constant term 1. It therefore covers the product series used below, but not an unnormalised series with a different constant term. All statements here are coefficientwise identities of formal series, with no analytic convergence assumption. For a general H, the coefficient in the lemma can vanish. The application below computes it for the chosen products and proves the required nonvanishing.

Lemma 3.5 (a coefficient of each transformed row)**.** *For $m \geq j \geq 0$ one has $D_j W_m(t) \in q^{E(m,j)}A$ and*

$$[q^{E(m,j)}]D_j W_m(t) = (-1)^j h_{j-t}.$$

Proof. Let e_u denote a formal basis vector indexed by $u \geq 0$, and suppress the argument q in $a_s(q)$. We use the transition rule

$$\Phi e_0 = \sum_{s\geq 1} a_s e_{s-1}, \qquad \Phi e_u = (q^u - 1)e_{u-1} + q^u \sum_{s\geq 1} a_s e_{u+s-1} \quad (u > 0).$$

In $\Phi^j e_t$, a coefficient means the sum of the weights of length-j paths from t to the specified endpoint. Each step lowers its index by at most one. A path ending at u therefore visits only indices at most $\max(t, u + j)$, so each coefficient is a finite sum. No action on the algebraic direct sum is assumed.

For $m \geq 1$, a direct expansion gives

$$W_m(u) - W_{m-1}(u) = q^m \sum_v (\Phi e_u)_v W_{m-1}(v),$$

since both sides equal

$$q^{mu} \prod_{r=1}^{m-1} H(q^r)(q^u H(q^m) - 1).$$

For the induction step take $m \geq j + 1$ and note that $E(m, j) - E(m - 1, j) = j$. Applying $I - q^j\mathcal{N}$ to $q^{E(m,j)}W_{m-j}(u)$ therefore gives $q^{E(m,j)}(W_{m-j}(u) - W_{m-j-1}(u))$. The preceding identity contributes one further factor q^{m-j}, and $E(m, j) + m - j = E(m, j + 1)$. Induction, starting with $D_0 = I$, gives

$$D_j W_m(t) = q^{E(m,j)} \sum_u (\Phi^j e_t)_u W_{m-j}(u). \tag{13}$$

The endpoint sum is also locally finite: $\operatorname{ord} W_{m-j}(u) = (m-j+1)u$, so only finitely many endpoints contribute to any fixed power of q.

Modulo q the operator simplifies: $\bar{\Phi}e_u = -e_{u-1}$ for $u > 0$, and $\bar{\Phi}e_0 = \sum_{s\geq 1} a_s(0)e_{s-1}$. Put $c_{j,t} = (\bar{\Phi}^j e_t)_0$. Then $c_{j+1,t} = -c_{j,t-1}$ for $t > 0$ and $c_{j+1,0} = \sum_{s=1}^{j+1} a_s(0)c_{j,s-1}$, the sum terminating because a state above j cannot reach 0 in j steps. Now $c_{0,t} = \delta_{t,0} = h_{-t}$, and if $c_{j,t} = (-1)^j h_{j-t}$ for all t then $c_{j+1,t} = (-1)^{j+1}h_{j+1-t}$ for $t > 0$ at once, while for $t = 0$ the identity $\bar{H}\bar{H}^{-1} = 1$ gives $h_{j+1} = -\sum_{s\geq 1} a_s(0)h_{j+1-s}$ and hence

$c_{j+1,0} = (-1)^j \sum_{s\geq 1} a_s(0) h_{j+1-s} = (-1)^{j+1} h_{j+1}$. So $c_{j,t} = (-1)^j h_{j-t}$ for all j, t. Since $\operatorname{ord} W_{m-j}(u) = (m-j+1)u$ and $m \geq j$, in (13) only the state $u = 0$ contributes at degree $E(m, j)$, which gives both assertions. □

Apply the lemma to a summand of the normalised remainder:

$$T_{m,t} = q^{(m+1)t} (q;q)_m^3 \frac{(q^{t+1};q)_m}{(q^{m+t+1};q)_{m+1}}.$$

Set

$$H_t(X) = \frac{(1-X)^3(1-q^tX)^2}{(1-q^tX^2)(1-q^{t+1}X^2)}.$$

Then $T_{m,t} = (1-q^{t+1})^{-1} W_m^{H_t}(t)$, by the telescoping identity

$$\prod_{r=1}^{m} H_t(q^r) = \frac{(q;q)_m^3 (1-q^{t+1})(q^{t+1};q)_m}{(q^{m+t+1};q)_{m+1}}.$$

The two denominator products contribute the consecutive factors $1-q^{t+2}, \dots, 1-q^{t+2m+1}$. Reducing modulo q gives $\bar{H}_t = (1-X)^3$ for $t > 0$, so $h_r^{(t)} = \binom{r+2}{2}$, and $\bar{H}_0 = (1-X)^4/(1+X)$, so $h_r^{(0)} = (r+1)(r+2)(2r+3)/6$. The scalar factor $(1-q^{t+1})^{-1}$ has constant term 1. The rows are $v_m^* = \sum_{t\geq 0} T_{m,t}$. In D_j only the shifts $m, m-1, \dots, m-j$ occur. Since $m \geq j$ and $\operatorname{ord} T_{m-r,t} = (m-r+1)t \geq (m-j+1)t$, only finitely many t can affect any fixed coefficient after any of these shifts. This justifies interchanging the sum and D_j, and then extracting its coefficient at degree $E(m, j)$. Terms with $t > j$ contribute $h_{j-t}^{(t)} = 0$, so Lemma 3.5 at $m = j + \ell$, where $E(j+\ell, j) = j(j+1)/2 + j\ell$, gives

$$D_j v_{j+\ell}^* = (-1)^j \frac{(j+1)^2(j+2)}{2} q^{j(j+1)/2+j\ell} + O(q^{j(j+1)/2+j\ell+1}) \qquad (j, \ell \geq 0). \tag{14}$$

The coefficient is obtained by summing

$$h_j^{(0)} + \sum_{t=1}^{j} h_{j-t}^{(t)} = \frac{(j+1)(j+2)(2j+3)}{6} + \binom{j+2}{3} = \frac{(j+1)^2(j+2)}{2}.$$

Proof of Theorem 3.4. The operators D_j act by lower unitriangular row operations, so they leave $\det(v_{i+j}^*)_{0\leq i,j<N}$ unchanged. By (14) the entry in row j and column ℓ has order $j(j+1)/2 + j\ell$. For $N = 2$, the matrix of entry orders is

$$\begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}.$$

The off-diagonal product is the unique term of order one. Its permutation sign cancels the negative leading coefficient of the second row, giving $6q + O(q^2)$. At arbitrary rank the same argument selects the reversed permutation. For a permutation ς, the corresponding Leibniz term has weight $\sum_j (j(j+1)/2 + j\varsigma(j))$. By the rearrangement inequality, $\sum_j j\varsigma(j)$ is uniquely minimised by the reversal $\varsigma(j) = N-1-j$, the values j being distinct. The minimum weight is $\sum_{j<N} j^2 = N(N-1)(2N-1)/6$, so exactly one

Leibniz term attains it and no cancellation is possible there. The sign of the reversal is $(-1)^{N(N-1)/2}$, which cancels $\prod_{j<N}(-1)^j$, and the surviving coefficient is

$$\prod_{j=0}^{N-1} \frac{(j+1)^2(j+2)}{2} = \frac{(N!)^2(N+1)!}{2^N}.$$

□

Formal order and analytic size are different questions. Theorem 3.4 fixes the first nonzero power of q and its coefficient at each fixed rank. It does not control the value at a fixed rational q as the rank grows. For the rest of this subsection, write $B_N = \sum_{j<N} j^2$ and $C_N = (N!)^2(N+1)!/2^N$ for the order and leading coefficient. The integer polynomials $f_N(q) = C_N q^{B_N}(1-q)^{N^3}$ have exactly these same two quantities, whereas $f_N(2/3) = C_N(2/3)^{B_N}3^{-N^3}$ carries a further cubic exponential factor that neither datum sees. The following separate positive-measure argument supplies the fixed-base estimate for V_N^*; it is not inferred from the formal order.

A separate positive-measure estimate. We seek positive moment weights comparable to $(k+1)^2(k+2)/2$, with constants depending only on q. In the determinant expansion these constants give factors exponential in N, so they do not change its cubic exponent of q. For fixed $0 < q < 1$ write $P = (q;q)_\infty$, $Q = (\sqrt{q};q)_\infty$, $T = (-1;q)_\infty^2$, and

$$G_q(w) = \frac{1}{(w;q)_\infty^3} \sum_{t\geq 0} \frac{w^t}{(q;q)_t} \frac{(q^t w^2;q)_\infty}{(q^t w;q)_\infty^2}, \qquad \gamma_k = [w^k]G_q(w), \qquad c_k = \frac{(k+1)^2(k+2)}{2}.$$

The choice $w = q^{m+1}$ collects all dependence on the moment index. Indeed, the three finite products in the tth remainder summand become

$$(q;q)_m = \frac{P}{(w;q)_\infty}, \quad (q^{t+1};q)_m = \frac{P}{(q;q)_t(q^t w;q)_\infty}, \quad (q^{m+t+1};q)_{m+1} = \frac{(q^t w;q)_\infty}{(q^t w^2;q)_\infty}.$$

Their substitution gives $v_m^* = P^4 G_q(q^{m+1})$ term by term. For fixed q and $0 < r < 1$, the tth summand on $|w| \leq r$ satisfies

$$\left| \frac{w^t (q^t w^2;q)_\infty}{(q;q)_t (w;q)_\infty^3 (q^t w;q)_\infty^2} \right| \leq \frac{(-r^2;q)_\infty}{P(r;q)_\infty^5} r^t.$$

Indeed, $(q;q)_t \geq P$, each denominator product in w has modulus at least $(r;q)_\infty > 0$, and the numerator product has modulus at most $(-r^2;q)_\infty$. The bound is independent of t, so the sum converges uniformly and absolutely on each such disk. The individual products converge there as well, since their tails are bounded by geometric series in q. Thus G_q is holomorphic for $|w| < 1$, and its Taylor series may be evaluated at $w = q^{m+1}$ to obtain the moment expansion. For formal series, the corresponding substitution is justified coefficient by coefficient, as in the short paper. The q-binomial theorem [28, Eq. 17.2.37]

$$\frac{(Aw;q)_\infty}{(w;q)_\infty} = \sum_{j\geq 0} \frac{(A;q)_j}{(q;q)_j} w^j \quad (0 \leq A \leq 1)$$

follows by comparing coefficients in $(1-w)R(w) = (1-Aw)R(qw)$ with $R(0) = 1$; the series converges for $|w| < 1$ since its coefficients are at most P^{-1}. They are nonnegative, and at least $(A;q)_\infty$ if $A < 1$. For $A = 1$ the series equals 1. Factor the numerator of the tth term using

$$(q^t w^2;q)_\infty = (q^{t/2}w;q)_\infty(-q^{t/2}w;q)_\infty(q^{(t+1)/2}w;q)_\infty(-q^{(t+1)/2}w;q)_\infty.$$

After pairing each positive-argument factor with one denominator, the tth summand of G_q becomes

$$\frac{w^t}{(q;q)_t}\frac{(q^{t/2}w;q)_\infty}{(w;q)_\infty}\frac{(q^{(t+1)/2}w;q)_\infty}{(w;q)_\infty} \times \frac{(-q^{t/2}w;q)_\infty(-q^{(t+1)/2}w;q)_\infty}{(w;q)_\infty(q^t w;q)_\infty^2}.$$

The q-binomial identity makes the two ratios nonnegative coefficientwise; the remaining product also has nonnegative coefficients. When $t = 0$, the first ratio is 1 and every coefficient of the second is at least Q. The last fraction has coefficients at least those of $(1-w)^{-3}$. Thus the $t = 0$ term alone bounds γ_k below by $Q\binom{k+3}{3}$.

For the upper bound, if R has nonnegative coefficients and $R(1) \le C < \infty$, convolution with a nondecreasing sequence d_k is bounded by Cd_k. At $t = 0$, bound the second ratio coefficientwise by $P^{-1}(1-w)^{-1}$. After extracting $(1-w)^{-3}$ from the last fraction, its remaining factor has value at $w = 1$ at most TP^{-3}. This gives $TP^{-4}\binom{k+3}{3}$. For $t \ge 1$, both ratios are bounded by $P^{-1}(1-w)^{-1}$. Extracting the one factor $(1-w)^{-1}$ from the last fraction leaves a factor with value at 1 at most TP^{-3}; also $(q;q)_t^{-1} \le P^{-1}$. The tth summand is therefore bounded coefficientwise by $TP^{-6}w^t(1-w)^{-3}$. Summing $t \ge 1$ gives $TP^{-6}\binom{k+2}{3}$ as the bound for its kth coefficient. Consequently

$$\frac{Q}{3}c_k \le \gamma_k \le T(P^{-4}+P^{-6})c_k.$$

Thus $\sum_{k\ge0} P^4\gamma_k q^k \delta_{q^k}$ is a finite positive measure on $[0,1]$, with infinitely many distinct support points and moments $v_m^*(q)$. In standard terminology, $(v_m^*(q))_{m\ge0}$ is a Hausdorff moment sequence. This statement fixes q; the measure is not a representing measure for the coefficient sequence $s_m(p)$ considered in the next subsection. A nonzero polynomial of degree less than N cannot vanish at all of its first N atoms; the associated Gram matrix is positive definite. Truncate the measure to its first $K+1$ atoms. At fixed rank N, each matrix entry converges as $K \to \infty$, and the determinant converges because it is a polynomial in those entries. Finite Cauchy–Binet and monotone convergence of the nonnegative tuple sums therefore give Heine's expansion, whose integral form is reproved in [10, Sec. 2, (2)–(5), pp. 2–3]:

$$V_N^* = \sum_{k_0<\cdots<k_{N-1}} \prod_i (P^4\gamma_{k_i}q^{k_i}) \prod_{i<j}(q^{k_i}-q^{k_j})^2.$$

Retaining the tuple $k_i = i$ and using $\prod_{d=1}^{N-1}(1-q^d)^{2(N-d)} \ge P^{2N}$ gives the lower bound below. For the upper bound put $k_i = i + \lambda_i$, where the λ_i are nonnegative and nondecreasing. Factoring the smaller power from each Vandermonde difference gives the

exact exponent

$$\sum_i k_i + 2\sum_{i<j} k_i = B_N + \sum_{i=0}^{N-1} (2N-1-2i)\lambda_i.$$

Every weight $2N-1-2i$ is at least 1. Since $0 < q < 1$, the remaining power of q is at most $q^{\sum_i \lambda_i}$. Also $c_{i+\lambda}/c_i \le (\lambda+1)^3$. Discard the remaining Vandermonde factors, each bounded by 1, and enlarge the nonnegative sum by dropping the ordering restriction on the λ_i. It now factors into N copies of $\sum_{\lambda\ge 0}(\lambda+1)^3 q^\lambda$, which gives

$$(P^6Q/3)^N C_N q^{B_N} \le V_N^*(q) \le \left[P^4T(P^{-4}+P^{-6})\frac{1+4q+q^2}{(1-q)^4}\right]^N C_N q^{B_N}.$$

Both constants are positive and finite, so $V_N^*(q) > 0$ and $\log(V_N^*(q)/(C_N q^{B_N})) = O_q(N)$, including $N = 0$ under the empty determinant convention. Here $N \to \infty$ with q fixed; the constants are not uniform as $q \uparrow 1$. The formal-order calculation instead fixes N and expands at $q = 0$. No joint uniform limit is asserted. The measure depends on q, not on the moment index or rank: one measure represents the whole sequence at that base. This is an analytic representation, not a denominator factor for the 2004 polynomial forms. Section 3.2 replaces the band $O_q(N)$ by an exact asymptotic with an explicit positive constant, using the same weights.

Formal verification and computations. The positive-measure argument, including its generating-function identification and infinite determinant expansion, is an ordinary proof and is not formalised. The formal-series argument has a checked recurrence for the leading coefficients. The Lean proof of Theorem 3.4 uses the first nonzero term of every transformed row, which for row 1 has order exactly $l+1$ and coefficient exactly -6 in every column. Independently of the proof, the order and the leading coefficient were computed exactly for $1 \le N \le 7$ (program, result), giving orders $0, 1, 5, 14, 30, 55, 91$ and leading coefficients $1, 6, 108, 4320, 324000, 40824000, 8001504000$, each matching the closed forms.

Earlier work and the unresolved value at $3/2$**.** The antecedent is the inequality of [9, Sec. 4, p. 7]; the equality and the leading coefficient are proved here. Lemma 3.5 and (14) give that row calculation. Nothing in this subsection decides the arithmetic nature of $F(3/2)$.

3.2 The size of V_N^* at a fixed base

The preceding bounds leave an error of order N in the logarithm. We now determine its linear, logarithmic and constant terms. There are two calculations: compare every node configuration with the first N nodes, then multiply the resulting moment weights. The first is independent of the particular weights under the hypotheses below; the second produces the Lambert-value correction for V_N^*. Throughout this subsection $0 < q < 1$ is fixed, and we write

$$P = (q;q)_\infty, \qquad \mathcal{M}(q) = \prod_{d\ge 1}(1-q^d)^{-d}, \qquad L = F(1/q) = \sum_{r\ge 1}\frac{q^r}{1-q^r},$$

with $B_N = \sum_{j<N} j^2$ and $C_N = \prod_{k<N} c_k = (N!)^2(N+1)!/2^N$ as above. Both products converge to positive finite values, and the abbreviation L is used only in this subsection; it is the value at q of the divisor series $\mathcal{L}$ of Section 10. The estimates in this subsection concern real determinants at fixed q. Denominator valuations for the 2004 forms were treated separately in Section 2.

For $a_k = 1$, the moments are $(1-q^{m+1})^{-1}$, so Cauchy's determinant formula gives the model calculation

$$D_N^{(0)} := \det\left(\frac{1}{1-q^{i+j+1}}\right)_{0\le i,j<N} = \frac{q^{B_N}\Delta_N}{\prod_{0\le i,j<N}(1-q^{i+j+1})}, \qquad \Delta_N = \prod_{d=1}^{N-1}(1-q^d)^{2(N-d)}. \tag{15}$$

The denominator has limiting multiplicity d at $1-q^d$, while

$$\frac{\Delta_N}{P^{2N}} = \prod_{d<N}(1-q^d)^{-2d}\prod_{d\ge N}(1-q^d)^{-2N} \longrightarrow \mathcal{M}(q)^2.$$

The convergence follows from $\sum d|\log(1-q^d)| < \infty$ and the bound $O_q(Nq^N)$ for the logarithm of the second product. Hence

$$D_N^{(0)} \sim \mathcal{M}(q)^3 q^{B_N} P^{2N}. \tag{16}$$

Two copies of $\mathcal{M}(q)$ come from the squared Vandermonde and one from the Cauchy denominator. We shall compare general weights with this exact model. Fixed displacements of the last few node indices have weight ratios tending to 1; the polynomial bound in the theorem controls the sum over all displacements.

Theorem 3.6 (determinants of geometric moments)**.** *Let $a_k > 0$ satisfy, for fixed constants C and κ,*

$$\frac{a_{k+h}}{a_k} \longrightarrow 1 \quad (k\to\infty) \text{ for each fixed } h\ge 0, \qquad \frac{a_{k+h}}{a_k} \le C(1+h)^\kappa \quad (k,h\ge 0).$$

Put $M_m = \sum_{k\ge 0} a_k q^{(m+1)k}$ and $D_N = \det(M_{i+j})_{0\le i,j<N}$. Then

$$D_N \sim \mathcal{M}(q)^3\, q^{B_N} P^{2N} \prod_{k=0}^{N-1} a_k \qquad (N\to\infty).$$

Proof of Theorem 3.6. We may increase C to at least 1 and κ to at least 0. The bound $a_h \le Ca_0(1+h)^\kappa$ makes $\sum_{k\ge 0} a_k q^k \delta_{q^k}$ a finite positive measure with moments M_m. Apply Cauchy–Binet to finite truncations. Entrywise convergence gives convergence of their determinants, and monotone convergence applies to the nonnegative tuple sums. Hence

$$D_N = \sum_{0\le k_0<\cdots<k_{N-1}} \prod_{i<N} a_{k_i} q^{k_i} \prod_{i<j<N} (q^{k_i}-q^{k_j})^2. \tag{17}$$

The tuple $k_i = i$ contributes $q^{B_N}\Delta_N \prod_{i<N} a_i$. It does not account for the whole asymptotic. For example, replace $(0,1,\dots,N-2,N-1)$ by $(0,1,\dots,N-2,N)$. The ratio of its contribution to that of the original tuple is

$$q\frac{a_N}{a_{N-1}}\prod_{d=1}^{N-1}\left(\frac{1-q^{d+1}}{1-q^d}\right)^2 = q\frac{a_N}{a_{N-1}}\left(\frac{1-q^N}{1-q}\right)^2 \longrightarrow \frac{q}{(1-q)^2}. \tag{18}$$

Only pairs involving the moved node change, so the product telescopes. At $q = 1/2$ the limiting ratio is 2: the displaced tuple contributes twice as much asymptotically as $(0, 1, \dots, N-1)$.

Subtracting $(0, 1, \dots, N-1)$ gives the shifts $\lambda_i = k_i - i$. Since $k_{i+1} \geq k_i + 1$, these are nonnegative and weakly increasing; reversing their order gives a partition. For $N = 4$, for example,

$$\underbrace{(0,1,3,5)}_{(k_i)} \longmapsto \underbrace{(0,0,1,2)}_{(\lambda_i)} \longmapsto \underbrace{(2,1,0,0)}_{(\mu_j)}.$$

In general, put $\mu_j = \lambda_{N-j}$ for $1 \leq j \leq N$, so that $\mu_1 \geq \cdots \geq \mu_N \geq 0$. Division by the contribution of $k_i = i$ turns the summand for this tuple into

$$\begin{aligned} W_N(\mu) = q^{\sum_{j\leq N}(2j-1)\mu_j} \prod_{j\leq N} \frac{a_{N-j+\mu_j}}{a_{N-j}} \\ \cdot \prod_{1\leq j<k\leq N} \left(\frac{1-q^{k-j+\mu_j-\mu_k}}{1-q^{k-j}} \right)^2. \end{aligned} \tag{19}$$

We order the indices increasingly, so their geometric nodes decrease towards zero. The jth index from the right contributes once through q^{k_i} and twice through each of its $j-1$ pairs with later indices in the squared Vandermonde. Its displacement therefore has exponent $1 + 2(j-1) = 2j-1$.

Set $W_N(\mu) = 0$ when μ has more than N positive parts, so that every sum has the same index set. Fix a partition with ℓ positive parts. It affects only the last ℓ indices of $(0, 1, \dots, N-1)$: for instance, $(2, 1)$ affects the last two for every $N \geq 2$. Those indices tend to infinity, so the weight ratios in (19) tend to 1 by the fixed-shift hypothesis.

The Vandermonde quotient still contains a growing number of factors, but its dependence on N telescopes. The factors with $j < k \leq \ell$ are fixed, and those with $j > \ell$ equal 1. For $N \geq \ell$ and each $j \leq \ell$, the remaining factors satisfy

$$\prod_{k=\ell+1}^{N} \frac{1-q^{k-j+\mu_j}}{1-q^{k-j}} = \frac{(q^{N+1-j};q)_{\mu_j}}{(q^{\ell+1-j};q)_{\mu_j}} \longrightarrow \frac{1}{(q^{\ell+1-j};q)_{\mu_j}}. \tag{20}$$

Thus each $W_N(\mu)$ has the same limit as for constant weights.

To pass this limit through the sum over partitions, we need a bound independent of N that remains summable as both the number and size of the parts grow. Only factors with $j \leq \ell$ can differ from 1. For each such j, the denominator product in the Vandermonde quotient is at least P, and its numerator product is at most 1. Together with the polynomial shift bound, this gives

$$W_N(\mu) \leq (CP^{-2})^{\ell} \prod_{j=1}^{\ell} (1+\mu_j)^{\kappa} q^{(2j-1)\mu_j}.$$

Let $A = \sum_{u\geq 1} (1+u)^{\kappa} q^{u-1} < \infty$. Dropping the ordering of the positive parts, we obtain

$$\begin{aligned} \sum_{\ell(\mu)=\ell} \sup_N W_N(\mu) &\leq (CP^{-2})^{\ell} \prod_{j=1}^{\ell} \sum_{u\geq 1} (1+u)^{\kappa} q^{(2j-1)u} \\ &\leq (CP^{-2}A)^{\ell} q^{\ell^2}. \end{aligned}$$

Here $q^{(2j-1)u} \le q^{2j-1}q^{u-1}$ and $1+3+\cdots+(2\ell-1) = \ell^2$. Since $q < 1$, the bound $(CP^{-2}A)^\ell q^{\ell^2}$ is summable over ℓ, independently of N.

Dominated convergence gives the same limiting partition sum as for $a_k = 1$. By (15), this sum tends to $\mathcal{M}(q)$, the reciprocal of the limiting Cauchy denominator. Together with $\Delta_N/P^{2N} \to \mathcal{M}(q)^2$, it accounts for all three copies of $\mathcal{M}(q)$. Restoring the principal tuple gives

$$\frac{D_N}{D_N^{(0)} \prod_{k<N} a_k} \longrightarrow 1.$$

The model asymptotic (16) now proves the theorem. The comparison explains why the normalised limit retains the geometric spacing of the nodes but loses the individual weights. □

We now return to $a_k = P^4\gamma_k$. A factorisation into two finite Gaussian multinomial sums supplies both the shift hypotheses and the first correction to a_k/c_k. The shift hypotheses allow application of the determinant theorem; the correction determines the product $\prod_{k<N} a_k$.

Proposition 3.7 (the moment weights as a product of two finite sums)**.** *For $r \ge 1$ and $k \ge 0$ set*

$$R_k^{(r)}(q) = \sum_{n_1+\cdots+n_r=k} \frac{(q;q)_k}{\prod_{j\le r}(q;q)_{n_j}},$$

the sum being over compositions into r nonnegative parts, so that $R_k^{(r)}$ is the sum of all q-multinomial coefficients of degree k and length r, equivalently the multivariate Rogers–Szegő polynomial at unit arguments [42]. These are polynomial identities in the indeterminate q; at a prime power, $r = 2$ gives the Galois number counting the subspaces of a k-dimensional space over the field of that order, and $r = 3$ counts the flags $V_1 \subseteq V_2$ in that space. The source is cited for those identifications; the factorisation below is proved here. Then

$$\gamma_k(q) = \frac{R_k^{(2)}(q)R_k^{(3)}(q)}{(q;q)_k}.$$

The product $R_k^{(2)}R_k^{(3)}$ lies in $\mathbb{Z}_{\ge 0}[q]$, has degree $\lfloor k^2/4\rfloor + \lfloor k^2/3\rfloor$, and has coefficient sum 6^k. Consequently $(q;q)_k\gamma_k \in \mathbb{Z}[q]$, and the weights $a_k = P^4\gamma_k$ satisfy

$$P^5 c_k \le a_k \le P^{-1}c_k \qquad (k \ge 0),$$

together with $a_{k+h}/a_k \le P^{-6}(1+h)^3$, so the hypotheses of Theorem 3.6 hold with $\kappa = 3$.

Proof. Write $\mathcal{E}(z) = (z;q)_\infty^{-1}$. Euler's expansion gives $\mathcal{E}(z)^r = \sum_{n\ge 0} R_n^{(r)} z^n/(q;q)_n$. We seek the product $R_k^{(2)}R_k^{(3)}$ at a common index k. Splitting $k = n + j$ keeps $R_k^{(3)}$ fixed while the sum over $((q;q)_n(q;q)_j)^{-1}$ supplies $R_k^{(2)}/(q;q)_k$. The operator $\partial_q f(z) = (f(z) - f(qz))/z$ produces the needed index shift: $\partial_q z^j = (1-q^j)z^{j-1}$ cancels the last factor of $(q;q)_j$. Thus

$$\partial_q \mathcal{E} = \mathcal{E}, \qquad \partial_q^n \mathcal{E}(z)^3 = \sum_{j\ge 0} \frac{R_{n+j}^{(3)}}{(q;q)_j} z^j.$$

The q-Leibniz rule, obtained by induction from the product rule, is

$$\partial_q^n(fg)(z) = \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix}_q (\partial_q^j f)(q^{n-j}z)\,(\partial_q^{n-j} g)(z).$$

Write $h_k(z) = \sum_{j=0}^{k} \begin{bmatrix} k \\ j \end{bmatrix}_q (z;q)_j$. Applying the rule to $\mathcal{E}\cdot\mathcal{E}$ and using $\mathcal{E}(q^m z) = (z;q)_m \mathcal{E}(z)$ gives $\partial_q^k \mathcal{E}(z)^2 = \mathcal{E}(z)^2 h_k(z)$; at $k=1$ both sides are $\mathcal{E}(z)^2(2-z)$. Applying it once more, to $\mathcal{E}\cdot\mathcal{E}^2$, gives

$$\partial_q^n \mathcal{E}(z)^3 = \mathcal{E}(z)^3 \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q (z;q)_k h_k(z),$$

Hence

$$\sum_{n\ge 0} \frac{w^n}{(q;q)_n} \partial_q^n \mathcal{E}(z)^3 = \mathcal{E}(w)\mathcal{E}(z)^3 \sum_{k\ge 0} \frac{w^k (z;q)_k}{(q;q)_k} h_k(z).$$

The left-hand side is $\sum_{n,j\ge 0} R^{(3)}_{n+j} w^n z^j / ((q;q)_n (q;q)_j)$. On setting $z = w$, all pairs with the same sum contribute to a single coefficient. Thus the coefficient of w^k is $R^{(3)}_k \sum_{j\le k} \big((q;q)_j (q;q)_{k-j}\big)^{-1} = R^{(2)}_k R^{(3)}_k / (q;q)_k$. On the right-hand side, put $z = w$, expand h_k, and write $k = j + t$. The identity $(w;q)_{j+t} = (w;q)_j (wq^j;q)_t$ gives

$$\mathcal{E}(w)^4 \sum_{j\ge 0} \frac{w^j (w;q)_j^2}{(q;q)_j} \sum_{t\ge 0} \frac{(wq^j;q)_t}{(q;q)_t} w^t.$$

The inner sum is evaluated by the q-binomial theorem:

$$\sum_{t\ge 0} \frac{(wq^j;q)_t}{(q;q)_t} w^t = \frac{(w^2 q^j;q)_\infty}{(w;q)_\infty}.$$

Finally, $(w;q)_j = (w;q)_\infty / (wq^j;q)_\infty$ cancels two of the five factors $\mathcal{E}(w)$, leaving

$$\mathcal{E}(w)^3 \sum_{j\ge 0} \frac{w^j}{(q;q)_j} \frac{(w^2 q^j;q)_\infty}{(wq^j;q)_\infty^2} = G_q(w).$$

Every step holds coefficientwise, and absolutely for $|w|$ small.

A q-multinomial coefficient has nonnegative integral coefficients and degree $\sum_{i<j} n_i n_j$. At $q = 1$ it equals the corresponding ordinary multinomial coefficient. Summing over compositions gives coefficient sum r^k for $R^{(r)}_k$ and hence 6^k for the product. The degree is maximised by balanced parts, which gives $\lfloor k^2/4 \rfloor$ for $r = 2$ and $\lfloor k^2/3 \rfloor$ for $r = 3$; no leading cancellation occurs because all coefficients are nonnegative.

For the bounds, write $b^{(r)}_k = [z^k](z;q)_\infty^{-r}$, so that $R^{(r)}_k = (q;q)_k b^{(r)}_k$ and $\gamma_k = (q;q)_k b^{(2)}_k b^{(3)}_k$. Euler's expansion exhibits $b^{(r)}_k$ as an r-fold convolution of the sequence $1/(q;q)_n$, whose terms lie in $[1, P^{-1}]$; the number of compositions of k into r nonnegative parts is $\binom{k+r-1}{r-1}$, so

$$\binom{k+r-1}{r-1} \le b^{(r)}_k \le P^{-r} \binom{k+r-1}{r-1}.$$

Combining these at $r = 2, 3$ with $P \le (q;q)_k \le 1$ and $c_k = (k+1)\binom{k+2}{2}$ gives $P^5 c_k \le a_k \le P^{-1} c_k$. The shift bound follows from $c_{k+h}/c_k \le (1+h)^3$.

It remains to show that $a_{k+h}/a_k \to 1$ for each fixed h. The two convolutions grow at least linearly and quadratically; it suffices to bound their increments by a constant and a linear function, respectively. Put $e_n = 1/(q;q)_n$. Since $(q;q)_{n+1} = (1-q^{n+1})(q;q)_n$, these terms satisfy $1 = e_0 \le e_1 \le \cdots \le P^{-1}$, and the convolutions read $b_k^{(2)} = \sum_{i \le k} e_i e_{k-i}$ and $b_k^{(3)} = \sum_{i \le k} e_i b_{k-i}^{(2)}$. Hence

$$b_{k+1}^{(2)} - b_k^{(2)} = \sum_{i \le k} e_i (e_{k+1-i} - e_{k-i}) + e_{k+1},$$
$$b_{k+1}^{(3)} - b_k^{(3)} = \sum_{i \le k} e_i (b_{k+1-i}^{(2)} - b_{k-i}^{(2)}) + e_{k+1}.$$

Every bracket is nonnegative. In the first line the brackets add up to $e_{k+1} - 1 < P^{-1}$ and each $e_i \le P^{-1}$, so $0 \le b_{k+1}^{(2)} - b_k^{(2)} \le P^{-2} + P^{-1} \le 2P^{-2}$. Inserting this bound into the $k+1$ terms of the second line gives $0 \le b_{k+1}^{(3)} - b_k^{(3)} \le 2(k+1)P^{-3} + P^{-1} \le 3(k+1)P^{-3}$. Against the lower bounds $b_k^{(2)} \ge k+1$ and $b_k^{(3)} \ge (k+1)(k+2)/2$ displayed above, the ratio $b_{k+1}^{(r)}/b_k^{(r)}$ lies in $[1, 1 + 2P^{-2}/(k+1)]$ for $r = 2$ and in $[1, 1 + 6P^{-3}/(k+2)]$ for $r = 3$. Both ratios tend to 1, and therefore so does $b_{k+h}^{(r)}/b_k^{(r)}$, a product of h consecutive ratios. Finally $(q;q)_{k+h}/(q;q)_k = \prod_{j=1}^{h}(1 - q^{k+j}) \to 1$, and $a_{k+h}/a_k = \gamma_{k+h}/\gamma_k$ is the product of these three ratios. The Lean proof obtains the fixed-shift limit by the same steps. □

The factorisation proves positivity at each fixed real q, not coefficientwise positivity in q: for example, $a_0 = P^4$ has q-coefficient -4. The cubic factors c_k multiply to C_N, but $a_k/c_k \to 1$ alone would not determine the product of N weights. The first relative correction has order $1/k$ and accumulates as a harmonic sum; the remaining $O_q(k^{-2})$ terms determine a finite constant. The two convolutions supply this first correction explicitly.

Theorem 3.8 (the size of V_N^* at a fixed base)**.** *Let γ_E be Euler's constant. The product*

$$\mathcal{A}(q) = e^{-8\gamma_E L} \prod_{k \ge 0} \left(\frac{a_k}{c_k} e^{8L/(k+1)} \right)$$

converges to a positive finite value. With $K(q) = \mathcal{A}(q)\mathcal{M}(q)^3$,

$$V_N^*(q) \sim K(q)\, C_N q^{B_N} P^{2N} N^{-8L},$$

equivalently

$$\log V_N^*(q) = B_N \log q + \log C_N + 2N \log P - 8L \log N + \log K(q) + o(1).$$

Proof of Theorem 3.8. Put $L = F(1/q)$, $e_k = (q;q)_k^{-1}$ and $E = P^{-1}$. Replacing every e_k by E in the two convolution sums gives $E^2(k+1)$ and $E^3\binom{k+2}{2}$. The factors $k+1$ and $\binom{k+2}{2}$ count the nonnegative pairs and triples with sum k, respectively. For a fixed index i, there is one pair $(i, k-i)$ and there are $k-i+1$ triples whose first coordinate is i. Thus a summable deficit in the sequence e_i produces a constant correction in the two-fold sum

and a linear correction in the three-fold sum. We calculate that deficit and bound terms with two or more deficits.

Writing $d_k = E - e_k$, the identity $e_{k+1} - e_k = q^{k+1}e_{k+1}$ gives

$$d_k = \sum_{j>k} q^j e_j, \qquad 0 \le d_k \le \frac{Eq^{k+1}}{1-q}.$$

Thus $\sum d_k$ and $\sum k d_k$ converge. To evaluate the former, use Euler's identity $\mathcal{E}(z) = \sum_{k\ge 0} e_k z^k = (z;q)_\infty^{-1}$ for $|z| < 1$. Tonelli's theorem and logarithmic differentiation at $z = q$ give

$$S := \sum_{k\ge 0} d_k = \sum_{j\ge 1} j q^j e_j = q\mathcal{E}'(q) = E\sum_{r\ge 1} \frac{q^r}{1-q^r} = EL.$$

In the double sum, $q^j e_j$ occurs once for each $k = 0, \dots, j-1$, which explains the factor j. Logarithmic differentiation is justified by locally uniform convergence on $|z| < 1$.

Let $b_k^{(r)} = [z^k]\mathcal{E}(z)^r$. The finite-sum formula in Proposition 3.7 becomes

$$a_k = P^4 (q;q)_k b_k^{(2)} b_k^{(3)}.$$

We expand each finite convolution using $e_k = E - d_k$. For two factors,

$$b_k^{(2)} = E^2(k+1) - 2E\sum_{i=0}^{k} d_i + \sum_{i=0}^{k} d_i d_{k-i} = E^2(k+1-2L) + O_q((k+1)q^k).$$

For three factors, the terms with one d_i contribute $-3E^2\sum_{i=0}^{k}(k-i+1)d_i$. Here the sum equals $(k+1)S + O_q(1)$ because $\sum i d_i < \infty$. Terms with two d_i are bounded by $3ES^2$, and the term with three is $O_q((k+1)^2 q^k)$. It follows that

$$b_k^{(3)} = E^3\left(\binom{k+2}{2} - 3L(k+1)\right) + O_q(1).$$

Dividing the correction $-3L(k+1)$ by the leading term $\binom{k+2}{2}$ gives the relative correction $-6L/(k+2)$. The two-fold convolution contributes $-2L/(k+1)$ in the same way. Since $(q;q)_k = P(1+O_q(q^k))$, the leading constants in $a_k = P^4(q;q)_k b_k^{(2)} b_k^{(3)}$ multiply to $P^5E^5 = 1$. Replacing $(k+2)^{-1}$ by $(k+1)^{-1}$ changes only the $O_q((k+1)^{-2})$ error. Thus the coefficients $2L$ and $6L$ add, and with $c_k = (k+1)^2(k+2)/2$ we obtain

$$\begin{aligned}\frac{a_k}{c_k} &= \left(1 - \frac{2L}{k+1} + O_q(q^k)\right)\left(1 - \frac{6L}{k+2} + O_q((k+1)^{-2})\right)\left(1 + O_q(q^k)\right)\\ &= 1 - \frac{8L}{k+1} + O_q((k+1)^{-2}).\end{aligned}$$

In particular $a_{k+1}/a_k \to 1$, and multiplication of finitely many adjacent ratios gives $a_{k+h}/a_k \to 1$ for each fixed h. The uniform shift bound follows from Proposition 3.7, so Theorem 3.6 applies.

For large k, taking logarithms preserves the first-order term and leaves the summable error $\log(a_k/c_k) + 8L/(k+1) = O_q((k+1)^{-2})$. The finitely many earlier factors are

positive, so the regularised product also has a positive finite limit: $0 < \mathcal{A}(q) < \infty$. Since $\prod_{k<N} c_k = C_N$, the harmonic contribution separates as

$$\log \frac{\prod_{k<N} a_k}{C_N} = -8L \sum_{k<N} \frac{1}{k+1} + \sum_{k<N} \left(\log \frac{a_k}{c_k} + \frac{8L}{k+1} \right).$$

The second sum tends to $\log \mathcal{A}(q) + 8\gamma_E L$. Using $\sum_{k<N} (k+1)^{-1} = \log N + \gamma_E + o(1)$, the Euler-constant terms cancel, giving $\prod_{k<N} a_k \sim \mathcal{A}(q) C_N N^{-8L}$. Theorem 3.6 now gives the stated determinant asymptotic. Taking logarithms yields

$$\log V_N^* = B_N \log q + \log C_N + 2N \log P - 8L \log N + \log K(q) + o(1).$$

□

The convolution proof explains $8L$ as $2L + 6L$, one contribution from each finite sum. An alternative reads the same correction from the generating function near $w = 1$. Its pole of order four accounts for the cubic growth of a_k; the next principal-part coefficient accounts for the relative term of order $1/k$.

An alternative coefficient calculation. Write $\Pi(w) = (qw;q)_\infty$, so that $(w;q)_\infty = (1 - w)\Pi(w)$, and for $t \geq 1$ put

$$f_t(w) = \frac{(q^t w^2; q)_\infty}{(q;q)_t (q^t w; q)_\infty^2}, \qquad \mathcal{S}(w) = \sum_{t \geq 1} w^t (f_t(w) - P^{-1}).$$

On each compact subset of $|w| < q^{-1}$ the bracket is $O_q(q^t)$ uniformly, so $\mathcal{S}$ is analytic there. Separating the summand $t = 0$, whose denominator carries the factor $(1 - w)^2$, and subtracting the constant part of the remaining summands gives

$$\mathcal{D}(w) := P^4 (1 - w)^4 G_q(w) = \frac{P^4}{\Pi(w)^3} \left((1 + w) \frac{(qw^2; q)_\infty}{\Pi(w)^2} + \frac{w}{P} + (1 - w)\mathcal{S}(w) \right).$$

Hence $\mathcal{D}$ is analytic on a disc of radius greater than 1.

At $w = 1$ one has $\Pi(1) = P$ and $\Pi'(1)/\Pi(1) = -L$, since $\Pi'/\Pi = -\sum_{j \geq 1} q^j/(1 - q^j w)$. Also $(qw^2; q)_\infty$ equals P at $w = 1$, and the logarithmic derivative of $(qw^2; q)_\infty / \Pi(w)^2$ at $w = 1$ is $-2L + 2L = 0$. Therefore the bracket has value $3/P$ and derivative $(2 - L)/P$ at $w = 1$; here $\mathcal{S}(1) = L/P$, because $f_t(1) = (P(1 - q^t))^{-1}$ and $\sum_{t \geq 1} ((1 - q^t)^{-1} - 1) = L$. Combining with the factor $P^4 \Pi^{-3}$, whose logarithmic derivative at 1 is $3L$, gives

$$\mathcal{D}(1) = 3, \qquad \mathcal{D}'(1) = 3 \left(3L + \frac{2 - L}{3} \right) = 2 + 8L.$$

The first two principal-part terms are determined by $\mathcal{D}(1)$ and $\mathcal{D}'(1)$. The terms from its second and third derivatives have coefficients of order at most k; after the full principal part is subtracted, the remainder is analytic on a larger disc and has exponentially decreasing coefficients. Keeping only the first two terms therefore gives

$$a_k = 3 \binom{k+3}{3} - (2 + 8L) \binom{k+2}{2} + O_q(k), \qquad \frac{a_k}{c_k} = 1 - \frac{8L}{k+1} + O_q((k+1)^{-2}),$$

the second line because $3\binom{k+3}{3} = c_k(k+3)/(k+1)$ and $\binom{k+2}{2} = c_k/(k+1)$.

The logarithmic error is again summable. The product argument in the first proof therefore gives $0 < \mathcal{A}(q) < \infty$ and, using $\sum_{k<N}(k+1)^{-1} = \log N + \gamma_E + o(1)$,

$$\prod_{k<N} a_k \sim \mathcal{A}(q)\, C_N N^{-8L}.$$

Proposition 3.7 supplies the uniform shift bound and the fixed-shift limit, so Theorem 3.6 applies to (a_k) and gives $V_N^* \sim \mathcal{M}(q)^3 q^{B_N} P^{2N} \prod_{k<N} a_k$. Substituting the product asymptotic proves both displays.

□

Comparison with the cubic bound. The estimate identifies the terms $2N\log P$, $-8F(1/q)\log N$ and $\log K(q)$ within the previous $O_q(N)$ error. It leaves the cubic coefficient unchanged and has no effect on prime-power valuations. For the denominator estimate at $3/2$, the difference of the relevant exponents is

$$\frac{39}{41}(4N^3 - 3N^2) - (2N^3 - N) = \frac{N(74N^2 - 117N + 41)}{41} > 0 \qquad (N \geq 2),$$

the positivity being immediate from $74N^2 - 117N + 41 = 74(N-2)^2 + 179(N-2) + 103$. Its leading size is $(74/41)N^3$. A logarithmic correction cannot offset this cubic gap in the stated denominator model. The factorisation refines the moment weights, not the denominator of an irrationality form; any such form still needs an estimate for its own rational coefficients.

An extension with limiting weight ratio

For $0 < \rho < q^{-1}$ set

$$\mathcal{M}(\rho; q) = \prod_{d\geq 1}(1 - \rho q^d)^{-d}, \qquad \mathcal{M}(1; q) = \mathcal{M}(q).$$

A limiting adjacent-weight ratio supplies an exponential shift bound, replacing the polynomial bound above. The condition $\rho q < 1$ makes that bound summable and changes one copy of the product constant from $\mathcal{M}(1; q)$ to $\mathcal{M}(\rho; q)$. The ordinary proof below has no Lean proof in the supplied record; this unformalised remark is not used in the checked fixed-base argument.

Remark (limiting weight ratios; unformalised). Let $0 < q < 1$ and $0 < \rho < q^{-1}$. Suppose that $a_k > 0$ for $k \geq 0$ and

$$\frac{a_{k+1}}{a_k} \longrightarrow \rho \qquad (k \to \infty).$$

Define

$$M_m = \sum_{k\geq 0} a_k q^{(m+1)k}, \qquad D_N = \det(M_{i+j})_{0\leq i,j<N}.$$

Then, as $N \to \infty$,

$$D_N \sim \mathcal{M}(1; q)^2 \mathcal{M}(\rho; q)\, q^{B_N} P^{2N} \prod_{k=0}^{N-1} a_k. \tag{21}$$

Ordinary argument for the ratio-limit remark. Choose $\hat{\rho}$ with $\rho < \hat{\rho} < q^{-1}$. There is a K such that $a_{j+1}/a_j \le \hat{\rho}$ for $j \ge K$. With

$$C = \prod_{j<K} \max\left(1, \frac{a_{j+1}}{\hat{\rho} a_j}\right),$$

we obtain, by multiplying adjacent ratios,

$$\frac{a_{k+h}}{a_k} \le C\hat{\rho}^h \qquad (k, h \ge 0). \tag{22}$$

This also gives $a_h \le C a_0 \hat{\rho}^h$, so every moment converges because $q\hat{\rho} < 1$. For finitely many nodes, Cauchy–Binet expands their moment determinant as a sum of squared Vandermonde products. We let the number of nodes tend to infinity. Entrywise convergence gives the limit of the determinant, while positivity gives the limit of the sum. Hence

$$D_N = \sum_{0 \le k_0 < \cdots < k_{N-1}} \prod_{i<N} a_{k_i} q^{k_i} \prod_{i<j<N} (q^{k_i} - q^{k_j})^2. \tag{23}$$

The tuple $k_i = i$ contributes

$$q^{B_N} \Delta_N \prod_{i<N} a_i, \qquad \Delta_N = \prod_{d=1}^{N-1} (1 - q^d)^{2(N-d)}.$$

Write $k_i = i + \lambda_i$ and reverse the displacements: $\mu_j = \lambda_{N-j}$ for $1 \le j \le N$. Then $\mu_1 \ge \cdots \ge \mu_N \ge 0$. After division by the least-tuple contribution, the summand becomes

$$\begin{aligned} W_N(\mu) = q^{\sum_{j \le N} (2j-1)\mu_j} \prod_{j \le N} \frac{a_{N-j+\mu_j}}{a_{N-j}} \\ \cdot \prod_{1 \le j < k \le N} \left(\frac{1 - q^{k-j+\mu_j-\mu_k}}{1 - q^{k-j}} \right)^2. \end{aligned} \tag{24}$$

For partitions of length greater than N, set $W_N(\mu) = 0$. All the sums now have the same countable index set.

Suppose that μ has ℓ positive parts. Only the factors with $j \le \ell$ can differ from 1. For a fixed such j, the denominators in the Vandermonde quotient multiply to at least P, and its numerators are at most 1. We therefore have a bound independent of N

$$W_N(\mu) \le (CP^{-2})^\ell \prod_{j=1}^{\ell} \hat{\rho}^{\mu_j} q^{(2j-1)\mu_j}.$$

We sum this majorant by first fixing ℓ and then dropping the ordering condition on the positive parts. This gives

$$\begin{aligned} \sum_{\ell(\mu)=\ell} \sup_N W_N(\mu) &\le (CP^{-2})^\ell \prod_{j=1}^{\ell} \sum_{u \ge 1} \hat{\rho}^u q^{(2j-1)u} \\ &\le C_*^\ell q^{\ell^2}, \qquad C_* = \frac{C\hat{\rho}P^{-2}}{1 - \hat{\rho}q}. \end{aligned}$$

Here $\sum_{j\le\ell}(2j-1)=\ell^2$ and $1-\hat\rho q^{2j-1}\ge 1-\hat\rho q>0$. The series $\sum_{\ell\ge0}C_*^{\ell}q^{\ell^2}$ converges.

For fixed h, telescoping the adjacent ratios shows that $a_{k+h}/a_k\to\rho^h$. Thus, for each fixed partition, the weight ratios in (24) have the same limit as for $a_k=\rho^k$. The Vandermonde quotient is independent of the weights, and its convergence follows from the same finite identity (20). Dominated convergence therefore reduces the limit of $\sum_\mu W_N(\mu)$ to the case $a_k=\rho^k$.

In that case $M_m=(1-\rho q^{m+1})^{-1}$, and Cauchy's determinant gives

$$D_N=\frac{\rho^{N(N-1)/2}q^{B_N}\Delta_N}{\prod_{0\le i,j<N}(1-\rho q^{i+j+1})}.$$

The multiplicity of $1-\rho q^d$ in the denominator increases to d, so the normalised sum tends to $\mathcal{M}(\rho;q)$. Finally,

$$\frac{\Delta_N}{P^{2N}}=\prod_{d<N}(1-q^d)^{-2d}\prod_{d\ge N}(1-q^d)^{-2N}\longrightarrow\mathcal{M}(1;q)^2,$$

since the logarithm of the second product is $O_q(Nq^N)$. Combining the two limits proves (21). □

For example, $a_k=\rho^k(k+1)^s$, with $s\in\mathbb{R}$, satisfies the ratio hypothesis, and $\prod_{k<N}a_k=\rho^{N(N-1)/2}(N!)^s$. For $s=1$ the moments are $(1-\rho q^{m+1})^{-2}$, giving

$$\det\left((1-\rho q^{i+j+1})^{-2}\right)_{i,j<N}\sim\mathcal{M}(q)^2\mathcal{M}(\rho;q)q^{B_N}P^{2N}\rho^{N(N-1)/2}N!.$$

The choice $a_k=\rho^k\exp(\sqrt{k})$ also satisfies the ratio hypothesis, since $\sqrt{k+1}-\sqrt{k}\to0$. For $\rho=1$, its relative growth at $k=0$ exceeds every polynomial in the shift. Thus this argument extends beyond the polynomial-bound hypotheses of Theorem 3.6.

3.3 Positive Hankel determinants and a rational special value

We construct a generating function with the rational special value 1, although its moment determinants have the same formal orders, leading coefficients and type of fixed-base power correction as Zudilin's. This shows why those determinant properties alone do not establish irrationality. The construction uses neither Zudilin's recurrence nor his polynomial factorisation or cyclotomic divisors, and gives no conclusion about $F(3/2)$. The arguments below have ordinary proofs only and are stated as unformalised remarks.

We choose the tail of the weight sequence to obtain the determinant asymptotic, and its first terms to fix the special value. A telescoping difference of the moments will then give the value 1. The cut-off below separates these two choices while leaving room to keep every weight positive.

Put

$$c_k=\frac{(k+1)^2(k+2)}{2},\qquad d_k=\frac{(k+1)(k+2)}{2},\qquad R(w)=\sum_{k\ge0}c_kw^k=\frac{1+2w}{(1-w)^4}.$$

For $0 < q \le 2/3$ define, with a single fixed cut-off 32,

$$a_k(q) = \begin{cases} c_k(1-q)^{32}, & 1 \le k < 32, \\ c_k - 12qd_k, & k \ge 32, \end{cases} \qquad a_0(q) = 1 - \sum_{k\ge1} a_k(q)q^k. \tag{25}$$

The weight generating function $A_q(w) = \sum_{k\ge0} a_k(q)w^k$ is rational in w. The value $G_q(q)$ concerns a different generating function, formed from the moments below. To obtain A_q, put

$$T_q(w) = R(w) - \frac{12q}{(1-w)^3} - \sum_{k=0}^{31}(c_k - 12qd_k)w^k,$$

so that $A_q(w)$ equals

$$a_0(q) + (1-q)^{32}\sum_{k=1}^{31} c_k w^k + T_q(w), \quad a_0(q) = 1 - (1-q)^{32}\sum_{k=1}^{31} c_k q^k - T_q(q).$$

Set $\mu_m(q) = A_q(q^{m+1})$, $\nu_m(q) = \mu_m(q) - q\mu_{m+1}(q)$, and $G_q(z) = \sum_{m\ge0}\nu_m(q)z^m$. The choice of a_0 makes $\mu_0 = 1$, while the factor q in ν_m makes the sum defining $G_q(q)$ telescope. Independently, for $k \ge 32$ the tail satisfies $a_k/c_k = 1 - 12q/(k+1)$; at $q = 2/3$ this becomes $1 - 8/(k+1)$, which produces the required power N^{-8} in the product of the weights. The proof below checks positivity, convergence and both determinant claims.

Remark (a rational special value). All the weights in (25) are positive for $0 < q \le 2/3$. The two moment sequences (μ_m) and (ν_m) have positive Hankel determinants of every size, G_q is not a rational function, and $G_q(q) = 1$. Their formal Hankel determinants both begin with

$$C_N q^{B_N}, \qquad B_N = \sum_{j<N} j^2, \quad C_N = \frac{(N!)^2(N+1)!}{2^N} \qquad (N \ge 1).$$

At the fixed base $q_0 = 2/3$, with $P(q) = \prod_{j\ge1}(1-q^j)$ and $M(q) = \prod_{j\ge1}(1-q^j)^{-j}$, there is an explicit $\mathcal{A} > 0$ such that

$$\det(\mu_{i+j}(q_0))_{0\le i,j<N} \sim \mathcal{A}M(q_0)^3 C_N q_0^{B_N} P(q_0)^{2N} N^{-8\xi}, \qquad \xi = G_{q_0}(q_0) = 1.$$

This countermodel and its denominator calculation have ordinary proofs below; they have not been Lean-checked or externally peer reviewed.

Proof. For $k \ge 32$ we have $a_k(q) = d_k(k+1-12q) > 0$, and the smaller-index weights are plainly positive. Write $M(q) = (1-q)^{32}\sum_{k=1}^{31} c_k q^k$. The generating function for the c_k gives $R(q) - 1 \le 6q(1-q)^{-4}$, whence $M(q) \le 6q(1-q)^{28} \le 6/29 < 1/4$. For $k \ge 32$,

$$\frac{d}{dq}(d_k(k+1-12q)q^k) = d_k(k+1)q^{k-1}(k-12q) > 0.$$

Direct substitution into the rational expression for T_q gives

$$T_{2/3}(2/3) = \frac{24794846199808}{205891132094649} < \frac{1}{4}.$$

Therefore $1/2 < a_0(q) = 1 - M(q) - T_q(q) < 1$.

The moments have positive atomic representations

$$\mu_m = \sum_{k\geq 0} a_k q^k (q^k)^m, \qquad \nu_m = \sum_{k\geq 0} a_k q^k (1-q^{k+1})(q^k)^m.$$

Infinitely many distinct atoms have positive mass. The Gram quadratic form of a nonzero polynomial of degree below N is strictly positive for either measure, proving positive definiteness of every N-by-N Hankel matrix. Summing the geometric series in m gives

$$G_q(z) = \sum_{k\geq 0} \frac{a_k q^k (1-q^{k+1})}{1-zq^k}.$$

Each $z = q^{-k}$ is a genuine pole: its residue is nonzero and the other terms are analytic there. Hence G_q is not rational. Telescoping shows $G_q(q) = \mu_0(q) = A_q(q) = 1$ and, for $S_{n-1} = \sum_{m<n} q^m \nu_m$, gives

$$1 - S_{n-1} = q^n \mu_n \in (0, q^n] \qquad (n \geq 0).$$

Every $a_k(q)$ belongs to $\mathbb{Z}[[q]]$ and has constant term c_k, including a_0, for which the defining sum is coefficientwise finite. The moments μ_m and ν_m also lie in $\mathbb{Z}[[q]]$, but their constant terms are 1: only a_0 contributes at $q = 0$. Their rational expressions have no poles for $|q| < 1$, so these formal series converge at q_0. For $H_N(q) = \det(\mu_{i+j}(q))_{i,j<N}$, the Gram determinant expansion is

$$H_N(q) = \sum_{0\leq k_0<\cdots<k_{N-1}} \left(\prod_{i<N} a_{k_i}(q) q^{k_i}\right) \prod_{i<j} (q^{k_i} - q^{k_j})^2.$$

It holds first for finitely many atoms and then coefficientwise. The lowest possible degree of one summand is $\sum_{i<N}(2N-1-2i)k_i$. Its unique minimum is attained at $k_i = i$ and equals B_N; the coefficient is $\prod_{i<N} c_i = C_N$. Replacing a_k by $a_k(1-q^{k+1})$ proves the same leading term for (ν_m).

At q_0 set $\varepsilon = 3^{-32}$ and $b_0 = a_0(q_0) > 0$. For $k \geq 32$, $a_k(q_0)/c_k = (k-7)/(k+1)$; all these ratios lie in $[\varepsilon, 1]$. Hence for fixed h, $a_{k+h}/a_k \to 1$ as $k \to \infty$, and uniformly $a_{k+h}/a_k \leq 3^{32}(1+h)^3$. These are precisely the weight hypotheses of Theorem 3.6, whose conclusion is

$$H_N(q_0) \sim \mathcal{M}(q_0)^3 q_0^{B_N} P(q_0)^{2N} \prod_{k<N} a_k(q_0).$$

The finite product has the exact form, for $N \geq 32$,

$$\frac{\prod_{k<N} a_k(q_0)}{C_N} = b_0 \varepsilon^{31} \frac{32!}{24!} \frac{(N-8)!}{N!}.$$

Thus $\mathcal{A} = b_0 \varepsilon^{31} 32!/24!$, and $N^8(N-8)!/N! \to 1$. The displayed exponent is exactly -8ξ because $\xi = 1$; it is asserted at this one fixed base, not as a uniform special-value formula for this family. □

The following calculation is retained with its ordinary proof. It has no Lean proof in the supplied record.

Remark (exact denominators of the calibrated moments). Let $q_0 = 2/3$ and $\delta_n = 3^{n+1} - 2^{n+1}$. For every $n \geq 1$, the reduced denominators are

$$\operatorname{den} \mu_n(q_0) = 3^{31n+62}\delta_n^4, \qquad \operatorname{den}(q_0^n \mu_n(q_0)) = 3^{32n+62}\delta_n^4.$$

The least integer clearing both rational coefficients in $\mu_n = q_0^{-n}\xi - q_0^{-n}S_{n-1}$ is $2^n 3^{31n+62}\delta_n^4$. At $n = 0$ the moment and tail equal one.

Proof. At q_0, split the rational seed into a pole part and a polynomial:

$$A_{q_0}(w) = \frac{-7 + 10w}{(1-w)^4} + Q(w), \qquad Q(w) = p_0 + \sum_{k=1}^{31} p_k w^k,$$

where $p_0 = b_0 + 7$ and $p_k = (3^{-32} - 1)c_k + 8d_k$. These coefficients have only powers of three in their denominators. The finite formulas give $v_3(p_0) \geq -63$, $v_3(p_k) \geq -32$ for $1 \leq k \leq 30$, and $v_3(p_{31}) = -31$: in the last calculation $c_{31} = 16896$ has $v_3(c_{31}) = 1$ and the other numerator terms are multiples of 3^{32}. At $w = (2/3)^{n+1}$, $n \geq 1$, the term $p_{31}w^{31}$ has valuation $-31n - 62$, uniquely lower than all other terms of $Q(w)$. The pole part equals

$$\frac{3^{3n+3}(-7 \cdot 3^{n+1} + 10 \cdot 2^{n+1})}{\delta_n^4}$$

and has positive 3-adic valuation. Hence $v_3(\mu_n) = -31n - 62$.

Every prime $\ell \mid \delta_n$ differs from 2, 3. Its numerator in the displayed pole part is congruent to $3 \cdot 2^{n+1} \pmod{\delta_n}$ and is coprime to δ_n. The polynomial part is ℓ-integral, so it cannot cancel the pole part's ℓ-denominator of order $4v_\ell(\delta_n)$. There are no other denominator primes. Multiplying by $q_0^n = 2^n/3^n$ adds exactly n factors of three, proving the second formula. In the affine error, $q_0^{-n} = 3^n/2^n$ has denominator 2^n, coprime to the odd denominator of μ_n. Subtraction does not reduce either negative valuation, proving the stated least common denominator. □

Each individual μ_n has logarithmic height linear in n: $\log \operatorname{den} \mu_n = (35n + 66)\log 3 + o(1)$. The determinant has a different scale.

The following calculation is retained with its ordinary proof. It has no Lean proof in the supplied record.

Remark (determinant height and row denominators). For $D_N = \operatorname{den} H_N(q_0)$ in lowest terms,

$$\liminf_{N\to\infty} \frac{\log D_N}{N^3} \geq \frac{\log(3/2)}{3}, \qquad \log D_N = O(N^3).$$

Already for $N = 2$, multiplying $H_2 = \mu_2 - \mu_1^2$ by $\operatorname{den}\mu_1 \operatorname{den}\mu_2$ leaves reduced denominator 625.

Proof. The positive Heine expansion above gives a bound independent of the asymptotic theorem. Since $0 < a_k(q_0) \le c_k$, write $k_i = i + s_i$. Dropping the ordering restriction and bounding $c_{i+s_i}/c_i \le (1+s_i)^3$ and each residual Vandermonde factor by one gives

$$0 < H_N(q_0) \le C_N q_0^{B_N} \prod_{j\ge 0} \sum_{s\ge 0} (s+1)^3 q_0^{(2j+1)s}.$$

For $A(t) = \sum_{s\ge 0}(s+1)^3 t^s = (1+4t+t^2)/(1-t)^4$, $\log A(t) \le A(t) - 1 \le 8t/(1-t)^4$. The product is therefore at most $\exp(3888/5)$. Since $D_N H_N$ is a positive integer,

$$\log D_N \ge B_N \log(3/2) - \log C_N - 3888/5.$$

Here $B_N \sim N^3/3$ and $\log C_N = O(N \log N)$. Conversely,

$$E_N = 3^{31N(N-1)+62N} \prod_{0\le i,j<N} \delta_{i+j}^4$$

clears every term in the determinant expansion, by the preceding denominator formula. Thus $D_N \mid E_N$; using $\log \delta_m \le (m+1)\log 3$ and $\sum_{i,j<N}(i+j+1) = N^3$ gives the asserted cubic upper bound.

For the assertion about H_2, $\operatorname{den} \mu_1 = 3^{93}5^4$ and $\operatorname{den} \mu_2 = 3^{124}19^4$. At five, $v_5(\mu_2) = 0$ and $v_5(\mu_1^2) = -8$, while the proposed product contributes only 5^4. At three and nineteen that product clears the denominator, and no other prime occurs. The remaining denominator is therefore $5^4 = 625$. □

Zudilin's determinantal criterion retains a divisibility condition on its successive clearing integers [11, Props. 1–2]; the calculation for H_2 rules out replacing it with their numerical ordering, not the criterion itself. The usual integer-polynomial height criterion likewise needs a common coefficient denominator for the *same* polynomial being evaluated. The model's exact partial errors alone cannot supply one cheaply. In fact, with $\alpha_m = q_0^{-m}$ and $\beta_m = \alpha_m - \mu_m(q_0)$, the pencil $\det(\alpha_{i+j}X - \beta_{i+j})_{i,j<N}$ is affine in X because (α_{i+j}) has rank one. At $X = 1$ it is the positive rational $H_N(q_0)$; after primitive integral clearing its value is a positive integer. Matching Hankel order, positivity, fixed-base asymptotic and a small *rational* value thus leaves the rational-base denominator problem exactly where it was. The Christoffel-type reweighting $a_k q^k \mapsto a_k q^k(1-q^{k+1})$ is standard moment machinery, not a new transform [31, Intro., (3)–(4)]; this example's claim is the explicit combination and its arithmetic obstruction.

3.4 Coefficient moments and cyclotomic content

We now study the coefficient sequence, not the remainder sequence. The identity $v_m^* = \alpha_m F(p) - \beta_m$ does not transfer a representing measure from (v_m^*) to (α_m): subtraction mixes two coefficient sequences. Their positivity, spectral properties and cyclotomic factors therefore require separate calculations. Fix $p > 1$ and define, using Gaussian binomial polynomials,

$$R_m(p) = \sum_{k=0}^{m} (-1)^{m+k} p^{k(k+1)/2} \begin{bmatrix} m \\ k \end{bmatrix}_p \begin{bmatrix} m+k \\ k \end{bmatrix}_p,$$
$$s_m(p) = ([m]_p!)^3 R_m(p),$$

where $[m]_p! = \prod_{j=1}^{m}(1+p+\cdots+p^{j-1})$. Each R_m is the signed value at $x = p^{m+1}$ of the little q-Legendre polynomial of degree m, with $q = p^{-1}$; this identification is discussed in the related-work section. The polynomial normalisation of [9, Sec. 3, p. 5], at $x = z = 1$, gives

$$\begin{aligned}\alpha_m &= p[p(p-1)^3]^m s_m(p),\\ \beta_m &= \left[\alpha_m \sum_{j\geq 1} \tau(j) p^{-j}\right]_+ - 1,\\ v_m^* &= \alpha_m F(p) - \beta_m.\end{aligned}$$

The brackets mean the polynomial part at infinity. Thus $\alpha_0 = p$ and $\beta_0 = 0$. This fixes the normalisation before any content or positivity test.

A positive expansion is not a moment representation. Substituting $x = p^{m+1}$ into Van Assche's alternative expansion [18, (16), p. 4] gives the identity

$$R_m(p) = \sum_{k=0}^{m} \begin{bmatrix} m \\ k \end{bmatrix}_p \begin{bmatrix} m+k \\ k \end{bmatrix}_p p^{(m-k)(m-k+1)/2} \prod_{j=m-k+1}^{m} (p^j - 1).$$

Indeed $(qx;q)_k = (-1)^k \prod_{j=m-k+1}^{m}(p^j-1)$, so the two signs cancel. This proves $R_m(p) > 0$ for $p > 1$, and nonnegative coefficients in the variable $t = p - 1$; it does not prove Hankel positivity. In fact,

$$\det(R_{i+j}(p))_{i,j<3} = -36(p-1)^2 + O((p-1)^3) \quad (p \downarrow 1).$$

A geometric factor $c\rho^m$ with $c, \rho > 0$ would change a Hankel matrix only by a positive scalar and an invertible diagonal congruence, so it would preserve positive definiteness. The factor $([m]_p!)^3$ is not geometric: its values at $m = 0, 1, 2$ are $1, 1, (1+p)^3$. It cannot be discarded in testing the moment condition. Nor does its positivity alone prove that multiplication by it repairs the failed condition for R_m. At $p = 1$, $s_m(1) = (m!)^3$ is the moment sequence of a product of three independent unit exponential variables, since their mth moments multiply. This is only the polynomial endpoint; $F(1)$ diverges. Berg's Theorem 5.1 states that $(m!)^c$ is Stieltjes indeterminate for $c > 2$ [36]. Thus a measure exists at the endpoint but is not unique. Neither uniqueness nor a canonical deformation is a premise of the question at $p > 1$.

Finite certificates and an all-rank degree check. Put $D_{N,h}(p) = \det(s_{i+j+h}(p))_{0\leq i,j<N}$, with $D_{0,h} = 1$. Exact polynomial calculations give strictly positive coefficients in $t = p-1$ for $D_{N,h}(1+t)$ when $h = 0, 1$ and $1 \leq N \leq 8$. Thus sixteen determinant polynomials are positive for every real $p \geq 1$. Their degrees, in increasing rank, are

$h = 0$	0	10	54	156	340	630	1050	1624
$h = 1$	2	26	96	236	470	822	1316	1976.

The calculation uses exact arithmetic in $\mathbb{Z}[p]$. Starting from $D_{0,h} = 1$ and $D_{1,h} = s_h$, the Desnanot–Jacobi identity gives

$$D_{N,h} = \frac{D_{N-1,h} D_{N-1,h+2} - D_{N-1,h+1}^2}{D_{N-2,h+2}} \qquad (N \geq 2).$$

The degree argument below shows that each denominator is nonzero. The computation checks that every division has zero remainder, then substitutes $p = 1 + t$ and tests every coefficient. All 8824 coefficients in the sixteen polynomials are strictly positive. A program computes the full lists. An independent checker evaluates each stored polynomial at more integer bases than its degree and compares the values with fraction-free determinants of the integer Hankel matrices, which identifies the stored list with $D_{N,h}(1+t)$ in $\mathbb{Z}[t]$. As a separate check of the normalisation, direct integer determinants at $p = 1, 2, 3, 5, 11$ agree with evaluations of all sixteen polynomials. Those evaluations are checks, not the proof of polynomial positivity. This is a finite computer-algebra calculation and gives no all-rank result.

For the eight unshifted determinants, positivity is also a Lean theorem, proved by a different certificate. Put $T = 2^{2929}$. An explicit bound shows that every coefficient of $D_{N,0}(1 + t)$, $N \le 8$, has absolute value less than $T/2$. An integer polynomial with this property is determined by its value at $t = T$, and its coefficients are the base-T digits of that value whenever every digit is less than $T/2$. The Lean kernel computes the eight integers $D_{N,0}(1 + T)$ exactly and checks this digit condition and that the lowest digit is nonzero. Every coefficient of $D_{N,0}(1 + t)$ is therefore nonnegative and the constant term is positive, so $D_{N,0}(p) > 0$ for every real $p \ge 1$. This is the positivity used in Proposition 3.9. The strict positivity of every coefficient, and the shifted determinants $D_{N,1}$, rest on the computation above. Neither the finite computation nor the Lean proof yields an all-rank coefficient-positivity theorem.

The degrees admit a separate all-rank proof. The Gaussian binomial $\begin{bmatrix} u \\ v \end{bmatrix}_p$ is monic of degree $v(u-v)$. In the signed sum for R_m, the degree increases by $2m-k$ from the kth to the $(k+1)$st term. The unique maximal term is $k = m$, with positive leading coefficient. Hence R_m is monic of degree $(3m^2 + m)/2$, and s_m is monic of degree $3m^2 - m$. In the determinant, the only permutation-dependent part of the degree is $6\sum_i i\sigma(i)$; strict rearrangement makes the identity its unique maximum. Therefore

$$\deg D_{N,h} = \sum_{i=0}^{N-1} (3(2i + h)^2 - (2i + h)), \qquad \operatorname{lc} D_{N,h} = 1.$$

More generally the same argument works for a fixed minor with distinct increasing row and column indices. Each such minor is positive for all sufficiently large p, with a threshold that may depend on the minor. At $p = 1$, both leading Hankel families are positive definite by the infinite-support product measure, so each fixed rank is also positive in some neighbourhood of 1. Neither argument provides a neighbourhood or large-base threshold uniform over every rank.

A Stieltjes moment representation requires positive semidefiniteness of both Hankel families at all ranks. The positive-definite formulation for nondegenerate sequences is stated in [33, Lemma 2.1, p. 4]; finite support requires allowing semidefinite matrices. The necessity of both conditions is visible from their quadratic forms: if a positive measure μ on $[0, \infty)$ satisfies $s_m = \int_0^\infty x^m \, d\mu(x)$, and $P(x) = \sum_{i=0}^{N-1} c_i x^i$ has real coefficients, then

$$\sum_{i,j=0}^{N-1} c_i c_j s_{i+j+h} = \int_0^\infty x^h P(x)^2 \, d\mu(x) \ge 0, \qquad h = 0, 1.$$

The shifted test records nonnegative support, not merely positivity of the measure. For example, the positive point mass at -1 has moments $(-1)^m$. Its unshifted quadratic form is $P(-1)^2$, whereas its shifted form is $-P(-1)^2$. Thus a Hamburger moment sequence, which permits support anywhere on $\mathbb{R}$, need not be a Stieltjes moment sequence. The two tests above do not assert a measure for the entire coefficient sequence considered here. Coefficientwise total positivity would be stronger than the sixteen certificates above. Positive production matrices and path constructions can supply such a mechanism in other families [38, Sec. 2][39, Thms. 9.7–9.8]; no such matrix for this moving-degree sequence has been established here.

A finite spectral consequence. Write $A_N = (\alpha_{i+j})_{0\le i,j<N}$ and $B_N = (\beta_{i+j})_{0\le i,j<N}$. We study the matrix pencil $YA_N - B_N$, where Y is a scalar variable, through the roots of its determinant. Let $D = \operatorname{diag}(1, p(p-1)^3, \dots, [p(p-1)^3]^{N-1})$. Then $A_N = pD(s_{i+j})D$. For $p > 1$ this is an invertible positive diagonal congruence. The remainder representation already proves $F(p)A_N - B_N = (v^*_{i+j}) > 0$ at every rank.

Proposition 3.9 (finite coefficient positivity and pencil roots)**.** *For every real $p > 1$ and $1 \le N \le 8$, A_N is positive definite and all roots of $\det(YA_N - B_N)$ are real and strictly less than $F(p)$. The roots at consecutive ranks $N, N+1 \le 8$ interlace non-strictly.*

The Lean proof checks all three conclusions for every real $p > 1$ and every rank $N \le 8$; it states nothing beyond rank eight.

Proof. The leading principal minors of $(s_{i+j})_{i,j<N}$ are exactly $D_{k,0}(p)$ for $1 \le k \le N$; their coefficient positivity proved above, together with Sylvester's criterion, gives positive definiteness. The diagonal congruence gives $A_N > 0$. The real symmetric matrix $A_N^{-1/2} B_N A_N^{-1/2}$ has these pencil roots as eigenvalues, and $F(p)I - A_N^{-1/2} B_N A_N^{-1/2} > 0$ puts them strictly below $F(p)$. The minimum–maximum principle for $x^{\mathsf{T}} B_N x / (x^{\mathsf{T}} A_N x)$ on nested coordinate subspaces gives non-strict interlacing. These are subspaces for the original pencils; the separately conjugated symmetric matrices need not be principal submatrices of one another. □

Proposition 3.9 uses only the eight unshifted certificates $D_{k,0}$, not an all-rank coefficient measure. The shifted certificates $D_{k,1}$ are used below for a truncated Stieltjes moment representation and the first fifteen continued-fraction coefficients, not for this application of Sylvester's criterion. An infinite-support coefficient measure would extend the pencil argument to all ranks. At the certified ranks, the largest root is a nondecreasing lower bound for $F(p)$; convergence to $F(p)$, simple roots, strict interlacing and denominator control are not established. At $p = 1$ the congruence degenerates and F diverges, so the proposition excludes that endpoint.

The pencil is different from the Jacobi matrix of the coefficient moments. This is already visible at rank two. The defining formulas give

$$\beta_0 = 0, \qquad \beta_1 = p^3(p-1)^2(p+2) > 0 \quad (p > 1).$$

Consequently $\det B_2 = -\beta_1^2 < 0$. Since $A_2 > 0$, the product of the two real pencil roots is $\det B_2 / \det A_2 < 0$: one root is negative and the other positive. In contrast, the finite Stieltjes construction below has only positive nodes. Positivity of the coefficient Hankel matrix therefore does not identify these two spectral constructions.

Formal continued fractions and finite quadrature. Write $D_n = D_{n,0}$, $E_n = D_{n,1}$, with $D_0 = E_0 = 1$. All these determinant polynomials are nonzero by the all-rank degree calculation above. The classical Stieltjes continued fraction is thus defined over the rational-function field $\mathbb{Q}(p)$, with the convention

$$\sum_{m\geq 0} s_m(p) z^m = \cfrac{1}{1 - \cfrac{\lambda_1 z}{1 - \cfrac{\lambda_2 z}{1 - \cdots}}},$$

$$\lambda_{2n-1} = \frac{E_n D_{n-1}}{D_n E_{n-1}}, \quad \lambda_{2n} = \frac{D_{n+1} E_{n-1}}{E_n D_n} \quad (n \geq 1).$$

Here $s_0 = 1$, and the identity is in $\mathbb{Q}(p)[[z]]$. The classical moment correspondence is recalled in [37, pp. 1–2]. Specialising a ratio at a fixed real p requires its displayed denominators to be nonzero. The sixteen certificates ensure this and $\lambda_1, \dots, \lambda_{15} > 0$ for every $p \geq 1$; they do not ensure it at all indices. Seeking positive formulas for all these ratios is therefore a specific version of the coefficient-moment question. The generating series has radius of convergence zero for every $p \geq 1$, so the displayed identity must be read formally. At $p = 1$ this follows from $s_m(1) = (m!)^3$. For $p > 1$, the $k = 0$ term of the positive expansion for R_m and the inequality $[m]_p! \geq p^{m(m-1)/2}$ give

$$s_m(p) \geq p^{2m^2 - m}.$$

Thus $s_m(p)^{1/m} \to \infty$. This concerns the moment power series; it does not decide convergence of the continued fraction at an individual nonzero value of z.

The same certificates also give a finite representing measure: for each fixed $p \geq 1$, eight positive atoms reproduce exactly $s_0, \dots, s_{15}$. The construction uses only the two certified 8×8 matrices, not positivity at higher ranks. Set

$$H_0 = (s_{i+j})_{i,j<8}, \qquad H_1 = (s_{i+j+1})_{i,j<8}, \qquad T = H_0^{-1} H_1,$$

and give $\mathbb{R}^8$ the inner product $\langle u, v\rangle = u^{\mathsf{T}} H_0 v$. The operator T is self-adjoint and positive definite because $H_0 T = H_1$ is symmetric and positive definite. If $e_0, \dots, e_7$ are the coordinate vectors, the Hankel identities give $Te_j = e_{j+1}$ for $0 \leq j < 7$. Hence e_0 is cyclic: its first eight iterates form a basis. The spectral theorem now gives eight distinct positive eigenvalues and a positive weight at each, namely the squared norm of the corresponding projection of e_0. Cyclicity ensures that no projection vanishes and that no eigenspace has dimension greater than one.

Let ν be the resulting atomic measure. For $0 \leq m \leq 14$, choose $i, j \leq 7$ with $i + j = m$. Self-adjointness gives

$$\int x^m \, d\nu(x) = \langle T^m e_0, e_0\rangle = \langle e_i, e_j\rangle = s_m.$$

For the last moment,

$$\int x^{15} \, d\nu(x) = \langle T^7 e_0, T T^7 e_0\rangle = e_7^{\mathsf{T}} H_1 e_7 = s_{15}.$$

The weights sum to $s_0 = 1$. Hence the measure agrees with the moment functional on every polynomial of degree at most 15; this is the degree range certified for the eight-node quadrature obtained here. In an orthonormal polynomial basis it is the classical Jacobi construction; Golub and Welsch describe the computation of its nodes and weights [41, Sec. 2, (2.6); Sec. 4]. Nothing here identifies the subsequent moments of ν with $s_{16}, s_{17}, \ldots$. This finite construction must also be distinguished from the coefficient pencil (A_N, B_N). For rational modifications of an already known moment functional, see Krattenthaler [32, Thm. 1; Sec. 6]; such a modification producing the entire sequence s_m has not been identified.

Testing for additional cyclotomic factors. Independently of the moment question, let v_{Φ_d} denote the multiplicity of the cyclotomic factor Φ_d in a nonzero polynomial in $\mathbb{Q}[p]$, and set $v_{\Phi_d}(0) = \infty$. For a 2×2 matrix with entry valuations $0, 2, 2, 5$, the two determinant products have valuations 5 and 4. The lower one is unique, so the determinant has valuation 4. A tie is different: all four entries of $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1+\Phi_d \end{smallmatrix}\right)$ have valuation zero, but its determinant is Φ_d. At general rank, we must find the least total valuation and then test whether the terms attaining it cancel. For $N, d \geq 1$, define

$$\begin{aligned} \mathcal{H}_N(Y; p) &= \det(\alpha_{i+j} Y - \beta_{i+j})_{i,j<N}, \\ t_{m,d} &= \min(v_{\Phi_d}(\alpha_m), v_{\Phi_d}(\beta_m)), \\ e_{N,d} &= \min_{\sigma \in S_N} \sum_{i=0}^{N-1} t_{i+\sigma(i),d}. \end{aligned}$$

The last expression is a minimum-cost assignment: row i is matched to column $\sigma(i)$ at cost $t_{i+\sigma(i),d}$. The valuation of a polynomial in Y means the minimum valuation of its coefficients in $\mathbb{Q}[p]$. Each determinant term is divisible by $\Phi_d^{e_{N,d}}$. Set

$$\Gamma_{N,d}(Y) = \sum_{\substack{\sigma \in S_N \\ \sum_i t_{i+\sigma(i),d} = e_{N,d}}} \operatorname{sgn}(\sigma) \prod_i \overline{\Phi_d^{-t_{i+\sigma(i),d}} (\alpha_{i+\sigma(i)} Y - \beta_{i+\sigma(i)})},$$

where the bar is reduction in $(\mathbb{Q}[p]/(\Phi_d))[Y]$. Then $v_{\Phi_d}\mathcal{H}_N = e_{N,d}$ exactly when $\Gamma_{N,d} \neq 0$. This isolates the cancellation that a genuine additional factor would require. The assignment bound records the smallest valuation available among the determinant terms. Excess divisibility is possible only if their leading residues cancel. To test that cancellation by one determinant, we distribute the minimum valuation between rows and columns. The integer costs are finite, since every α_m is nonzero. Minimum-cost assignment duality gives integer row and column potentials with

$$u_i + v_j \leq t_{i+j,d}, \qquad \sum_i u_i + \sum_j v_j = e_{N,d}.$$

Here is a direct construction, so their existence is not an additional hypothesis. Choose a minimum-cost permutation σ. On the column indices put a directed edge from $\sigma(i)$ to j of length $t_{i+j,d} - t_{i+\sigma(i),d}$. A negative directed cycle would improve σ by reassigning the corresponding rows along that cycle; there is therefore no such cycle. Add a new

vertex with an edge of length zero to every column. The shortest-path distances v_j are finite integers and satisfy

$$v_j - v_{\sigma(i)} \le t_{i+j,d} - t_{i+\sigma(i),d}.$$

Putting $u_i = t_{i+\sigma(i),d} - v_{\sigma(i)}$ gives the inequalities above, with equality on the chosen assignment and hence equality of totals.

Divide row i by $\Phi_d^{u_i}$ and column j by $\Phi_d^{v_j}$ over $\mathbb{Q}(p)$. The inequalities ensure that every resulting entry is in $\mathbb{Q}[p, Y]$, even if some potentials are negative. Reduce these entries modulo Φ_d. Only those with $u_i + v_j = t_{i+j,d}$ survive. The determinant of the reduced matrix is $\Gamma_{N,d}$, because every non-minimal permutation vanishes on reduction. Thus the valuation exceeds $e_{N,d}$ exactly when the reduced matrix is singular over $(\mathbb{Q}[p]/(\Phi_d))(Y)$. A kernel relation valid at just one value of Y is insufficient: the determinant must vanish as a polynomial in Y. Conversely, one nonzero value $\Gamma_{N,d}(Y_0)$ proves equality with the assignment bound. Neither assignment duality nor the existence of the potentials forces the reduced determinant to vanish.

An exact calculation outside Lean gives the following monic contents. Here cont_Y means the monic gcd in $\mathbb{Q}[p]$ of the coefficients in Y:

N	$\mathrm{cont}_Y \mathcal{H}_N$
1	p
2	$p^5(p-1)^4$
3	$p^{14}(p-1)^{15}(p+1)^4$
4	$p^{30}(p-1)^{32}(p+1)^8(p^2+p+1)^4$
5	$p^{55}(p-1)^{55}(p+1)^{19}(p^2+1)^4(p^2+p+1)^8$.

These factorisations attain the assignment bound for every d at these five ranks. The residue certificates verify equality for $d \le 8$; the displayed complete factorisations contain no other cyclotomic factors, so the nonnegative assignment bound is zero at all remaining d. The power of p is not cyclotomic content.

For the table, an exact program computes the full polynomials $\mathcal{H}_N(Y_0; p)$ at $Y_0 = 0, \dots, N$ by fraction-free elimination, checking every polynomial division. Their monic gcd in $\mathbb{Q}[p]$ is $\mathrm{cont}_Y \mathcal{H}_N$. A divisor of all coefficients divides every evaluated polynomial; conversely, the $N+1$ values recover those coefficients by rational interpolation in Y. This proves the reverse divisibility over $\mathbb{Q}[p]$, where the interpolation denominators are units. The calculation includes every factor, not just a prescribed list of cyclotomic candidates.

The same program produces a nonzero residue witness for every pair

$$1 \le N \le 8, \qquad 1 \le d \le \max(8, 2N-2).$$

There are 76 such pairs. For each, the certificate gives an optimal assignment, integer dual potentials, an integer $Y_0 \in \{0, \dots, N\}$ and the nonzero residue $\Gamma_{N,d}(Y_0)$. The signed minimum-cost subset recurrence and the determinant of the reduced matrix give the same residue. A separate check computes the full polynomial $\mathcal{H}_N(Y_0; p)$ and verifies

directly that division by $\Phi_d^{e_{N,d}}$ leaves exactly this nonzero residue. At $N = 2, d = 1$, for example, the entry valuations are $\left(\begin{smallmatrix} 0 & 2 \\ 2 & 5 \end{smallmatrix}\right)$, $e_{2,1} = 4$, and $\Gamma_{2,1}(0) = -9$.

The source coefficients, determinant-value polynomials and witnesses are computed by that program, which also repeats the direct determinant checks; its recorded result lists every witness. At ranks six to eight these certificates cover only the stated cyclotomic-index window; complete coefficient contents are computed only through rank five.

These computations distinguish systematic common factors from cancellation beyond entry valuations, a distinction also important in fraction-free matrix decompositions [40, Sec. 2; Thm. 6]. In particular the gcd argument at the points $0, \dots, N$ is over $\mathbb{Q}[p]$: the rational Vandermonde inverse is not generally integral, so it must not be used to assert the corresponding coefficient gcd in $\mathbb{Z}[p]$.

Proposition 4 of Krattenthaler–Rochev–Väänänen–Zudilin [13, pp. 14–15; (4.10), p. 17] obtains cyclotomic factors through root-of-unity annihilation for a different first-order tail recurrence. No corresponding recurrence has been established for this moving-diagonal pencil. The finite witnesses prove equality with the assignment bound at these 76 pairs. They do not exclude excess at larger ranks or, at ranks six to eight, cyclotomic indices outside the stated window. Whether minimum-valuation residues ever cancel systematically, and whether s_m is a Stieltjes sequence, remain separate questions. The remainder measure answers neither.

4 Rescaling integer rows

For a single irrationality form, the essential conditions are integral coefficients and a nonzero remainder tending to zero. Dividing a nonzero integer row (U, V) by $\gcd(|U|, |V|)$ preserves integrality and divides its remainder by the same amount; the resulting row is primitive. A useful content division removes a factor already shared by the two entries. Multiplying a row first merely introduces a factor that also multiplies its remainder. For two rows, the same factors multiply their determinant. The identities below separate this rescaling from genuine cancellation. A height estimate is needed for an irrationality measure, not for the elementary single-form criterion. For a real target S, put

$$L_S(U, V) = US - V, \qquad \Delta((U_n, V_n), (U_m, V_m)) = U_n V_m - U_m V_n.$$

Here $L_S(U, V)$ is the *error* of the integer linear form, not the rational approximation error $|S - V/U|$ when $U \neq 0$. The determinant eliminates S exactly:

$$\Delta = U_m L_S(U_n, V_n) - U_n L_S(U_m, V_m).$$

If this integer determinant does not vanish, then

$$1 \leq |\Delta| \leq |U_m|\,|L_S(U_n, V_n)| + |U_n|\,|L_S(U_m, V_m)|,$$

and the two errors cannot both be smaller than $1/(|U_n| + |U_m|)$. The next theorem compares the divisor introduced by rescaling with the resulting change in the determinant's absolute value.

Theorem 4.1 (rescaling rows and their determinant)**.** *Let S be real, let (U_n, V_n) and (U_m, V_m) be pairs of integers, and let c_n, c_m be integers. Then*

$$L_S(c_n U_n, c_n V_n) = c_n L_S(U_n, V_n),$$

$$\Delta(c_n(U_n, V_n), c_m(U_m, V_m)) = c_n c_m \Delta((U_n, V_n), (U_m, V_m)),$$

and consequently

$$|\Delta(c_n(U_n, V_n), c_m(U_m, V_m))| = |c_n||c_m|\,|\Delta((U_n, V_n), (U_m, V_m))|.$$

In particular $c_n c_m$ divides the scaled determinant. Multiplication by these scalars therefore introduces a divisor whose absolute value is exactly the factor multiplying the determinant's absolute value.

Proof. Expand the linear form and use the bilinearity of the determinant. The original integer determinant is the quotient witnessing divisibility; it need not be primitive. □

Example 4.2. Take $(U_n, V_n) = (1, 2)$ and $(U_m, V_m) = (3, 5)$, so that $\Delta = 1 \cdot 5 - 3 \cdot 2 = -1$. Multiplying the first row by $c_n = 6$ and the second by $c_m = 10$ gives the rows $(6, 12)$ and $(30, 50)$, whose determinant is $6 \cdot 50 - 30 \cdot 12 = -60$. That determinant is now divisible by 60, which looks like a local gain of 60; and its absolute value has risen from 1 to 60, which is a cost of exactly the same size.

Lean checks the error identity in error scaling, the determinant identity in content factorisation, the exact absolute-height identity in absolute determinant scaling, the divisor statement in content-product divisibility, and the elimination identity in exterior determinant identity.

The identities allow zero scalars, but cancelling the scalar factors requires them to be nonzero. Multiplication alone gives no improvement in the comparison between a divisor and the determinant's absolute value. The theorem neither asserts that the determinant is nonzero nor estimates the original approximation error. Cancelling polynomial factors before evaluation, finding factors common to several minors, and adding different rows remain separate operations.

5 Congruences after evaluation at $3/2$

We use congruences modulo powers of 2 and 3 to find combinations of evaluated polynomials with prescribed divisibility. The argument applies to arbitrary integral polynomials, including the forms of [8, Sections 3–5]. The denominator-clearing map must be the same for every row that will be added. We therefore fix a common truncation index $W \geq 0$ and, for $P(X) = \sum_i p_i X^i \in \mathbb{Z}[X]$, put

$$H_W(P) = \sum_{i=0}^{W} p_i\, 3^i 2^{W-i}.$$

This is the denominator-cleared evaluation at 3/2. Each term $p_i X^i$ with $i \leq W$ contributes $p_i 3^i 2^{W-i}$. Keep the index W fixed when polynomials are added. When $W \geq$

deg P, this is exactly the cleared numerator, since

$$\sum_{i=0}^{W} p_i 3^i 2^{W-i} = 2^W \sum_{i=0}^{W} p_i \left(\tfrac{3}{2}\right)^i = 2^W P\left(\tfrac{3}{2}\right).$$

Here a unit coefficient in $\mathbb{Z}$ means 1 or -1. Thus a monic polynomial of degree W with constant term ± 1 satisfies both unit conditions below. Those conditions are sufficient, not necessary: the congruences themselves only require that the relevant coefficient not be divisible by the relevant prime.

Theorem 5.1 (endpoint residues)**.** *Let $P = \sum_i p_i X^i \in \mathbb{Z}[X]$ and let $W \geq 0$. Then*

$$H_W(P) \equiv p_0\, 2^W \pmod 3, \qquad H_W(P) \equiv p_W\, 3^W \pmod 2.$$

Consequently a unit constant coefficient prevents divisibility by 3*, and a unit coefficient at index W prevents divisibility by* 2*.*

Proof. Modulo 3, every summand with $i > 0$ vanishes; modulo 2, every summand with $i < W$ vanishes. The remaining powers are units in the corresponding residue fields. □

Since 2^W is invertible modulo 3 and 3^W is invertible modulo 2, the two congruences say more than the stated consequence: divisibility of $H_W(P)$ by 3 is decided by p_0 alone, and divisibility by 2 by p_W alone. The rest of the coefficient vector is invisible to both primes. The coefficient p_W is the top coefficient of P exactly when $\deg P = W$, and is zero when $\deg P < W$. The identity $H_W(P) = 2^W P(3/2)$ is guaranteed when $\deg P \leq W$. Without a degree bound, increasing the truncation index gives

$$H_{W+1}(P) = 2H_W(P) + p_{W+1} 3^{W+1},$$

so the truncated sum doubles precisely when the extra coefficient is zero.

Example 5.2. Take $W = 2$. The three polynomials below differ only at an endpoint.

P	$H_2(P)$	$3 \mid H_2(P)$	$2 \mid H_2(P)$
$X^2 + 1$	$1 \cdot 4 + 0 \cdot 6 + 1 \cdot 9 = 13$	no	no
$X^2 + 3$	$3 \cdot 4 + 0 \cdot 6 + 1 \cdot 9 = 21$	yes	no
$2X^2 + 1$	$1 \cdot 4 + 0 \cdot 6 + 2 \cdot 9 = 22$	no	yes

The first has both endpoints equal to 1 and its evaluation, 13, is divisible by neither prime; as a check, $2^2((3/2)^2 + 1) = 13$. The second and third show that each hypothesis is used: spoiling the constant endpoint admits 3, and spoiling the top endpoint admits 2.

Formal proofs cover the congruences modulo 3 and modulo 2, together with the consequences for a unit constant coefficient and a unit coefficient of X^W.

In the following proposition, a unit top endpoint means that the coefficient at index W is 1 or -1, with no degree bound imposed. The two conditions apply to different entries of the polynomial pair.

Proposition 5.3 (common divisor)**.** *Let $U, V \in \mathbb{Z}[X]$ and let $W \geq 0$. If U has unit top endpoint, V has unit constant endpoint, and an integer c divides both $H_W(U)$ and $H_W(V)$, then*

$$2 \nmid c \qquad \text{and} \qquad 3 \nmid c.$$

Proof. If $2 \mid c$, then $2 \mid H_W(U)$, contrary to the unit coefficient at index W and Theorem 5.1. Similarly, $3 \mid c$ would imply $3 \mid H_W(V)$, contrary to the unit constant coefficient. □

Lean checks this common-divisor exclusion. The proposition does not say that the two evaluations are coprime; it says only that every common divisor is coprime to 6. It gives no restriction on the other prime factors of that divisor.

Example 5.4. Take $W = 2$, $U = X^2 + 3$ and $V = 5X^2 + 1$. The top endpoint of U and the constant endpoint of V are both 1, and

$$H_2(U) = 3 \cdot 4 + 1 \cdot 9 = 21, \qquad H_2(V) = 1 \cdot 4 + 5 \cdot 9 = 49.$$

Here $\gcd(21, 49) = 7$. The endpoint assumptions therefore allow a nontrivial common divisor, but that divisor is coprime to 6.

Corollary 5.5 (limits of rescaling and common-divisor cancellation at 3/2)**.** *Under the endpoint hypotheses of Proposition 5.3, every common divisor of the unscaled evaluations $H_W(U)$ and $H_W(V)$ is coprime to 6. Multiplying two integer rows by nonzero integers c_n, c_m multiplies their determinant by $c_n c_m$ and its absolute value by $|c_n c_m|$. Cancelling this introduced scalar factor therefore leaves the original comparison between divisor and determinant size unchanged.*

Proof. The common-divisor assertion is Proposition 5.3. By Theorem 4.1, rescaling the two rows by c_n, c_m multiplies their errors by c_n, c_m, respectively, and their 2×2 determinant by $c_n c_m$. For nonzero scalars, the introduced divisor has the same absolute value as the introduced factor in that determinant. Dividing out that factor restores the original determinant, including its original divisor-to-size comparison. □

One further consequence of Theorem 5.1 is worth stating, because it bears on the most natural way one might hope to import an existing denominator reduction. Write Φ_m for the mth cyclotomic polynomial and, for coprime $a > b \geq 1$, put $\Phi_m(a, b) = b^{\varphi(m)} \Phi_m(a/b)$ for its homogenisation, with exponent $\varphi(m) = \deg \Phi_m$.

Proposition 5.6 (coprimality of homogenised cyclotomic values)**.** *Let $a > b \geq 1$ with $\gcd(a, b) = 1$ and let $m \geq 1$. Then $\gcd(\Phi_m(a, b), ab) = 1$. In particular $\gcd(\Phi_m(3, 2), 6) = 1$ for every m.*

Proof. The polynomial Φ_m is monic and $\Phi_m(0) = \pm 1$. The endpoint argument in Theorem 5.1 also works at (a, b): modulo a prime dividing b, only $a^{\varphi(m)}$ survives, and modulo a prime dividing a, only $\Phi_m(0) b^{\varphi(m)}$ survives. Coprimality of a and b makes each surviving term nonzero. □

Rhin and Viola's factorial-coset quotients [17] reduce denominators of rational forms in $\zeta(2)$. Zudilin's q-analogue has an order-twelve symmetry group for the normalised series; its denominator cancellation uses the order-six subgroup preserving the required sign condition [8, Sec. 4, pp. 159–161]. Proposition 5.6 applies to these cyclotomic factors: each homogenised value at $(3, 2)$, and hence any product of them, is divisible by neither 2 nor 3. It makes no such assertion about an arbitrary integer or factorial factor, which can contain both primes. Cancelling a common cyclotomic factor can still reduce the real absolute values. That reduction is distinct from producing the powers of 2 or 3 sought in Section 10.

Scalar multiplication can introduce powers of 2 and 3, but does not improve this comparison. Under the endpoint hypotheses, the unscaled pair has no common factor at either prime. Neither observation shows that a gain at these primes is necessary for a proof by linear forms at $3/2$.

The operation studied next is different: take an integer combination of several rows and ask that the combination be divisible where the individual rows are not. The proposed divisor must come from cancellation between rows, not from multiplying a row by that divisor. The endpoint congruences are the first case of a divisibility condition that can be imposed to any depth, and it is that condition, read additively, which is counted below.

We first raise the two congruences to prime powers. Fix depths $R, S \geq 0$. For $P \in \mathbb{Z}[X]$, let $J_{3,R}(P)$ and $J_{2,S}(P)$ be the residues of $H_W(P)$ modulo 3^R and 2^S, respectively. Theorem 5.1 computes the depth-one residues. At depth R, terms with $i \geq R$ are invisible modulo 3^R; at depth S, terms with $W - i \geq S$ are invisible modulo 2^S. Thus deeper congruences inspect longer segments of the two ends of the coefficient list, always at the same fixed W.

Lemma 5.7 (a residue and divisibility by 3^R)**.** *Vanishing of the residue modulo 3^R is exactly divisibility by the corresponding power of three: $J_{3,R}(P) = 0$ if and only if $3^R \mid H_W(P)$.*

Proof. By definition $J_{3,R}(P)$ is the image of the integer $H_W(P)$ in $\mathbb{Z}/3^R\mathbb{Z}$. The kernel of this quotient map is $3^R\mathbb{Z}$, which proves the equivalence, including $R = 0$. □

The same argument gives $J_{2,S}(P) = 0 \iff 2^S \mid H_W(P)$ for $S \geq 0$.

The *vector of four residues* of a coefficient pair (U, V) is then the quadruple

$$(J_{3,R}(U), J_{3,R}(V), J_{2,S}(U), J_{2,S}(V)) \in (\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

two residues for each of the two primes, one from each entry of the pair. By the displayed criteria it vanishes exactly when 3^R divides both specialised entries and 2^S divides both.

Instead of requiring each input row to have a common divisor, we seek a small integer combination whose two entries are both divisible by 3^R2^S. Since H_W is additive, the four residues of the combination are the corresponding sums of the input residues. This allows a pigeonhole argument on subset sums.

Theorem 5.8 (equal residues for two subset sums)**.** *Fix a truncation index W and depths R, S, and let $(U_j, V_j)_{j<M}$ be any M pairs of integral polynomials. Represent each subset of*

$\{0,\dots,M-1\}$ *by its indicator vector in* $\{0,1\}^M$. *If the* 2^M *subsets outnumber the possible residue vectors in*

$$(\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

then two distinct subsets have the same residue vector. Subtracting their indicator vectors gives a nonzero coefficient vector in $\{-1,0,1\}^M$ *cancelling all four residues. The target has exact cardinality*

$$(3^R)^2(2^S)^2.$$

In particular, if $R>0$ *and* $4R+2S\le M$, *such a collision exists.*

Proof. Send each subset to the sum of the residue vectors of its members. The pigeonhole principle gives two distinct subsets with the same residue vector. The number of possible vectors is the product of the four moduli. For $R>0$,

$$(3^R)^2(2^S)^2 < (4^R)^2(2^S)^2 = 2^{4R+2S} \le 2^M,$$

which proves the stated sufficient threshold. □

The power bracket improves the generic coefficient $4R$ when the depth R is a positive integer multiple of 41. All depths and row counts in the following corollary, including T, are integers.

Corollary 5.9 (the exact count at depth 41)**.** *Let* $T>0$. *At modulus* 3^R *with* $R=41T$, *any family of* $M \ge 130T+2S$ *integral polynomial pairs has two distinct binary vectors with the same residue vector. For* $T=1$ *the coefficient* 130 *is exact for this counting argument:*

$$2^{129+2S} < |(\mathbb{Z}/3^{41}\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2|.$$

No exact-optimality assertion is made here for $T>1$.

Proof. The upper power inequality gives

$$(3^{41T})^2(2^S)^2 < (2^{65})^{2T}(2^S)^2 = 2^{130T+2S} \le 2^M,$$

so Theorem 5.8 applies. For $T=1$, direct integer evaluation gives $2^{129} < 3^{82}$; multiplying by $(2^S)^2$ gives the displayed reverse count at $129+2S$. □

The formal statements check the sufficient count $130T+2S$ and, when $T=1$, the failure of the count at $129+2S$.

For all depths at once the ambient-cardinality inequality $2^M > 3^{2R}2^{2S}$ holds exactly when

$$M \ge \lfloor 2R\log_2 3 + 2S \rfloor + 1,$$

by the definition of the floor function; irrationality of $\log_2 3$ is not needed for this strict-inequality reformulation. At $R=41$ this is $130+2S$, and at $R=41\cdot 31$ it is $4029+2S$, one below the uniform bound $4030+2S$ of Corollary 5.9; the corollary trades exactness for a certificate that is a single integer comparison. Here 41 is the exponent of 3 in the modulus, not the number of input rows; at that depth the counting bound is $130+2S$ rows.

The count produces two distinct subsets, not necessarily two distinct row sums or remainder values. Repeated rows already illustrate the first failure. To prevent the second, we bound how many subsets can have any specified remainder. The following finite-set statement isolates that requirement.

Theorem 5.10 (equal residues with different values)**.** *Let A and B be finite sets, let $f : A \to B$, and let $g : A \to C$ be any map into a set C. Suppose every fibre of g has at most k elements. If*

$$|B|k < |A|,$$

then there exist distinct $x, y \in A$ such that

$$f(x) = f(y) \qquad \text{and} \qquad g(x) \neq g(y).$$

Thus, when f records the four residues and g records the real remainder, a bound on the multiplicities of equal remainders guarantees a pair with equal residues and different remainders.

Proof. If every pair in a common f-fibre also had the same g-value, each f-fibre would lie in one g-fibre and hence have size at most k. Summing over the at most $|B|$ fibres of f would give $|A| \leq |B|k$, contrary to the hypothesis. □

If g is injective, the hypothesis holds with $k = 1$. It need not hold with a useful small k for subset sums: repeated rows produce many equal sums, even when every row is primitive. The theorem assumes a bound on every fibre of g, not merely on the fibres of (f, g). It gives a nonzero difference but no upper bound on its real size. To bound the size as well, we partition the range of g into small intervals. The count now uses residue class, interval and multiplicity. Only repetitions within one residue class matter, so the quantitative version assumes a bound for fibres of (f, g) rather than for all fibres of g.

Theorem 5.11 (equal residues with close, distinct values)**.** *Let A, B, J be finite sets and let $f : A \to B$, $g : A \to \mathbb{R}$ and $\iota : A \to J$. Suppose that each simultaneous fibre of (f, g) has at most k elements and that, for some $\delta > 0$,*

$$\iota(x) = \iota(y) \quad \Longrightarrow \quad |g(x) - g(y)| < \delta.$$

If $|B||J|k < |A|$, then some distinct $x, y \in A$ satisfy

$$f(x) = f(y), \qquad 0 < |g(x) - g(y)| < \delta.$$

Proof. Suppose that no such pair exists. Two points in a fibre of (f, ι) have equal f-values and g-values less than δ apart, so their g-values must be equal. Thus each fibre of (f, ι) lies in a fibre of (f, g) and contains at most k points. Summing over its at most $|B||J|$ fibres gives $|A| \leq |B||J|k$, a contradiction. □

For an application, let M integral coefficient rows (A_j, B_j) have real remainders e_j at the same target. Let D be a positive integer, and suppose that the coefficient-pair subset sums $\sum_{j \in I}(A_j, B_j)$ take Q values in $(\mathbb{Z}/D\mathbb{Z})^2$. Put $T = \sum_j |e_j|$. The least and greatest subset remainders are respectively

$$\sum_{e_j < 0} e_j \quad \text{and} \quad \sum_{e_j > 0} e_j,$$

so the interval between them has length T. Divide it into half-open intervals of width D/n, using a final interval for the right endpoint if necessary. At most $\lfloor nT/D \rfloor + 1$ intervals are needed. If each simultaneous residue-and-remainder fibre has size at most k, the inequality

$$2^M > Qk(\lfloor nT/D \rfloor + 1)$$

therefore gives equal residues with remainder difference strictly between 0 and D/n in absolute value. Subtracting the two subset sums and dividing both coefficients by D gives an integral form of nonzero absolute value less than $1/n$. For a prescribed family, the number Q, the width T and the multiplicity bound k must all be justified. We have no such bounds giving the required forms for the Zudilin family at $3/2$.

The next improvement needs a much stronger hypothesis than primitivity: adjacent determinants must vanish modulo the chosen modulus. Unit multiples of one unimodular row satisfy it. Arbitrary primitive rows do not; for example, $(1,0)$ and $(0,1)$ have determinant 1. Under this hypothesis the rows lie in the free rank-one submodule generated by a unimodular row. A change of coordinates makes their second coordinates zero, so only one residue coordinate must be counted.

Theorem 5.12 (vanishing minors and a residue count)**.** *Let R_0 be a commutative ring and let $w_n = (A_n, B_n) \in R_0^2$. Suppose that each row is unimodular, meaning that $u_n A_n + v_n B_n = 1$ for some $u_n, v_n \in R_0$, and that every adjacent minor vanishes:*

$$A_n B_{n+1} - B_n A_{n+1} = 0 \qquad (n \geq 0).$$

Then every pairwise minor $A_i B_j - B_i A_j$ vanishes. In particular, take $R_0 = \mathbb{Z}/(2^S 3^R)\mathbb{Z}$ with $R > 0$. If $S + 2R \leq k$, there are two distinct binary vectors $s, t \in \{0,1\}^k$ such that

$$\sum_{i<k} s_i w_i = \sum_{i<k} t_i w_i.$$

Thus $S + 2R$ rows suffice. The ambient two-coordinate argument gives the sufficient bound $2S + 4R$.

Proof. If (a, b) is unimodular, say $ua + vb = 1$, and $ay - bx = 0$, then

$$(x, y) = (ux + vy)(a, b).$$

Indeed, the first coordinate follows by replacing ay with bx, and the second by the reverse substitution. Apply this identity to consecutive rows: each next row is a scalar multiple of the current one. Induction places the entire tail on the line through w_0 and proves the pairwise-minor assertion. Right multiplication by

$$\begin{pmatrix} u_0 & -B_0 \\ v_0 & A_0 \end{pmatrix}$$

has determinant $u_0 A_0 + v_0 B_0 = 1$ and sends w_0 to $(1,0)$. All transformed subset sums therefore have second coordinate zero and occupy at most $2^S 3^R$ values. Finally

$$2^S 3^R < 2^S 4^R = 2^{S+2R} \leq 2^k,$$

so the pigeonhole principle gives two distinct subsets with the same sum. □

No particular coordinate needs to be invertible: modulo six, $(2, 3)$ is unimodular because $-2 + 3 = 1$, although neither entry is a unit. Some nondegeneracy is essential. The rows $(1, 0)$, $(0, 0)$, $(0, 1)$ have zero adjacent minors but outer minor one. For the counting conclusion, vanishing is required only in the quotient ring. The integer rows $(1, 0)$, $(1, 6)$ used in Section 10 are independent over $\mathbb{Q}$, with determinant 6, but coincide modulo 6. Thus dependence modulo the modulus neither follows from primitivity nor implies dependence of the integer rows.

This conditional theorem is stronger than the ambient count of possible residue vectors only after its minor-vanishing hypothesis has been established. No such all-tail hypothesis is proved here for an actual q-Apéry or Zudilin family, and the theorem says nothing about whether the difference of the subset sums has nonzero analytic remainder.

The count in Corollary 5.9 is sharp at $T = 1$ for comparison with the full residue space. To obtain different remainders, Theorem 5.10 additionally needs a bound on how often the same real value occurs. Such a bound is not proved here for the q-Apéry or Zudilin remainder family.

Lean checks the cardinality of the residue space, the pigeonhole argument for the residue map and the sufficient row count for equal residues. The pigeonhole argument gives divisibility without an independence hypothesis. Further information about the family is needed to ensure that the nonzero difference of the indicator vectors gives a nonzero polynomial pair and a nonzero analytic remainder. None of these statements supplies such a family or proves either nonvanishing conclusion.

Example 5.13. At depths $R = S = 1$ the space of four residue coordinates is $(\mathbb{Z}/3\mathbb{Z})^2 \times (\mathbb{Z}/2\mathbb{Z})^2$, of cardinality $9 \cdot 4 = 36$, and the threshold reads $M \geq 4 \cdot 1 + 2 \cdot 1 = 6$. With six pairs there are $2^6 = 64$ subsets and only 36 possible residue vectors. Two subsets therefore have equal residues. The difference of their indicator vectors is a nonzero vector in $\{-1, 0, 1\}^6$ whose sum vanishes in all four coordinates.

The following elementary comparison concerns only the two scalar exponents of Zudilin's construction; it makes no assertion about the coefficient polynomials or their real remainders.

Theorem 5.14 (a restriction on the two scalar exponents)**.** *Let $C_1 > 0$. If $C_0 \leq 0$ or $2C_0 \leq C_1$, then*

$$C_0 \log 3 - C_1 \log 2 < 0.$$

In the positive branch $C_0 > 0$ and $2C_0 \leq C_1$, the stronger estimate is

$$C_0 \log 3 - C_1 \log 2 < -\frac{17}{41} C_0 \log 2.$$

Proof. If $C_0 \leq 0$, then $C_0 \log 3 - C_1 \log 2 \leq -C_1 \log 2 < 0$. If $C_0 > 0$ and $2C_0 \leq C_1$, use $3^{41} < 2^{65}$ to obtain

$$C_0 \log 3 - C_1 \log 2 \leq C_0(\log 3 - 2 \log 2) < -\frac{17}{41} C_0 \log 2 < 0.$$

□

Written multiplicatively, the conclusion is $3^{C_0} < 2^{C_1}$. Lean checks the inequality and the 17/41 margin of the positive branch. The scope of this elementary inequality matters. The primary Zudilin theorem [8, Theorem 1, p. 154; Sec. 5, pp. 161–162] supplies an integer-base irrationality-exponent estimate on its parameter cone, and the elementary inequality $\mu \geq 2$ then forces $2C_0 \leq C_1$ whenever $C_0 > 0$; Lean checks that implication separately. The primary 2004 theorem is stated for an integer inverse base. The rational specialisation proved earlier in this record has separate proofs for the constructed forms. The scalar statement in this paragraph is only the displayed inequality; it neither constructs a new family at 3/2 nor formalises a universal exclusion of linear-form methods.

6 Failure of the stated clearing conditions at $3/2$

At an integer base, congruences for divisor coefficients can produce long blocks of zero digits. We recall that argument before comparing it with the coordinatewise clearing conditions below.

The integer-base argument. For an integer $b \geq 2$, Erdős uses the expansion $F(b) = \sum_{n\geq 1} \tau(n) b^{-n}$ and the Chinese remainder theorem to produce arbitrarily long blocks of divisor coefficients with prescribed powers of b. After controlling carries from the tail, he obtains arbitrarily long blocks of zero digits and proves that the expansion does not terminate. An eventually periodic expansion with arbitrarily long zero blocks must terminate, so these two facts imply irrationality [1, pp. 63–66]. Positivity of the original summands alone would not prove nontermination; for example, $\sum_{n\geq 1}(b-1)b^{-n} = 1$.

Direct clearing of a partial sum. A more naive attempt is to suppose $F(b) = p/q$, cut at N, and multiply the remaining identity by qb^N. This does not by itself trap a positive integer below 1: the first uncleared tail contribution is $q\tau(N+1)/b$. The clearing conditions below describe one bounded-window version of this attempt. They fail at 3/2.

Write $\beta = r/s$ for the base and $c(n)$ for the coefficient of β^{-n}, so that $c = \tau$ in the case at hand. The term $c(n)\beta^{-n}$ is $c(n)s^n/r^n$. Clearing the power of r leaves the numerator factor s^n in place. That factor is invisible when $s = 1$ and grows geometrically when $s \geq 2$. At 3/2 it is 2^n.

We return to the attempt to clear successive terms of the series. We retain just two necessary demands: the last coordinate in the window must clear, and the coefficient-free lower bound from the first omitted positive term must be smaller than 1. Even these weaker tests are incompatible at 3/2. The argument below uses six natural-number parameters; it excludes the stated clearing scheme, not other approximations to $F(3/2)$.

Definition 6.1 (clearing conditions)**.** For natural numbers a, b, N, K, Q, D, the clearing conditions are

$$a > 0, \qquad Q > 0, \qquad D > 0, \qquad D \leq N + K, \qquad a^K \mid QD, \qquad Q\,b^{N+K+1} < a^{K+1}.$$

The reading is: a and b are the numerator and denominator of the base, playing the roles of r and s in the partial-sum calculation above, so that $(a, b) = (3, 2)$ is the case

of interest; N is the shift, K is the width of the cleared window, Q is the accumulated clearing factor, and D is the final coefficient being cleared. The bound $D \le N + K$ is the only property of the coefficient used; for the divisor-counting coefficient it holds because $\tau(n) \le n$. The divisibility $a^K \mid QD$ tests the final coordinate of the cleared window; it does not by itself clear every earlier coordinate. For a reduced base a/b, the denominator of $Qa^N D(b/a)^{N+K}$ is cleared exactly when $a^K \mid QD$, since a and b are coprime. For example, at $a = 2$, $b = 1$, $N = 21$, $K = 3$ and $Q = 1$, the final term clears because $2^3 \mid \tau(24) = 8$, but the preceding scaled term is $\tau(23)/2^2 = 1/2$ and is not integral.

The final inequality is a necessary smallness test, not a sufficient bound for the entire tail. For $a > b \ge 1$ and $Q > 0$, positivity and $\tau(m) \ge 1$ give

$$Qa^N \sum_{m \ge N+K+1} \tau(m)(b/a)^m > \frac{Qb^{N+K+1}}{a^{K+1}}.$$

A tail smaller than 1 must therefore satisfy the final inequality. The divisibility gives $a^K \le Q(N + K)$, while the smallness test requires $Qb^{N+K+1} < a^{K+1}$. The same Q must satisfy both. These conditions are defined in Lean by the six clearing conditions.

Theorem 6.2 (a necessary inequality for clearing)**.** *If (a, b, N, K, Q, D) satisfies the clearing conditions, then*

$$b^{N+K+1} < a\,(N + K).$$

Proof. Since $QD > 0$ and $a^K \mid QD$, we have $a^K \le QD$, and $D \le N+K$ gives $a^K \le Q(N+K)$. Hence

$$Q\,b^{N+K+1} < a^{K+1} = a^K \cdot a \le Q(N + K) \cdot a,$$

and cancelling the positive factor Q gives the claim. □

The necessary inequality in Theorem 6.2 shows the difference between these clearing tests at integer and noninteger rational bases. At $b = 1$ it reads $1 < a(N + K)$, which holds for every $a \ge 2$ and every nonempty window; this necessary inequality imposes no obstruction. The other clearing conditions still have to be checked. At $b \ge 2$ the left side is exponential in $N + K$ and the right side is linear, so the conditions can hold only for bounded $N + K$ at each fixed base. At $(a, b) = (3, 2)$ the necessary inequality already fails when $N = K = 1$.

Example 6.3. Take the smallest window with $N, K \ge 1$, namely $N = K = 1$, and numerator $a = 3$. At $b = 1$ the tuple $(3, 1, 1, 1, 3, 1)$ satisfies the conditions: $D = 1 \le 2$, the divisibility reads $3 \mid 3$, and the last inequality reads $3 \cdot 1^3 = 3 < 3^2 = 9$. At $b = 2$ no choice works. The last inequality becomes $Q \cdot 2^3 < 3^2$, which forces $Q = 1$; the divisibility then reads $3 \mid D$, and the only candidates are $D = 1$ and $D = 2$, neither divisible by 3. This is the proof of Theorem 6.2 in miniature: it derives $a^K \le Q(N + K)$, here $3 \le 2$.

Feasibility of these necessary conditions is not a successful tail estimate. For $(a, b, N, K, Q, D) = (2, 1, 1, 1, 1, 2)$ all six conditions hold, and $D = \tau(N + K)$. Nevertheless,

$$2 \sum_{m \ge 3} \frac{\tau(m)}{2^m} > 2\left(\frac{2}{2^3} + \frac{3}{2^4} + \frac{2}{2^5}\right) = 1.$$

The failure at $3/2$ is useful because it already occurs at the weaker, necessary tests; their feasibility at an integer base proves no bound for the full remainder.

Proposition 6.4. *For every natural number $x \geq 2$ we have $3x < 2^{x+1}$.*

Proof. Induct on x, starting from $6 < 8$ at $x = 2$. For the step, $2^{x+1} \geq 2^2 > 3$ when $x \geq 1$, so $3(x+1) = 3x + 3 < 2^{x+1} + 2^{x+1} = 2^{x+2}$. □

Theorem 6.5 (failure of the stated clearing conditions at $3/2$)**.** *For all $N \geq 1$ and $K \geq 1$ and all natural Q, D, the tuple $(3, 2, N, K, Q, D)$ does not satisfy the clearing conditions.*

Proof. The clearing conditions would give $2^{N+K+1} < 3(N+K)$ by Theorem 6.2, contradicting Proposition 6.4 applied to $x = N + K \geq 2$. □

Theorem 6.5 excludes the divisibility pattern and necessary tail test in Definition 6.1 at $3/2$. It gives no conclusion about the rationality of $F(3/2)$ or about clearing schemes with different conditions.

7 Successive scaled remainders

Consecutive partial sums differ by one term. After clearing the powers of a rational base, this identity gives a recurrence for the scaled differences from a rational number. We derive it for arbitrary coefficients; the differences are tails only when the series converges to the chosen rational number.

Let r, s, B, ξ be rationals with $r \neq 0$ and let $c : \mathbb{N} \to \mathbb{Q}$ be arbitrary. Define the prefix and the scaled remainder by

$$P_N = \sum_{m=0}^{N-1} \frac{c(m+1)\, s^{m+1}}{r^{m+1}}, \qquad U_N = B\, r^N (\xi - P_N). \tag{$*$}$$

Thus P_N is the partial sum through index N, and U_N is its scaled difference from ξ. These are the rational-base partial sum and the scaled remainder.

Theorem 7.1 (recurrence for the scaled remainder)**.** *Let $r, s, B, \xi \in \mathbb{Q}$ with $r \neq 0$, let $c : \mathbb{N} \to \mathbb{Q}$, and let P_N and U_N be as in ($*$). Then for every N,*

$$U_{N+1} = r\, U_N - B\, c(N+1)\, s^{N+1}.$$

Proof. Expanding $P_{N+1} = P_N + c(N+1)s^{N+1}/r^{N+1}$ and $r^{N+1} = r^N \cdot r$ in the definition of U_{N+1} and clearing the denominator r^{N+1}, which is nonzero, gives the identity. □

For a reduced positive base $r/s > 1$, the forcing term $Bc(N+1)s^{N+1}$ contains the denominator power absent at integer bases. Its size gives a useful bound on two consecutive remainders, although it need not bound each remainder separately.

Theorem 7.2 (the forcing term)**.** *Let s, B be natural numbers and $c : \mathbb{N} \to \mathbb{N}$.*

1. *If $s \geq 2$, $B \geq 1$ and $c(N+1) \geq 1$, then $2^{N+1} \leq B\, c(N+1)\, s^{N+1}$.*
2. *If $s = 1$, then $B\, c(N+1)\, s^{N+1} = B\, c(N+1)$.*

Proof. For the first part, $2^{N+1} \le s^{N+1} = 1 \cdot s^{N+1} \le Bc(N+1)s^{N+1}$, using $B\,c(N+1) \ge 1$. The second part is the definition with $s = 1$. □

At $s = 1$ the forcing term is $Bc(N+1)$, so it grows only as fast as the coefficient; for the divisor function this is $O(N^{\varepsilon})$ for every $\varepsilon > 0$. This comparison does not itself construct a bounded sequence of remainders. At $s \ge 2$ the same term is at least 2^{N+1} whenever the coefficient is nonzero.

Example 7.3. Take $B = 1$, $c = \tau$ and $N = 9$, so that the coefficient is $\tau(10) = 4$. At $s = 1$ the forcing term is 4. At $s = 2$ it is $4 \cdot 2^{10} = 4096$, and part (1) of Theorem 7.2 already guarantees at least $2^{10} = 1024$ without knowing the coefficient at all.

A large forcing term cannot be balanced by two small consecutive remainders. For a natural index N, with B, s positive integers, $s \ge 2$ and $c(N+1) \ge 1$, the recurrence and triangle inequality give

$$2^{N+1} \le Bc(N+1)s^{N+1} = |rU_N - U_{N+1}| \le (1+|r|)\max\{|U_N|, |U_{N+1}|\}.$$

If these assumptions hold at every index, the full sequence is unbounded. They do not imply $|U_N| \to \infty$ under the stated algebraic hypotheses. For example, take $r = -2$, $s = 2$, $B = 1$, $c(n) = 1$ and $\xi = 0$. Directly from the defining partial sums,

$$U_{2j} = 0, \qquad U_{2j+1} = -2^{2j+1} \qquad (j \ge 0).$$

The forcing term grows exponentially while every even remainder is zero. This is an algebraic example, not a convergent Lambert series: it shows that the adjacent-pair bound does not control every subsequence. It concerns the displayed normalisation only and supplies no small nonzero integer forms.

8 The height criterion at $7/2$

The base $7/2$ already lies in the irrationality region of Bundschuh and Väänänen [6, Theorem 2, p. 177; hypotheses pp. 175–176]. We check their height condition in their notation, with q denoting the base. At $\alpha = -1$ it reads

$$\lambda < \left(\frac{1}{2} + \frac{1}{\pi^2}\right)^{-1}.$$

At $q = 7/2$ the Archimedean parameter is $\lambda = \log 7/\log(7/2)$. The criterion therefore applies once the following strict inequality is checked.

Theorem 8.1 (the $7/2$ height condition)**.**

$$\frac{\log 7}{\log(7/2)} < \left(\frac{1}{2} + \frac{1}{\pi^2}\right)^{-1}.$$

Proof. Since $2^{18} = 262144 < 823543 = 7^7$, we have $\log 2/\log 7 < 7/18$. Also $\pi > 3$ gives $1/2 + 1/\pi^2 < 11/18$. Hence

$$\frac{\log 7}{\log(7/2)} = \frac{1}{1 - \log 2/\log 7} < \frac{18}{11} < \left(\frac{1}{2} + \frac{1}{\pi^2}\right)^{-1}.$$

Every denominator is positive, so the reciprocal inequalities have the displayed directions. □

Numerically the two sides are 1.5533 ... and 1.6630 ..., so the condition holds with a margin of about 0.11. The Lean proof factors the estimate through the explicit height-region predicate and the integer certificate $2^{18} < 7^7$, the resulting logarithmic ratio bound $\log 2/\log 7 < 7/18$, the π-bound $1/\pi^2 < 1/9$, and the strict margin

$$\frac{\log 2}{\log 7} < \frac{1}{2} - \frac{1}{\pi^2}.$$

The final height inequality then rewrites $\log(7/2) = \log 7 - \log 2$ and closes the displayed condition.

This formalises the complete elementary parameter check at $q = 7/2$; it does not formalise Bundschuh and Väänänen's analytic irrationality theorem, whose proof occupies pp. 189–193 of the source. The conclusion that $F(7/2)$ is irrational is consequently cited from that theorem, not claimed as a Lean theorem here.

The criterion does not cover 3/2, and the results of Sections 6 and 7 give no evidence either way about whether $F(3/2)$ is irrational.

8.1 The 81/200 logarithmic region

Consider the rational-height region

$$\frac{\log b}{\log a} < \frac{81}{200} \qquad (a > b > 0).$$

The number 81/200 is the rational lower bound on θ^* proved earlier. The bracket $81/200 < \theta^* < 1/2$ in Section 2.5 puts this entire smaller region inside Theorem 2.2. Membership of the smaller region is an integer comparison. The Lean module of this library treats the displayed inequality as a definition and proves elementary memberships and exclusions. In particular,

$$\frac{2}{5} < \frac{\log 4}{\log 31} < \frac{81}{200} < \frac{\log 2}{\log 3}.$$

The first two comparisons come respectively from $31^2 < 4^5$ and $4^{200} < 31^{81}$; the last comes from $3^{81} < 2^{200}$. Lean checks the lower bound. Consequently 31/4 lies in the enlarged region, and so does every positive power $(31/4)^r$. The same base lies strictly outside the earlier Bundschuh–Väänänen region, so the 81/200 inequality defines a strict set-theoretic enlargement of the earlier recorded logarithmic region. Combined with the bracket $81/200 < \theta^*$, these memberships are exactly the finite input to Theorem 2.3, which is where the irrationality of $F(31/4)$ and of every $F((31/4)^r)$ is proved.

It still does not approach 3/2. The exact comparison

$$3^{81} < 2^{200} \quad \Longrightarrow \quad \frac{81}{200} < \frac{\log 2}{\log 3}$$

is Lean-checked, as is the conclusion that 3/2 belongs to neither height region (comparison with $\log 2/\log 3$, 3/2 fails the condition). Thus 81/200 is the larger of the two explicitly defined cutoffs used below, while a cutoff that includes 3/2 must be strictly larger than $\log 2/\log 3 \approx 0.6309$.

9 A denominator-exponent model for Padé-type forms

For a sum of Padé-type terms, a common denominator must clear each summand. The two exponent expressions below are bounded by $(3n^2 - n)/2$, as their differences factor over $\mathbb{Z}$. We double all exponents to avoid fractions and write $\widetilde{E}_n = 3n^2 - n$. These are algebraic comparisons only: no approximation family is specified, so integral coefficients and a nonzero remainder estimate would still be needed for an arithmetic application.

Proposition 9.1 (exponent model: summand bound and exact gap)**.** *Let* $\widetilde{E}_n = 3n^2 - n$ *and put*

$$\widetilde{P}(n,k) = 2(k(n-k) + nk) + k(k-1),$$

$$\widetilde{Q}(n,m) = 2(n^2 - n) + j^2 + 2jm + j - m^2 + 3m, \qquad j = n - m - 1.$$

Then, for integers n, k, m*:*

1. *if* $0 \le k \le n$*, then* $\widetilde{P}(n,k) \le \widetilde{E}_n$*, and the gap factors as* $\widetilde{E}_n - \widetilde{P}(n,k) = (n-k)(3n-k-1)$*;*
2. $\widetilde{E}_n - \widetilde{Q}(n,m) = 2(n + m(m-1))$ *identically;*
3. *if* $n \ge 0$ *and* $m \ge 1$*, then* $\widetilde{Q}(n,m) \le \widetilde{E}_n$*.*

Part (1) is the summand exponent bound, part (2) the exact gap identity, and part (3) the maximal exponent bound. The gap in (2) is an identity in $\mathbb{Z}[n,m]$, so (3) follows from $m(m-1) \ge 0$ and $n \ge 0$. The range $m \ge 1$ is sufficient, not necessary for this polynomial inequality: for integer m, one also has $m(m-1) \ge 0$ when $m \le 0$. No claim about a summand of an unspecified approximation is needed.

Example 9.2. At $n = 2$ the proposed doubled exponent is $\widetilde{E}_2 = 10$, and $\widetilde{P}(2,k)$ takes the values $0, 6, 10$ at $k = 0, 1, 2$. The three gaps are $10, 4, 0$, matching the factorisation $(2-k)(5-k)$ of part (1); the summand at $k = 2$ is the one that saturates the proposed denominator. For part (2), at $m = 1$ we have $j = 0$ and $\widetilde{Q}(2,1) = 6$, with gap $4 = 2(2 + 1 \cdot 0)$.

These are inequalities between polynomials in the exponents. They establish that the proposed exponent $\widetilde{E}_n$ dominates the two displayed summand exponent expressions, and nothing further. Positivity of the remainder, its rate of decay, and the comparison of that rate against the denominator height are the analytic obligations, and none of them is treated here, so nothing in this section is an irrationality measure.

10 Complements and further questions

An unrestricted construction of small nonzero integer forms is another formulation of irrationality, not yet a method of proving it. The next problem remains equivalent to irrationality at $3/2$ even with its quadratic degree bounds, and even after imposing the exponential coefficient-height bound discussed below. We prove the equivalence before asking what a specified approximation family can supply.

Problem 10.1 (a divided linear form with small nonzero remainder)**.** Exhibit an integer constant $C \geq 1$ and, for every sufficiently large positive integer n, positive integers W_n, R_n, S_n, M_n such that

$$n^2 \leq W_n, R_n, S_n \leq Cn^2, \qquad 4R_n + 2S_n \leq M_n \leq Cn^2,$$

together with polynomial pairs $(U_{n,j}, V_{n,j}) \in \mathbb{Z}[X]^2$ for $0 \leq j < M_n$, each of degree at most the common degree bound W_n, whose specialised integer rows are primitive:

$$\gcd(H_{W_n}(U_{n,j}), H_{W_n}(V_{n,j})) = 1.$$

Find a nonzero vector $\lambda^{(n)} \in \{-1, 0, 1\}^{M_n}$ for which, on putting

$$U_n = \sum_{j<M_n} \lambda_j^{(n)} U_{n,j}, \qquad V_n = \sum_{j<M_n} \lambda_j^{(n)} V_{n,j},$$

the pair (U_n, V_n) is not $(0, 0)$, all four residues vanish,

$$J_{3,R_n}(U_n) = J_{3,R_n}(V_n) = 0, \qquad J_{2,S_n}(U_n) = J_{2,S_n}(V_n) = 0,$$

where every residue in this display is formed using the common degree bound W_n, and the resulting divided integer linear form

$$A_n = \frac{H_{W_n}(U_n)}{3^{R_n}2^{S_n}}, \qquad B_n = \frac{H_{W_n}(V_n)}{3^{R_n}2^{S_n}}, \qquad \rho_n = A_n F(3/2) - B_n$$

satisfies the explicit analytic condition

$$0 < |\rho_n| < \frac{1}{n}.$$

The residue equations make A_n, B_n integers. A solution would prove irrationality: if $F(3/2) = a/b$ in lowest terms, every nonzero ρ_n has absolute value at least $1/b$, contradicting the displayed bound for $n > b$. Theorem 5.8 supplies only a nonzero signed relation with the four residue equations once the pairs and size inequality are present; it does not supply primitive input rows, a nonzero combined polynomial pair, or the nonvanishing and decay of ρ_n.

Why the choice of family matters. The converse can start from an arbitrary small integer form and build the polynomial rows afterwards. Pigeonhole approximation supplies the small form; Bézout's identity lifts its coordinates to polynomial evaluations. Thus the degree and residue conditions do not constrain how the approximation was obtained. Let ξ be any irrational real number. Comparing the fractional parts of $0, \xi, \dots, (n+1)\xi$ in $n+1$ equal half-open intervals gives integers $A \geq 1$ and B with

$$A \leq n + 1, \qquad 0 < |A\xi - B| < \frac{1}{n+1} < \frac{1}{n}.$$

Set $w = n^2$, $W_n = R_n = S_n = w$, $M_n = 6w$, and $D = 6^w$. The three integer rows

$$(DA, 1), \qquad (1, DB - 1), \qquad (-1, 0)$$

are primitive and sum to (DA, DB). Pad them to M_n rows with $(1, 0)$, and use the coefficient vector $(1, 1, 1, 0, \dots, 0)$.

To realise every coordinate as a polynomial evaluation of degree at most w, choose integers u, v such that $u2^w + v3^w = 1$. For any integer h, the polynomial $h(u + vX^w)$ satisfies $H_w(h(u + vX^w)) = h$. Lift all row coordinates this way. The combined polynomial pair is nonzero because its first evaluation is $DA \neq 0$. Its two evaluations are divisible by $3^w 2^w$, and its divided remainder is $A\xi - B$. All the conditions of Problem 10.1, with $F(3/2)$ replaced by ξ, hold with $C = 6$.

Even a coefficient-height bound of the form $\exp(O_\xi(n^2))$ would not exclude these lifts. Choose $0 \leq v < 2^w$, which gives $|u| < 3^w$. Since $|B| \leq (n + 1)|\xi| + 1$, every lifted coefficient has absolute value at most $(|\xi| + 2)(n + 1)18^w$. Together with the preceding rationality contradiction, this proves that the unrestricted construction problem for a real target ξ is equivalent to the irrationality of ξ. It does not establish either assertion for $F(3/2)$.

The short paper refers to this formulation and to its equivalence with irrationality. To pose a more specific question, one must replace the free choice of polynomial pairs by an actual set given by coefficient formulas, admissible parameters and permitted normalisations. The restriction must exclude the freedom to choose the rows from an already good approximation. Merely giving the resulting family a name, an explicit formula or an exponential height bound does not do so: the lifts above have all three. For a fixed indexed family, the distinction between all sufficiently large indices and a sparse subsequence can matter. Arbitrary padding and reindexing remove it in the unrestricted problem.

Corollary 5.5 gives no improvement from integer rescaling and, under its endpoint hypotheses, no common factor 2 or 3 in the unscaled evaluations. It makes no general assertion about polynomial cancellation before specialisation, combinations of rows, determinant-specific divisibility or other approximation families.

A prescribed family must supply nonvanishing and decay for the same signed combinations. A residue collision alone supplies neither; separate examples of the two properties would not suffice either. A coefficient-height estimate is needed when the approximation argument calls for it, not as an extra hypothesis of the elementary irrationality criterion above.

10.1 A congruence relation with nonzero remainder

Fixing the approximation family is not enough: the residue map must also use a fixed normalisation. Polynomial coefficient content, content of an evaluated row, and content of a combined row are three different quantities. We keep them separate as follows. For each n, let W_n be a common degree bound and let

$$(U_{n,j}, V_{n,j}, \mathcal{R}_{n,j})_{j<M_n}$$

be the chosen forms, with $U_{n,j}, V_{n,j} \in \mathbb{Z}[X]$ of degree at most W_n and

$$\mathcal{R}_{n,j}(t) = U_{n,j}(t)F(t) - V_{n,j}(t) \qquad (t > 1).$$

This identity uses the same normalisation as the evaluated rows and all subsequent residue equations. Fix all divisions before forming the residue map. Require each polynomial pair to have coefficient gcd 1, by dividing its common integer coefficient factor if necessary. This is different from making the two evaluated integers coprime, as required in Problem 10.1. The two operations need not preserve the same polynomial family. Coefficient content 1 alone gives no coprimality with 6 after evaluation: at $W = 1$, the pairs $(X, 3)$ and $(2X-1, 2)$ have coefficient content 1 but give the integer rows $(3, 6)$ and $(4, 4)$, respectively.

The gcd

$$\gcd(H_{W_n}(U_{n,j}), H_{W_n}(V_{n,j}))$$

of one evaluated row can differ from the gcd of the final sum. Neither is divided out without also dividing the corresponding remainder and measuring the height after that division. If, in addition, the coefficient of X^{W_n} in $U_{n,j}$ and the constant coefficient of $V_{n,j}$ are both units, Proposition 5.3 makes this row gcd coprime to 6. These endpoint assumptions are extra conditions, not consequences of coefficient primitivity. Dividing by such a gcd preserves divisibility by every power of 2 and 3 for that individual row. It need not preserve a relation with a fixed coefficient vector when different rows are divided by different contents. For example, at $W = 1$ the pairs $(X-1, X-1)$ and $(X+1, X+1)$ give rows $(1, 1)$ and $(5, 5)$. Their sum is zero modulo 6, whereas the sum of their primitive normalisations is $(2, 2)$, not zero modulo 6. Both pairs satisfy the two unit conditions just stated. Each division is invertible modulo 6, but different rows are multiplied by different units, so their sum need not stay zero. The residue map must therefore be formed again after row-by-row normalisation.

Dividing a specialised row by its content need not preserve the chosen polynomial family, and lifting the divided row back is a constrained problem. For fixed W, the map $P \mapsto H_W(P)$ on integer polynomials of degree at most W is surjective onto $\mathbb{Z}$, since 3^W and 2^W are coprime, so a lift always exists. The issue is not an arbitrary exponential height bound, as the construction above shows. It is whether the lift belongs to the chosen approximation family and satisfies that family's remainder identity and quantitative height estimate. With $W = 1$, for instance, $(X+1, X+1)$ specialises to $(5, 5)$, whose primitive normalisation $(1, 1)$ lifts to $(X-1, X-1)$, but not to an integer scalar multiple of the original pair. Any lift used below is therefore supplied together with its degree bound, its height bound and its exact remainder.

For target depths R_n, S_n, let $J_n(\lambda)$ be the vector of four residues of the pair $\sum_j \lambda_j (U_{n,j}, V_{n,j})$, and define

$$C_n = \{\lambda \in \{-1, 0, 1\}^{M_n} \setminus \{0\} : J_n(\lambda) = 0\},$$

$$K_n^{\mathrm{poly}} = \left\{ \lambda \in \{-1, 0, 1\}^{M_n} : \sum_j \lambda_j U_{n,j} = 0,\ \sum_j \lambda_j V_{n,j} = 0 \right\},$$

$$K_n^{\mathrm{rem}} = \left\{ \lambda \in \{-1, 0, 1\}^{M_n} : \sum_j \lambda_j \mathcal{R}_{n,j}(3/2) = 0 \right\}.$$

The second set consists of signed relations whose remainder vanishes at $3/2$. It also contains every relation with an identically zero remainder function. These are sets of restricted coefficient vectors, not assertions that the signed cube is a vector space.

The exact pigeonhole condition is

$$2^{M_n} > 3^{2R_n} 2^{2S_n}, \qquad \text{equivalently} \qquad M_n > 2R_n \log_2 3 + 2S_n. \tag{26}$$

Thus the least integer number of rows satisfying this counting test is

$$\lfloor 2R \log_2 3 + 2S \rfloor + 1.$$

The checked condition $M \geq 4R + 2S$ for $R > 0$ is a convenient sufficient corollary, not the exact threshold. Applying Cauchy–Schwarz to the sizes of the residue classes shows that there are at least

$$\frac{1}{2}\left(\frac{2^{2M_n}}{3^{2R_n} 2^{2S_n}} - 2^{M_n} \right) \tag{27}$$

unordered pairs of distinct subsets with equal residues. It would therefore suffice to prove that fewer than this many pairs give a zero polynomial combination or a zero remainder at $3/2$.

Theorem 5.8 supplies only $C_n \neq \emptyset$; it does not rule out zero polynomial pairs or zero real remainders. For a useful approximation family, bounding the multiplicities in (27) remains a possible way to obtain nonvanishing. Without restrictions on the family, however, nonvanishing alone is elementary.

Example 10.2 (nonvanishing without decay)**.** For $n \geq 1$, set $W_n = R_n = S_n = n^2$ and $M_n = 6n^2$. Take

$$\begin{aligned} (U_{n,0}, V_{n,0}) &= ((1 + 2^{W_n}) X^{W_n}, 1), \\ (U_{n,j}, V_{n,j}) &= (X^{W_n}, 1) \qquad (1 \leq j < M_n), \end{aligned}$$

and define $\mathcal{R}_{n,j}(t) = U_{n,j}(t) F(t) - V_{n,j}(t)$ for $t > 1$. All pairs have coefficient gcd 1 and primitive evaluated rows. The coefficient vector $(1, -1, 0, \dots, 0)$ belongs to $C_n \setminus (K_n^{\mathrm{poly}} \cup K_n^{\mathrm{rem}})$, but the divided integer form equals $F(3/2)$ for every n.

Indeed, the evaluated rows are $(3^{W_n} + 6^{W_n}, 2^{W_n})$ and $(3^{W_n}, 2^{W_n})$; their first coordinates are odd, so both rows are primitive. Their difference is $(6^{W_n}, 0)$, while the polynomial difference is $(2^{W_n} X^{W_n}, 0)$. Thus all four residues vanish. The unscaled remainder difference is $3^{W_n} F(3/2) > 0$. Clearing the evaluation denominators multiplies it by 2^{W_n}, giving $6^{W_n} F(3/2)$. Dividing this cleared integer form by $3^{R_n} 2^{S_n} = 6^{W_n}$ leaves the same positive constant $F(3/2)$. The counting condition (26) holds because $\log_2 3 < 2$.

Example 10.2 meets the two nonvanishing requirements in Problem 10.1 without any irrationality assumption. What it fails is the smallness condition. The useful question is therefore to obtain nonvanishing and decay in the same prescribed approximation family, not merely to exhibit some family with a nonzero congruence relation.

The range of subset sums. The residue condition and the real-error condition can be counted together. Within each residue class, subdivide the range of subset remainders into intervals of width D/n; a bound on repeated remainder values then forces a nonzero small difference. Fix a positive integer n. For M integer rows (A_j, B_j) and a modulus $D \geq 1$, set $e_j = A_j F(3/2) - B_j$. For each attained residue vector b, let T_b be the difference between the largest and smallest remainders $\sum_j \varepsilon_j e_j$ with $\varepsilon_j \in \{0, 1\}$ and $\sum_j \varepsilon_j (A_j, B_j) \equiv b \pmod{D}$. Let the integer k_b bound the number of subsets attaining any one remainder value in that residue class. Then

$$2^M > \sum_b k_b \left(\left\lfloor \frac{nT_b}{D} \right\rfloor + 1 \right)$$

produces two subsets with equal residues and distinct remainders less than D/n apart. Within each residue fibre, subtract the least remainder, multiply by n/D, and take floors. These integer indices range from 0 to $\lfloor nT_b/D \rfloor$, including a separate final index when T_b is a positive integral multiple of D/n. Equal indices give a remainder difference strictly less than D/n. If each index corresponded to only one remainder value, it would account for at most k_b subsets, contradicting the displayed inequality. Subtracting the two subset sums and dividing their coefficients by D gives an integer form with nonzero absolute value less than $1/n$. The common width bound $T = \sum_j |e_j|$ and a common multiplicity bound k give the coarser count $Qk(\lfloor nT/D \rfloor + 1)$, where Q is the number of attained residue vectors. All rows must have the same normalisation. Primitive rows can still have repeated subset sums, and positivity need not survive subtraction.

The lattice after evaluation. For example, $(1, 0), (1, 6)$ generate $\mathbb{Z} \oplus 6\mathbb{Z}$. They give two residues modulo 2. Within that lattice, the vectors whose two coordinates are even form $2\mathbb{Z} \oplus 6\mathbb{Z}$; dividing them by 2 gives $\mathbb{Z} \oplus 3\mathbb{Z}$, of index 3. These are two different counts: the number of attainable residues before division, and the index of the lattice left after division. Smith normal form computes both for any rank-two lattice. The first count describes congruences available before division; the second describes the integrality restriction that survives it. Neither count estimates the real remainders of the rows.

Let primitive integer rows $u_j \in \mathbb{Z}^2$ span a rank-two lattice L, and let g be the gcd of their 2×2 minors. Since at least one row is primitive, the Smith invariants are $1, g$ [34, Thms. 2.3–2.4, p. 3]. Hence, for $D \geq 1$,

$$|\operatorname{im}(L \longrightarrow (\mathbb{Z}/D\mathbb{Z})^2)| = \frac{D^2}{\gcd(g, D)}, \qquad [\mathbb{Z}^2 : (L \cap D\mathbb{Z}^2)/D] = \frac{g}{\gcd(g, D)}.$$

Indeed an integral unimodular change of coordinates takes L to $\mathbb{Z} \times g\mathbb{Z}$; reduction modulo D and intersection with $D\mathbb{Z}^2$ give the two formulas. This is Smith normal form over $\mathbb{Z}$, not over $\mathbb{Z}[p]$. It measures the actual image, rather than the ambient residue space. If two independent divided rows have coordinate height at most H and remainders of absolute value at most ε, their nonzero integer determinant gives the necessary inequality $2H\varepsilon \geq g/\gcd(g, D)$. That is not a necessary condition for producing a single nonzero form. If the specialised rows are not primitive, let $d_1 \mid d_2$ be the positive Smith invariants

instead. For $D \geq 1$ the corresponding formulas are

$$|\operatorname{im}(L \bmod D)| = \frac{D^2}{\gcd(d_1, D)\gcd(d_2, D)},$$
$$[\mathbb{Z}^2 : (L \cap D\mathbb{Z}^2)/D] = \frac{d_1 d_2}{\gcd(d_1, D)\gcd(d_2, D)}.$$

They follow by applying the one-dimensional calculation to each summand $d_i\mathbb{Z}$. Polynomial coefficient content alone does not determine either integer Smith invariant after specialisation.

The lattice spanned by the actual tails. The formulas above apply to any rank-two lattice. For the rows that the approximation problem actually supplies, the invariant can be computed outright, and it is the smallest one compatible with the base. Let $a > b \geq 1$ be coprime, write the partial sum $S_m(b/a) = \sum_{r \leq m} (b/a)^r/(1 - (b/a)^r)$ in lowest terms as P_m/Q_m with $Q_m > 0$, and set $P_0 = 0$, $Q_0 = 1$. The primitive integer row of the mth tail is (Q_m, P_m), which represents $Q_m F(a/b) - P_m$.

Proposition 10.3 (the prefix lattice of the tails)**.** *Every Q_m is coprime to ab, and b divides every P_m. For every prefix containing $m = 0$ and $m = 1$,*

$$\operatorname{span}_{\mathbb{Z}}\{(Q_m, P_m) : 0 \leq m < M\} = \mathbb{Z} \times b\mathbb{Z} \qquad (M \geq 2),$$

so the Smith invariants are $1, b$, the gcd of the 2×2 minors is b, and the image modulo D has cardinality $D^2/\gcd(b, D)$. Multiplying an individual tail by a nonzero rational weight leaves its primitive row unchanged up to sign.

Proof. A common denominator of $S_m(b/a)$ is a product of the integers $a^r - b^r$ for $r \leq m$, each coprime to ab; over that denominator every summand has numerator divisible by b^r. Reduction to lowest terms removes no prime factor of b, which gives the first two assertions. The rows at $m = 0, 1$ are $(1, 0)$ and $(a - b, b)$, and subtracting $a - b$ copies of the first from the second gives $(0, b)$; every later row already lies in $\mathbb{Z} \times b\mathbb{Z}$ by the first assertions. The Smith and image statements are the specialisation of the formulas above at $g = b$. Finally the rational line through a primitive pair meets $\mathbb{Z}^2$ in exactly the multiples of that pair, so clearing denominators in a nonzero rational multiple and dividing by the coordinate gcd returns the same pair up to sign. □

At $3/2$ the first three partial sums are $0, 2$ and $14/5$, with rows $(1, 0)$, $(1, 2)$ and $(5, 14)$. Their span is already $\mathbb{Z} \times 2\mathbb{Z}$. There is no three-adic collapse, and the only local saving is the single factor of two that the image formula displays. For a consecutive window containing $m = s$ and $m = s + 1$ and no earlier row, every pairwise minor is divisible by b^{s+1}. The gcd has valuation $(s + 1)v_p(b)$ at each prime $p \mid b$ and valuation zero at each prime $p \mid a$. The adjacent-row identity

$$Q_s P_{s+1} - P_s Q_{s+1} = \frac{Q_s Q_{s+1} b^{s+1}}{a^{s+1} - b^{s+1}}$$

attains both, since its denominator is a unit at every prime dividing ab.

Rescaling an individual tail and then making its row primitive returns the same row up to sign. Thus it preserves the lattice spanned by the full family. Integer combinations remain in that lattice, but a selected family of combinations need not span all of it and can have a larger minor gcd. At $3/2$, write $u_0 = (1, 0)$ and $u_1 = (1, 2)$. The primitive rows u_0 and $3u_1 - 2u_0 = (1, 6)$ span $\mathbb{Z} \times 6\mathbb{Z}$, with minor gcd 6, although u_0, u_1 span $\mathbb{Z} \times 2\mathbb{Z}$. The selection replaces the original lattice by an index-three sublattice, accounting for the change from 2 to 6. Containment therefore preserves divisibility by the original gcd, not equality of gcds or endpoint valuations. The prefix and window equalities above concern the full indicated families.

Rational combinations can also leave the original lattice: the half-sum of u_0, u_1 is the primitive row $(1, 1)$, whose minor against u_0 is 1. Coefficient denominators must then be accounted for; likewise, primitive normalisation after taking combinations requires a new lattice calculation. These operations may change the minor gcd in either direction. None by itself controls the real error: a useful selected family must still have nonzero remainders small enough after the proposed division.

10.2 Estimating the divided remainder

Once a construction is fixed, let $H_n \geq 1$ be its height, $L_n \neq 0$ its undivided remainder, and $D_n = 3^{R_n} 2^{S_n}$ its proved divisor, counted once. We ask for the additional height and remainder estimate below. With a different divisor, the two logarithmic terms should be replaced by $\log D_n$. The estimate must use the height of the objects that enter the comparison with a nonzero integer; scalar and exterior-determinant heights cannot be interchanged. For scalar forms, integral coefficients and $0 < |L_n|/D_n \to 0$ already suffice for irrationality, without a height factor. The proposed estimate implies that decay because $H_n \geq 1$, although it is not necessary for the scalar criterion. Allowing arbitrarily small positive heights would lose this implication.

Problem 10.4 (an additional height and remainder estimate)**.** Prove the explicit estimate

$$\limsup_{n\to\infty} \frac{\log H_n + \log |L_n| - R_n \log 3 - S_n \log 2}{n^2} < 0. \tag{28}$$

Every denominator, row content and final-combination content must already be included in H_n and L_n.

Nonvanishing alone does not address (28). Conversely, a formal decay estimate cannot supply a nonzero form if every combination with the required residues has zero remainder. A proposed divisor saving must be compared with the height and remainder of the same explicitly normalised objects.

10.3 A restriction on Mahler functional equations

For the divisor generating series $\mathcal{L}(z) = \sum_{n\geq 1} \tau(n) z^n$, the classification of Bell and Smertnig excludes a k-Mahler equation for every $k \geq 2$ [27, Thm. 1.3 and the consequences on p. 3]. We first explain its application to τ. We then prove the weaker simultaneous $2/3$ statement by combining the Adamczewski–Bell theorem with the functional nonrationality from Section 2.1. The cited 2026 works have the status of the identi-

fied preprints. These functional statements leave the arithmetic nature of an individual rational-base value undecided.

To apply the classification, note that τ is multiplicative: for coprime integers m, n, each divisor of mn has a unique factorisation into a divisor of m and a divisor of n. Suppose that $\mathcal{L}$ were k-Mahler. The classification would give a prime p, an integer $r \geq 0$ and an eventually periodic function χ with $\tau(m) = m^r \chi(m)$ whenever $p \nmid m$. For a prime $\ell \neq p$, this forces

$$\chi(\ell^j) = \frac{j+1}{\ell^{jr}} \qquad (j \geq 0).$$

If $r = 0$, these values are unbounded. If $r > 0$, they are nonzero and tend to zero. Both alternatives contradict the finite range of an eventually periodic function.

Here the ambient space is $\mathbb{Q}((z))$, the field of formal Laurent series, viewed as a vector space over $\mathbb{Q}(z)$. Stability means that substituting z^k for z sends each member of the subspace back into that subspace; this substitution is not a $\mathbb{Q}(z)$-linear map.

Proposition 10.5 (no finite simultaneous 2/3-system)**.** *Let*

$$\mathcal{L}(z) = \sum_{n \geq 1} \frac{z^n}{1 - z^n}.$$

There is no finite-dimensional $\mathbb{Q}(z)$-vector space that contains $\mathcal{L}$ and is stable under both $z \mapsto z^2$ and $z \mapsto z^3$.

Proof. Suppose V were such a space, of dimension d. Stability under $z \mapsto z^2$ places the $d+1$ elements $\mathcal{L}(z), \mathcal{L}(z^2), \dots, \mathcal{L}(z^{2^d})$ in V, so they are linearly dependent over $\mathbb{Q}(z)$; clearing denominators gives polynomials $P_0, \dots, P_d$, not all zero, with $\sum_{i=0}^{d} P_i(z)\mathcal{L}(z^{2^i}) = 0$.

Linear dependence need not give a nonzero coefficient of the unshifted function, as a Mahler equation requires. One cannot simply replace z by $z^{1/2^j}$, since the polynomial coefficients need not have exponents divisible by 2^j. Instead, separate their exponent classes. Let j be the least index with $P_j \neq 0$. Write each polynomial uniquely as $P_i(z) = \sum_{r=0}^{2^j-1} z^r Q_{i,r}(z^{2^j})$. Every series $\mathcal{L}(z^{2^i})$ with $i \geq j$ has exponents divisible by 2^j, so the relation splits by exponent residues modulo 2^j. Choose r with $Q_{j,r} \neq 0$ and put $u = z^{2^j}$. The corresponding relation is

$$\sum_{i=j}^{d} Q_{i,r}(u)\mathcal{L}(u^{2^{i-j}}) = 0,$$

with a nonzero coefficient of $\mathcal{L}(u)$. Thus $\mathcal{L}$ is 2-Mahler. The same argument with 3 in place of 2 makes it 3-Mahler. Since 2 and 3 are multiplicatively independent, a theorem of Adamczewski and Bell [16, Thm. 1.1, p. 6] then forces $\mathcal{L}$ to be a rational function. Since $\mathcal{L}(z) = F(1/z)$ for $0 < z < 1$, this contradicts the nonrationality of F proved in Section 2.1. Rivin's periodic-coefficient corollary [24, Cor. 6.4, p. 9] gives the same nonrationality conclusion. □

Proposition 10.5 uses no property of the point 2/3: the obstruction is functional and appears before regularity at a particular point is considered. For this scalar function,

Bell and Smertnig's single-base classification already implies the proposition. The proof above instead derives it from simultaneous closure and elementary functional nonrationality, using the Adamczewski–Bell theorem. A construction using additional functions or functional relations must specify those functions and its closure conditions; the single-base statement is not an obstruction to every approximation method. The classification and the Adamczewski–Bell theorem are cited, not proved here; the needed nonrationality has the elementary proof given earlier.

The Lean proof of Proposition 10.5 (no finite simultaneous 2/3-system) uses neither cited theorem, and it uses only stability under $z \mapsto z^2$. As in the first step of the proof above, that stability gives polynomials $P_0, \dots, P_d$, not all zero, with $\sum_{i=0}^{d} P_i(z)\mathcal{L}(z^{2^i}) = 0$, and this relation also holds between the values of these functions on the open unit disc. If ζ is a primitive nth root of unity, then $(1-r)\mathcal{L}(r\zeta)/\log(1/(1-r)) \to 1/n$ as $r \uparrow 1$. These limits, taken at roots of unity of suitable orders, force every P_i to vanish. The same argument excludes, for every $k \geq 2$, a finite-dimensional space containing $\mathcal{L}$ and stable under $z \mapsto z^k$ alone (single-base statement).

10.4 A limitation of the rectangular exponent model

We conclude the comparison of exponent models by bounding Θ_{HP}, defined in Section 3. For $\rho \geq 0$ and $\sigma \geq 1 + \rho$ its denominator

$$\frac{(1+\rho)^2}{2} + \sigma(1+\rho) + \frac{1+\rho^2}{2} + \sigma$$

is positive. Put $u = \sigma - 1 - \rho \geq 0$. Multiplying the difference $\Theta_{\mathrm{HP}}(\rho, \sigma) - (1/2 - 1/\pi^2)$ by $2\pi^2$ times this denominator gives exactly

$$-\pi^2\rho^2 - \pi^2\rho u - 2\pi^2\rho - 2\rho^2 - 10\rho u - 4\rho - 6u^2 - 8u.$$

Every term is nonpositive. The original difference has the same sign, and it vanishes exactly when $\rho = 0$ and $\sigma = 1$ (exact expansion, nonpositivity, equality case). Equivalently, within that model the displayed threshold never exceeds the classical one-function margin, with equality only at the classical endpoint (bound, equality). This bounds only the displayed exponent model. Applying it to an approximation family would require the polynomial construction, integrality and asymptotic estimates connecting that family to Θ_{HP}. The rational cutoff 81/200 discussed above satisfies

$$\frac{81}{200} < \frac{\log 2}{\log 3}.$$

For a sufficient criterion of the form $\log b/\log a < T$, including 3/2 requires $T > \log 2/\log 3$, not merely an improvement on 81/200. This numerical target applies to that form of criterion, not to every irrationality argument.

A concrete optimisation question is available within the 2004 construction itself. Write a source parameter direction as $\alpha = (\alpha_0, \alpha_1, \alpha_2; \beta)$, with positive integer entries satisfying

$$\alpha_1 \leq \alpha_2, \qquad \alpha_1 + \alpha_2 < \beta \leq \alpha_0 + \alpha_2, \qquad \gcd(\alpha_0, \alpha_1, \alpha_2, \beta) = 1.$$

The source parameters are $a_j = \alpha_j n + 1$ and $b = \beta n + 2$; here b is not a rational-base denominator. The gcd condition is a normalisation of the parameter direction, not a primitivity assertion about evaluated rows. Its invariance under dilation is checked below. For these directions set

$$c_{00} = \alpha_0 + \alpha_1 + \alpha_2 - \beta, \quad c_{01} = \alpha_0, \quad c_{11} = \alpha_1, \quad c_{21} = \alpha_2,$$
$$c_{12} = \beta - \alpha_1, \quad c_{22} = \beta - \alpha_2, \qquad m = \max(c_{00}, c_{01}, c_{11}, c_{21}, c_{12}, c_{22}),$$

and define the periodic step function

$$\omega_\alpha(u) = \max\left\{\begin{matrix}0,\\ \lfloor c_{21}u\rfloor + \lfloor c_{22}u\rfloor - \lfloor c_{11}u\rfloor - \lfloor c_{12}u\rfloor,\\ \lfloor c_{01}u\rfloor + \lfloor c_{21}u\rfloor - \lfloor c_{00}u\rfloor - \lfloor c_{12}u\rfloor\end{matrix}\right\}.$$

Zudilin's constants in (25) and (26), already used in the finite direction scan, are

$$C_1(\alpha) = (\alpha_0 + \alpha_1 + \alpha_2)\beta - \frac{\alpha_1^2 + \alpha_2^2 + \beta^2}{2},$$
$$C_0(\alpha) = \frac{\alpha_1^2}{2} + \alpha_0\alpha_1 + (\beta - \alpha_2)(\alpha_2 - \alpha_1) - \frac{3}{\pi^2}\left(m^2 - \int_0^1 \omega_\alpha(u)\, d(-\psi_1(u))\right).$$

Here ψ_1 is the trigamma function used earlier. The step function is bounded, nonnegative, periodic with period 1, and zero near zero, so the integral is finite.

The gcd normalisation does not change the ratio being optimised. Indeed, periodicity and $-\psi_1'(u) = 2\sum_{j\ge 0}(u+j)^{-3}$ give, by Tonelli's theorem,

$$\int_0^1 \omega_\alpha(u)\, d(-\psi_1(u)) = \int_0^\infty \frac{2\omega_\alpha(u)}{u^3}\, du.$$

For any positive integer k, the floor formulas give $\omega_{k\alpha}(u) = \omega_\alpha(ku)$. Substitution in the last integral therefore multiplies it by k^2. All other terms in C_0 and C_1 are quadratic in the direction, including m^2. Hence $C_i(k\alpha) = k^2 C_i(\alpha)$ for $i = 0, 1$, so passing to a primitive direction preserves both positivity and C_0/C_1. Let $\mathcal{A}$ be exactly the displayed primitive directions with $C_0(\alpha) > 0$ and $C_1(\alpha) > 0$.

Problem 10.6 (optimising the published parameter directions)**.** Determine

$$\sup_{\alpha\in\mathcal{A}} \frac{C_0(\alpha)}{C_1(\alpha)},$$

or improve its bounds. In particular, does an admissible direction give a ratio strictly greater than the value θ^* in Theorem 2.2, attained at $(14, 12, 14; 27)$? A finite numerical scan does not establish optimality over $\mathcal{A}$.

The immediate bounds are $\theta^* \le \sup_{\mathcal{A}} C_0/C_1 \le 1/2$. Indeed, at any integer base $p \ge 2$ the source gives $\mu_{\mathrm{irr}}(F(p)) \le C_1(\alpha)/C_0(\alpha)$, whereas pigeonhole approximation gives $\mu_{\mathrm{irr}}(F(p)) \ge 2$. Thus even the optimal reciprocal criterion in this class cannot exceed

$1/2$, which is below $\log 2/\log 3$. Optimising these directions cannot include $3/2$ by that criterion. For other noninteger bases, a new direction would still require the polynomial degree and coefficient-height estimates used in the rational-base transfer; the integer-base source alone does not supply that application. The unrestricted question of which rational bases give irrational values is not settled here and is not reduced to any one of these problems.

Related work and comparison of constructions

The relevant comparisons require agreement of both target and construction. A theorem about the Lambert function need not decide a special value; a theorem at integer bases need not survive rational specialisation; and a shared orthogonal-polynomial kernel need not give the same recurrence on different diagonals. The following examples distinguish these three questions.

Functional nonrationality is not value irrationality. At the level of functions, Rivin proves that if a sequence γ and its divisor-sum sequence are both eventually linearly recurrent, then γ is finitely supported [24, Theorem 1.1, p. 2; proof pp. 6–7]. His periodic-coefficient corollary therefore shows that

$$\sum_{n\geq 1} \frac{z^n}{1-z^n}$$

is not a rational function [24, Corollary 6.4, p. 9]. This is an exact structural statement about the function underlying Problem #1049, but it gives no irrationality statement for a special value at $z = 1/t$: a nonrational function may take rational values at particular rational points.

Padé and orthogonal-polynomial constructions. In 1991 Borwein proved the irrationality of shifted series $\sum_{n\geq 1}(t^n + w)^{-1}$ at integer bases $t \geq 2$, for every nonzero rational w with $w \neq -t^m$ for all $m \geq 1$, by Padé approximation rather than by digit clearing; his estimates also show that these values are not Liouville numbers [3, Thm. 4, pp. 257–258]. In 2001 Van Assche recovered the integer-base irrationality and the bound $\mu_{\mathrm{irr}}(F(p)) \leq 2\pi^2/(\pi^2 - 2) = 2.50828\ldots$ using little q-Legendre Padé approximants [18, Thm. 1, p. 10; proof pp. 10–11]. His more general Theorem 3 proves irrationality of $\sum_{k\geq 1}(cp^k - 1)^{-1}$ for an integer $p > 1$ and a fixed nonzero rational c with $cp^k \neq 1$ for every $k \geq 1$ [18, Thm. 3, p. 14]; it does not cover a rational noninteger base t in $F(t)$, because the multiplier needed to write t^k over an integer base varies with k.

Related series and their different recurrences. The same Lambert value was already the target of Amdeberhan and Zeilberger's q-WZ construction [5]. Here $p > 1$ is the integer-base parameter, $q = p^{-1}$, and $P_n(x \mid q)$ denotes the little q-Legendre polynomial used by these authors, normalised by $P_n(0 \mid q) = 1$. The two constructions share that bivariate little-q-Legendre Padé kernel, but Van Assche's diagonal does not satisfy the Amdeberhan–Zeilberger scalar recurrence. Their Theorems 1 and 2 do prove, respectively, the irrationality of the non-alternating $h_p(1)$ and alternating q-logarithm values for their stated integer parameters, with reported irrationality measure 4.80; those are

external integer-parameter results and do not transfer to the rational noninteger base 3/2 studied here.

Amdeberhan–Zeilberger use the moving diagonal $P_n(p^{n+1} \mid p^{-1})$, whereas Van Assche uses $P_n(p^n \mid p^{-1})$. Van Assche also records, citing Borwein's 1992 Lemma 2, the neighbouring evaluation $P_{n-1}(cp^{n+1} \mid p^{-1})$ [18, (23) and the following paragraph, p. 6]; see [4, Lemma 2, p. 143]. The shared kernel therefore does not license transfer of recurrence, endpoint, lattice, or valuation claims between the diagonals. Indeed, if $A_n(p) = P_n(p^n \mid p^{-1})$, exact substitution at $n = 0$ into the Amdeberhan–Zeilberger operator [5, Sec. 1.5, p. 2] leaves

$$-p(p-1)^2(p+1)(p^5+2p^4+2p^3+2p^2+2),$$

which is nonzero for every real $p > 1$. One nonzero residual is decisive for non-transfer of that recurrence.

Lean checks the exact residual factorisation and its nonvanishing for $p > 1$. These are finite statements at $n = 0$; they do not supply a recurrence for either diagonal. No recurrence, endpoint, lattice, or valuation statement for one diagonal is used for the other; the displayed exact residual is the sole claim made here about their incompatibility.

Coussement and Smet use little q-Jacobi polynomials for further Lambert series whose base parameter is the reciprocal of an integer [19, Thms. 1.2–1.3, p. 2; Sec. 2], and Koizumi and Yokoi identify one of their three-parameter Apéry-type approximations with the Coussement–Smet Padé approximants [20, Sec. 7, Prop. 7.1, p. 33]. That identification includes the source's moving evaluation point $z_n = p_2^{n+1}$ and its normalization; it is not an identification of all little-q-Jacobi diagonals or their scalar recurrences.

At a noninteger rational base, convergence of the rational approximants is not enough. If $P_n/Q_n \to \xi$, the cleared error is $Q_n\xi - P_n = Q_n(\xi - P_n/Q_n)$, whose size also depends on Q_n. For example, $1/n \to 0$, but the corresponding cleared error is always -1. Thus convergence alone does not give the small integer linear forms required for irrationality.

In 2013 Vandehey proved that $\sum_{n\geq 1} d(n)a_n/b^n$ is irrational whenever $b > 1$ is an integer and (a_n) ranges in a finite integer alphabet excluding zero; taking $a_n = (-1)^n$ completes the elementary digit method for integer bases $b \leq -2$ [21, Thm. 1.2, p. 2]. His companion theorem permits a finite alphabet of nonnegative integers containing zero, provided the coefficient sequence is not eventually zero [21, Thm. 1.1, p. 2]. All of these statements require an integer base. Luca and Tachiya's periodic-coefficient theorem likewise proves irrationality at every integer base of absolute value greater than one; their full-support example strengthens this to joint linear independence for finite families of iterated divisor functions [22, Theorem A, p. 139; Example 1, p. 140]. Duverney and Tachiya refine the Chowla–Erdős method to linear independence results for Lambert series at an integer base q with $|q| > 1$ [23, Thms. 1.1–1.2, pp. 2–3]; their Corollary 1.1 generalises Vandehey's Theorem 1.2 [23, p. 3]. Each of these results assumes an integer base. For the underlying small-integral-tail criterion, see also Duverney [35, Thm. 2, p. 4 of the author manuscript]. We use that elementary criterion only as methodological context, not his separate square-index linear-independence argument.

In 2004 Zudilin obtained the uniform bound $\mu_{\mathrm{irr}}(F(p)) \leq 2.46497868\ldots$ for every integer $p \geq 2$, by way of Heine's basic transform and a permutation group [8, Thm. 1,

p. 154; Secs. 4–5, pp. 159–162]. His source theorem also covers negative integer inverse bases; our notation $F(t)$ and positive-remainder arguments are restricted to $t > 1$. The ordinary-hypergeometric antecedent is Rhin and Viola's S_5 action and twelve-coset denominator reduction for rational forms in $\zeta(2)$ [17, Sec. 3, pp. 38–42; Sec. 4, pp. 46–51].

These sources suggest looking for common polynomial factors before specialisation. The congruence lemmas in Section 5 describe what the first and last coefficients can imply at $3/2$. Their hypotheses have not been proved there for either cited coefficient family.

The neighbouring problem of Lambert subseries $\sum_{n\in A}(t^n-1)^{-1}$ over a restricted index set A is treated by Kovač and Tao [25, Sec. 2.1.2 and Thm. 2.3, pp. 4–5].

The rational non-integer progress relevant here is instead the height criterion of Bundschuh and Väänänen [6, Thm. 2, p. 177], which gives the strongest explicit numerical threshold for this value among the previously published sources compared here; Theorem 2.2 enlarges the region that criterion covers. Zudilin's later Padé-and-Hankel treatment of the generalized q-logarithm also comments on non-integral rational bases: it remarks that its results can be given for non-integer $p = r/s$ with $|p| > 1$, under an assumption $\log|r| > c\log|s|$ for some computable $c > 0$ [9, Sec. 2, p. 4]. No value of c is computed there. That remark announces the shape of a rational-base extension for the generalized q-logarithm of that paper. Section 2 proves an explicit extension for F using the different, 2004 construction. Its polynomial cancellation and denominator calculation show that $c = \mu = 2.464978683574975\ldots$, the 2004 bound quoted above, is admissible, and gives the resulting region and its application to $31/4$.

A third published rational-base region for this same series is Duverney's Théorème 2, whose hypothesis is $\log|s|/\log|r| < \frac{1}{3}(1-3/\pi^2) = 0.2320\ldots$ [14, Théorème 2, p. 174]; his Théorème 1 for general numerators is weaker still. Both cutoffs lie strictly inside the Bundschuh and Väänänen region. Matala-aho, Väänänen and Zudilin's combined treatment of q-logarithms keeps the integer hypothesis $p = 1/q \in \mathbb{Z}\setminus\{0,\pm1\}$ throughout, and states in print that its methods do not sharpen the q-harmonic case of [8] [15, abstract p. 879; introduction p. 880]. So none of the printed regions for F compared here reaches $\log 4/\log 31$.

Statements and declarations

Funding and competing interests. This work received no external funding. The author declares no competing interests.

Acknowledgements. The problem numbering and status follow the Erdős Problems catalogue maintained by Thomas Bloom [26].

A Verification and reproducibility

The evidence record identifies the supplied Lean declarations and recorded Comparator comparisons; no fresh execution or independent review of this revision is reported. Theorem 3.6 retains its checked polynomial shift-bound hypotheses, and its proof does not use the ratio-limit remark. That broader argument, the rational-value model, its denominator calculation and the height bound have ordinary proofs here but no Lean proofs

in the supplied record. The unformalised material is kept separate from the checked fixed-base theorem. The source revision includes the Lean sources, toolchain and library manifest.

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

Formal verification and computations. The printed proofs of Theorems 2.2 and 2.3 use Zudilin's construction: his Lemma 7 in its polynomial reading, together with the inputs of that lemma's own proof, namely his Lemma 3, the exponent $M(a;b)$ of (16) proved by his Lemma 4, the group stability of his Lemma 5, and the identity (9)–(11), all in [8, pp. 156–161], and his direction and constants [8, p. 162]. Proved here are the nonrationality of F as a function, the identification of the polynomial coefficients in Lemma 7, the exact degrees of U_n and V_n, positivity, the Archimedean size estimate, and the limit $(K_n - W_n)/n^2 \to C_0$. That last limit rests on Zudilin's Lemmas 1 and 2 [8, p. 155], whose method goes back to [7, Lemma 1, p. 466]; Section 2.3 proves them by that method with an explicit truncation estimate. Lean proves both theorems and the measure bound of Corollary 2.4; it constructs the polynomials and proves their remainder estimates. The Lean sources also contain the finite comparisons, the rational bracket in Section 2.5 and the degree identities in Section 2.2. Appendix A distinguishes these proofs from computer algebra.

Statements for every rank or every base. Lean constructs the cancelled polynomial forms of Section 2 and proves irrationality in the stated region, irrationality at every positive integral power of $31/4$ and the irrationality-exponent bound. It proves the first nonzero term of every transformed row of the 2016 determinant, deduces the order and leading coefficient of the determinant at every rank, and proves the fixed-base asymptotic of Section 3.2. Both constructions therefore supply their mathematical objects; neither is assumed to exist. Proposition 10.5 is checked with no additional hypothesis; its Lean proof uses stability under $z \mapsto z^2$ alone, without the theorem of Adamczewski and Bell [16, Thm. 1.1, p. 6] cited in the printed proof, as explained after that proof.

Statements through rank eight. Proposition 3.9 is checked in Lean for every real $p > 1$ and every rank $N \leq 8$. The Lean proof includes the identity $F(p)A_N - B_N = (v^*_{i+j})$, the positivity of the eight unshifted determinants $D_{k,0}$ by the certificate described in Section 3.4, the reality of the pencil roots, their position below $F(p)$ and their interlacing. It has no all-rank coefficient-positivity or root-convergence conclusion.

Computations outside Lean. The shifted determinants $D_{k,1}$ and the strict positivity of every coefficient rest on the sixteen-polynomial computation of Section 3.4. The reconstruction of U_n, V_n for $n \leq 4$ in Section 2.7, the five complete polynomial contents and the seventy-six cyclotomic residue witnesses have exact reproduction scripts and direct determinant checks. Separate exact computations certify the rational interval bounds

and the finite parameter search. The historical high-precision real remainder evaluations are separate numerical reports, not reproduced by these exact computations. None of these finite computations is a Lean declaration or is needed for the proof of Theorem 2.2.

The Lambert identity and the integer case. The cited *Formal Conjectures* file contains unproved declarations for the conjecture and the integer-base irrationality theorem, as well as a proved identity between the two series [29]. For $t > 1$, that identity is an equality of convergent series. The file also treats a different range using Lean's convention that an unsummable series has totalised sum zero; that convention is not used in the present argument. The proved identity should therefore not be confused with a proved irrationality statement. The rational-base formal proofs used here are those described above.

References

[1] P. Erdős, *On arithmetical properties of Lambert series*, J. Indian Math. Soc. (N.S.) **12** (1948), 63–66.

[2] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[3] P. B. Borwein, *On the irrationality of* $\sum 1/(q^n + r)$, J. Number Theory **37** (1991), no. 3, 253–259, doi:10.1016/S0022-314X(05)80041-1.

[4] P. B. Borwein, *On the irrationality of certain series*, Math. Proc. Cambridge Philos. Soc. **112** (1992), no. 1, 141–146, doi:10.1017/S030500410007081X. Van Assche cites Lemma 2 for the neighbouring little-q-Legendre evaluation used above.

[5] T. Amdeberhan and D. Zeilberger, *q-Apéry irrationality proofs by q-WZ pairs*, Adv. Appl. Math. **20** (1998), no. 2, 275–283, doi:10.1006/aama.1997.0565; arXiv:math/9804122v1. Page references are to arXiv:math/9804122v1.

[6] P. Bundschuh and K. Väänänen, *Arithmetical investigations of a certain infinite product*, Compositio Math. **91** (1994), no. 2, 175–199.

[7] W. Zudilin, *Remarks on irrationality of q-harmonic series*, Manuscripta Math. **107** (2002), no. 4, 463–477, doi:10.1007/s002290200249.

[8] W. Zudilin, *Heine's basic transform and a permutation group for q-harmonic series*, Acta Arith. **111** (2004), no. 2, 153–164, doi:10.4064/aa111-2-4. Page references are to the printed journal pages.

[9] W. Zudilin, *On the irrationality of generalized q-logarithm*, arXiv:1601.02688; Res. Number Theory **2** (2016), Art. 15, doi:10.1007/s40993-016-0042-x. Page references are to arXiv:1601.02688v2. The remark that the results extend to non-integer $p = r/s$, $|p| > 1$, under an assumption $\log|r| > c \log|s|$ for a computable $c > 0$, is in Section 2, p. 4, in the paragraph beginning "Finally, we remark"; no value of c is computed there, and the remark is made for the generalized q-logarithm of that paper.

[10] W. Zudilin, *A determinantal approach to irrationality*, Constr. Approx. **45** (2017), no. 2, 301–310, doi:10.1007/s00365-016-9333-7; arXiv:1507.05697v1. Page and equation references are to arXiv:1507.05697v1.

[11] W. Zudilin, *A determinantal approach to irrationality*, arXiv:1507.05697v2. Propositions 1–2 are cited from v2; the preceding Heine-expansion locators retain their separately cited v1 edition.

[12] P. Bundschuh and W. Zudilin, *Rational approximations to a q-analogue of π and some other q-series*, in H. P. Schlickewei, K. Schmidt and R. F. Tichy (eds.), *Diophantine Approximation*, Dev. Math. **16**, Springer, 2008, pp. 123–139, doi:10.1007/978-3-211-74280-8_6.

[13] C. Krattenthaler, I. Rochev, K. Väänänen and W. Zudilin, *On the non-quadraticity of values of the q-exponential function and related q-series*, Acta Arith. **136** (2009), no. 3, 243–269, doi:10.4064/aa136-3-4; arXiv:0812.2921v1. Page references are to arXiv:0812.2921v1.

[14] D. Duverney, *À propos de la série $\sum_{n\geq 1} x^n/(q^n-1)$*, J. Théor. Nombres Bordeaux **8** (1996), no. 1, 173–181. Théorème 2 on p. 174 gives the rational-base region $\log|s|/\log|r| < \frac{1}{3}(1-3/\pi^2) = 0.2320\ldots$ for this series; Théorème 1, for a general numerator, is weaker.

[15] T. Matala-aho, K. Väänänen and W. Zudilin, *New irrationality measures for q-logarithms*, Math. Comp. **75** (2006), no. 254, 879–889, doi:10.1090/S0025-5718-05-01812-0. The hypothesis $p = 1/q \in \mathbb{Z}\setminus\{0,\pm 1\}$ is carried in the abstract on p. 879 and in both theorem statements on p. 880, where the authors also record that their methods do not sharpen the q-harmonic case of [8].

[16] B. Adamczewski and J. P. Bell, *A problem about Mahler functions*, Ann. Sc. Norm. Super. Pisa Cl. Sci. **17** (2017), no. 4, 1301–1355; arXiv:1303.2019v1, 2013. Theorem 1.1 on p. 6 of arXiv:1303.2019v1: over a field of characteristic zero, a power series is both k- and ℓ-Mahler for multiplicatively independent k, ℓ if and only if it is a rational function. Bell and Smertnig cite it as Theorem 1.3 of the journal version [27, Thm. 2.5, p. 5].

[17] G. Rhin and C. Viola, *On a permutation group related to $\zeta(2)$*, Acta Arith. **77** (1996), no. 1, 23–56, doi:10.4064/aa-77-1-23-56.

[18] W. Van Assche, *Little q-Legendre polynomials and irrationality of certain Lambert series*, Ramanujan J. **5** (2001), no. 3, 295–310, doi:10.1023/A:1012930828917. Page references are to arXiv:math/0101187v1.

[19] J. Coussement and C. Smet, *Irrationality proof of certain Lambert series using little q-Jacobi polynomials*, J. Comput. Appl. Math. **233** (2009), no. 3, 680–690, doi:10.1016/j.cam.2009.02.036; arXiv:math/0701345v1. Page references are to arXiv:math/0701345v1.

[20] J. Koizumi and A. Yokoi, *Apéry-type approximations and irrationality measures for certain q-series*, arXiv:2608.26918v1, 27 August 2026.

[21] J. Vandehey, *On an incomplete argument of Erdős on the irrationality of Lambert series*, Integers **13** (2013), Paper A58. Page references are to arXiv:1206.0340v1 (2012).

[22] F. Luca and Y. Tachiya, *Linear independence results for the values of divisor functions series*, RIMS Kôkyûroku No. 2014 (2017), 138–150. Theorem A on p. 139 restates the periodic-coefficient irrationality theorem; Example 1 on p. 140 gives the divisor-function specialization.

[23] D. Duverney and Y. Tachiya, *Refinement of the Chowla–Erdős method and linear independence of certain Lambert series*, Forum Math. **31** (2019), no. 6, 1557–1566, doi:10.1515/forum-2018-0299. Page references are to the authors' version.

[24] I. Rivin, *Zero Coefficients of Rational Power Series and Rational Lambert Series*, arXiv:2604.25151v1, 28 April 2026. Theorem 1.1 is on p. 2 and proved on pp. 6–7; the periodic-coefficient Corollary 6.4 is on p. 9.

[25] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, Acta Math. Hungar. **175** (2025), no. 2, 572–608, doi:10.1007/s10474-025-01528-0; arXiv:2406.17593v4. Page references are to arXiv:2406.17593v4.

[26] T. F. Bloom, *Erdős Problem #1049*, `erdosproblems.com/1049`. Accessed 28 July 2026, when the page displayed "last edited 28 September 2025".

[27] J. Bell and D. Smertnig, *Mahler series with multiplicative coefficient sequences*, arXiv:2603.23456v1, 24 March 2026. Theorem 1.3 is on pp. 2–3; its stated consequences on p. 3 include that the divisor and totient generating series are not k-Mahler for any $k \geq 2$.

[28] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, `dlmf.nist.gov/17.2.E37`, Eq. 17.2.37 in §17.2(iii), accessed 15 September 2026.

[29] The Formal Conjectures Authors, *FormalConjectures.ErdosProblems.1049*, Lean source at commit `f776d2f`, 2026, accessed 28 July 2026. The irrationality declarations are unproved; the Lambert-series identity is proved.

[30] K. Postelmans and W. Van Assche, *Irrationality of $\zeta_q(1)$ and $\zeta_q(2)$*, J. Number Theory **126** (2007), no. 1, 119–154, doi:10.1016/j.jnt.2006.11.011. Page references are to arXiv:math/0604312v1 (2006).

[31] R. Kozhan and M. Vaktnäs, *Christoffel transform and multiple orthogonal polynomials*, arXiv:2407.13946v1 (2024). The moment-transform comparison uses the introduction, (3)–(4), of v1.

[32] C. Krattenthaler, *A determinant identity for moments of orthogonal polynomials that implies Uvarov's formula for the orthogonal polynomials of rationally related densities*, arXiv:2103.03969v1 (2021). Page references are to this version.

[33] Y. Wang and B.-X. Zhu, *Log-convex and Stieltjes moment sequences*, Adv. Appl. Math. **81** (2016), 115–127, doi:10.1016/j.aam.2016.06.008. Page references are to arXiv:1612.04114v1.

[34] R. P. Stanley, *Smith normal form in combinatorics*, J. Combin. Theory Ser. A **144** (2016), 476–495, doi:10.1016/j.jcta.2016.06.013; arXiv:1602.00166v1. Page references are to arXiv:1602.00166v1.

[35] D. Duverney, *Arithmetical functions and irrationality of Lambert series*, AIP Conf. Proc. **1385** (2011), 5–16, doi:10.1063/1.3630035. Theorem and page references are to the 12-page author manuscript.

[36] C. Berg, *On powers of Stieltjes moment sequences, II*, J. Comput. Appl. Math. **199** (2007), 23–38; arXiv:math/0412340v1. Theorem references use the preprint; Theorem 5.1 treats factorial powers.

[37] A. D. Sokal and J. Walrad, *Continued-fraction characterization of Stieltjes moment sequences with support in $[\xi, \infty)$*, arXiv:2404.12131v1, 2024. The classical Stieltjes criterion is recalled on pp. 1–2.

[38] H. Liang, J. Remmel and S. Zheng, *Stieltjes moment sequences of polynomials*, arXiv:1710.05795v1, 2017.

[39] M. Pétréolle, A. D. Sokal and B.-X. Zhu, *Lattice paths and branched continued fractions: An infinite sequence of generalizations of the Stieltjes–Rogers and Thron–Rogers polynomials, with*

coefficientwise Hankel-total positivity, Mem. Amer. Math. Soc. **291** (2023), no. 1450, doi:10.1090/memo/1450; arXiv:1807.03271v2. Theorem and page references use that preprint version.

[40] J. Middeke, D. J. Jeffrey and C. Koutschan, *Common Factors in Fraction-Free Matrix Decompositions*, Math. Comput. Sci. **15** (2021), no. 4, 589–608, doi:10.1007/s11786-020-00495-9; arXiv:2005.12380v1. Section and theorem references use the preprint.

[41] G. H. Golub and J. H. Welsch, *Calculation of Gauss Quadrature Rules*, Math. Comp. **23** (1969), no. 106, 221–230, doi:10.1090/S0025-5718-69-99647-1.

[42] C. R. Vinroot, *Multivariate Rogers–Szegő polynomials and flags in finite vector spaces*, arXiv:1011.0984v1, 3 November 2010, abstract accessed 20 September 2026. Cited for the identification of the sum of all q-multinomial coefficients of fixed degree and length with a flag count, and for its recursion generalising the Galois numbers. The factorisation of the moment weights is proved here.

Verification concordance

Statement	Formal proof	Statement	Formal proof
Lemma 2.1	Lean · Comparator	Theorem 2.2	Lean · Comparator
Theorem 2.3	Lean · Comparator	Corollary 2.4	Lean · Comparator
Theorem 2.5	Lean · Comparator	Corollary 2.6	Lean · Comparator
Lemma 2.7	Lean · Comparator	Theorem 3.1	Lean · Comparator
Corollary 3.2	Lean · Comparator	Theorem 3.3	Lean · Comparator
Theorem 3.4	Lean · Comparator	Lemma 3.5	Lean · Comparator
Theorem 3.6	Lean · Comparator	Proposition 3.7	Lean · Comparator
Theorem 3.8	Lean · Comparator	Remark 1, p. 35	Verification record
Remark 2, p. 38	Verification record	Remark 3, p. 39	Verification record
Remark 4, p. 40	Verification record	Proposition 3.9	Lean · Comparator
Theorem 4.1	Lean · Comparator	Theorem 5.1	Lean · Comparator
Proposition 5.3	Lean · Comparator	Corollary 5.5	Lean · Comparator
Proposition 5.6	Lean · Comparator	Lemma 5.7	Lean · Comparator
Theorem 5.8	Lean · Comparator	Corollary 5.9	Lean · Comparator
Theorem 5.10	Lean · Comparator	Theorem 5.11	Lean · Comparator
Theorem 5.12	Lean · Comparator	Theorem 5.14	Lean · Comparator
Theorem 6.2	Lean · Comparator	Proposition 6.4	Lean · Comparator
Theorem 6.5	Lean · Comparator	Theorem 7.1	Lean · Comparator
Theorem 7.2	Lean · Comparator	Theorem 8.1	Lean · Comparator
Proposition 9.1	Lean · Comparator	Proposition 10.3	Lean · Comparator
Proposition 10.5	Lean · Comparator		